MATHEMATICS

GRADE 7

STUDENT TEXT BOOK

AUTHORS

Bezuneh Mekonnen
Melesse Abate
Sewalem Alameneh

Editors and Evaluators

Mustefa Kedir
Gesset Assefa
Rodas Diriba

Lay out and Illustration design

Girma Darge
HileGiorgis Temesgen

Coordinator

Getachew Talema Atinafu

ADDIS ABABA CITY ADMINISTRATION EDUCATION BUREAU

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Unit 1
Basic Concepts of Sets

Learning Outcomes:

At the end of this unit, you should able to:

- *understand the concept of set.*
- *describe the relation between two sets.*
- *perform two operations (intersection & union) on sets.*

Introduction

In mathematics the concept of set is important to describe collection of objects. In this unit you will learn about the concept of set, relationship between two sets and perform the operations union and intersection on sets.

1.1. Introduction to sets

By the end of these section you should be able to:

- *explain what is meant by "set".*
- *explain what is meant by "element".*

Activity 1.1.1

Discuss with your friends/partners

1. Write the collection of all vowels in English alphabet.
2. Write the collection of all tall students in your class.
3. Define the term “set” in your own words.
4. If the set of natural numbers less than 5 is described using the braces symbol "{ }" as
   {1,2,3,4}, then
   a. list each of the elements of the given set.
   b. Is 5 the member the given set?
5. List all students in your class, whose age is 100 years.
6. What do we call a set which has no element?

Definition 1.1: Set is a collection of well-defined objects.

- The individual objects in a set are called members or elements of the set.
- The braces "{ }" are used to enclose the members of the set.
- Each elements of a set separated by comma are listed between a pair of braces { } without repetition and without concern about the order of their appearance.
- We use capital letters like A, B, C, etc to denote sets. Small case letters are used to denote elements of the set. For example, the set of all vowels in English alphabet can be represented by a single letter A and described as

$$A = \{a, e, i, o, u\}$$

- If a is a member of set A, then we write $a \in A$ (read as a is an element of set A).
- If a is not a member of set A, then we write $a \notin A$ (read as a is not an element of set A).

Example1: If $N = \{2,4,6,8,10\}$, then which of the following are true and which of them are false.

a. $6 \in N$ b. $12 \notin N$ c. $3 \in N$ d. $2 \notin N$

Solution:

a. True, because 6 is found in the set N.
b. True, because 12 is not found in set N.
c. False, because 3 is not an element of set N.
d. False, because 2 is an element of set N.

Example 2: Identify whether each of the following are sets or not sets.

a. The collection of all whole numbers less 4.
b. The collection five most talented writers of Ethiopia.

Solution:

a. The collection of all whole numbers less 4 contains four elements, namely 0,1,2 and 3. So, this collection is well defined and therefore it is a set.
b. The collection five most talented writers of Ethiopia is not a set, since no rule has been given for deciding whether a given writer is talented or not.

Example 3: How many elements does each of the following sets contain?

a. $B = \{1,2,4,1\}$
b. A = The set of whole numbers less than 0.

Solution:

a. Set B contains 3 elements and the elements are 1, 2 and 4. Repeating the same element in the set do not increase the number of elements.
b. There is no whole number less than 0. So, set A does not have element.

Note: A set which do not have an element is called empty set. We use the symbol { } or Ø to represent empty set.

Example 4: which of the following sets are empty set?

a. A = The set of natural numbers less than 1.
b. B = The set of cats that can fly.
c. C = The set of whole numbers between 4 and 5
d. D = The set of rivers in Ethiopia.

Solution:

a. There is no natural number less than 1. So, set A is empty set.
b. There are no cats that can fly. So, set B is empty set.
c. There is no whole number between 4 and 5. So set C is empty set.
d. Set D is not empty set. Since there are rivers like Abay river, Baro river.

Exercise 1.1.1

1. For the following questions write true if the statement is correct and write false if the statement is incorrect.
   a. $0 \in \{0.5,1.2,4,10\}$
   b. $10 \in \{0,2,4, \ldots\}$
   c. $1 \notin \{1,2,5,7\}$
   d. $8 \notin \{1,3,5, \ldots\}$
2. Identify whether each of the following are sets or not sets.
   a. The collection of intelligent students in your class.
   b. The set of months of the year which starts with letter "J".
   c. The set of natural numbers between 20 and 21.
3. How many elements does each of the following sets contain.
   a. $A = \{2,2,2,2\}$
   b. $C = \{\ \}$
   c. $B = \{a, b\}$
   d. D = the set of whole numbers less than 6.
4. Which of the following are empty set?
   a. A = the set of trees in Addis Ababa
   b. M = the set of students in your class whose age is 30 years.

1.2. Relations among sets

By the end of these section you should be able to:

- *describe relationship among sets such as proper subset, subset, equal and equivalent sets.*

A. Equal sets

Definition 1.2 Two non-empty sets A and B are said to be equal, if they have exactly the same elements and write, $A = B$.

Example 1: Let $A = \{p, q, r, s\}$ and $B = \{q, r, p, s\}$. Are A and B equal?

Solution: A and B have exactly the same elements, so $A = B$

Example 2: Let $C = \{1,1,2,2,3\}$ and $D = \{1,2,3\}$. Are C and D equal?

Solution: repetition of elements in a set has no meaning, so

$C = \{1,1,2,2,3\} = \{1,2,3,\}$. Therefore, $C = D$

Example 3: Let $E = \{u, v, x\}$ and $F = \{x, , u, 3\}$. Are E and F equal?

Solution: elements in set E are not exactly the same as the elements in set F, so E and F are not equal (i.e. $E \neq F$).

B. Equivalent sets

Definition 1.3 Two sets A and B are said to be equivalent, written $A \leftrightarrow B$, if and only if they have equal number of elements.

Example 4: Let $E = \{u, v, x\}$ and $F = \{x, , u, 3\}$. Are E and F equivalent?

Solution: E and F have equal number of elements, so $E \leftrightarrow F$.

Example 5: Let $A = \{p, q, r, s\}$ and $B = \{q, r, p\}$. Are A and B equivalent?

Solution: Set A and B are not equivalent, since the number of elements in A and B are not equal.

Example 6: Let $C = \{1,2,3\}$ and $D = \{3,2,1\}$. Are C and D equivalent?

Solution: C and D have equal number of elements, so $C \leftrightarrow D$.

Note: All equal sets are equivalent, but all equivalent sets are not equal sets.

C. Subsets

Definition 1.4 A set A is said to be a subset of set B if every element of A is also an element of B, and write, $A \subseteq B$

Example 7: Let $A = \{2,3,5\}$ and $B = \{2,3,5,7,9\}$. Is A subset of B?

Solution: every element of A is an element of B. so, $A \subseteq B$

Example 8: Let A $= \{a, c, e\}$ and B $= \{a, e, i, o, u\}$. Is A subset of B?

Solution: there is an element in A which is not B, for example $c \in$ A, but $c \notin$ B.therefore A is not subset B (i.e. A $\not\subset$ B).

Example 9: let M $= \{\quad\}$ and N $= \{a, d\}$, then which of the following is true?

a. M $\subseteq$ N b. N $\subseteq$ N

Solution: a. $\{\quad\}$ contains no element at all, so there is no element of $\{\quad\}$ which is not contained in N, hence M $\subseteq$ N is true.

b. All elements in N are also in N, hence N $\subseteq$ N

Example 10: Let A $= \{1,2\}$, then list all subsets of set A

Solution: $\{1\} \subseteq$ A, $\{2\} \subseteq$ A, $\{1,2\} \subseteq$ A and $\{\quad\} \subseteq$ A, therefore all subsets of set A are $\{1\}, \{2\}, \{1,2\}$ *and* $\{\quad\}$.

Note:

1. Empty set is a subset of any set (i.e. for any set A, $\{\quad\} \subseteq$ A).
2. Any set is a subset of itself (i.e. for any set A, A $\subseteq$ A).

D. Proper subsets

Definition 1.5 A set A is said to be a proper subset of set B, written A $\subset$ B, if and only if A is a subset of B and B has at least one more element.

Example 11: Let A $= \{a, e, c\}$ and B $= \{a, c, d, e, \}$. Is A proper subset of B?

Solution: all elements in A are also in B, and B has additional element, so A $\subset$ B.

Example 12: let D $= \emptyset$ and E $= \{a, d\}$, then which of the following is true?

a. D $\subset$ E b. E $\subset$ E

Solution: a. since empty set is a proper subset of any

Note:

1. Empty set is a proper subset of any other set (i.e. for any set A different from empty set, $\{\quad\} \subset$ A).
2. Any set is not a proper subset of itself (i.e. for any set A, A $\not\subset$ A).

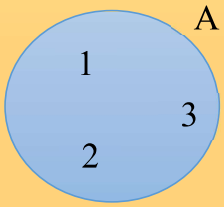
other set, D ⊂ E is true.

b. Since a set is not a proper subset of itself, E ⊂ E is false.

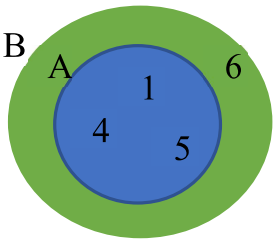
Example 13: Let A = $\{a, c\}$, then list all proper subsets of set A

Solution: $\{a\}$ ⊂ A, $\{c\}$ ⊂ A, and ∅ ⊆ A, therefore all proper subsets of set A are $\{a\}, \{c\}, and\ \emptyset$.

Note: Venn diagram is a diagram that shows collection of numbers or objects in a circle. For example, A = {1,5,6}, using Venn diagram it is represented as



Example 14: let A = {1,4,5} and B = {1,4,5,6}. A ⊂ B, this relation in Venn diagram is represented as follows:



Exercise 1.2.1

1. State whether each of the following statements is true or false.
   a. {1,4,3} ⊆ {3,4,1} d. $\{a, b, c\} \subset \{c, a, b\}$ g. { } ⊂ {3,4}
   b. {2,6,8} ⊆ {6,8} e. {7,4} ⊂ {7,4,5,6} h. ∅ ⊆ {2}
   c. $\{2, a, 4\} = \{a, 2,4\}$ f. $\{5,7,9\} \leftrightarrow \{x, y, z\}$ i. $\{m, a, c\} = \{a, m, d\}$
2. Let A = $\{0, b\}$, then list all subsets of set A.
3. Let B = $\{m, 2\}$, then list all proper subsets of set B.
4. Let M = {0,4,6,8} and N = {6,8}, then show the relationship between M and N in Venn diagram.
5. Let E = $\{a, x, m, 4\}$ and F = $\{a, 4,8,9\}$, then show the relationship between E and F in Venn diagram.
6. State whether each of the following statements is true or false.
   a. All equal sets are equivalent.
   b. Empty set is a subset of any set.
   c. A set is a proper subset of itself.
   d. If A = B,then A ⊆ B.

1.3. Operations on sets

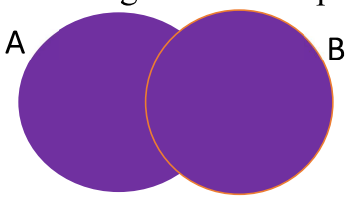
By the end of these section you should be able to:

- *determine the intersection of two given sets.*
- *determine the union of two given sets.*
- *use Venn diagram to represent union and intersection of two set.*

1.3.1 The union of sets

Definition 1.6 The union of two sets A and B, denoted by A ∪ B and read "A union B", is the set of all elements which belong to A or B or both.

The shaded region in the figure below represents A ∪ B



Example 1: If $A = \{3,4,5,6\}$ and $B = \{4,6,8,10\}$, then find A ∪ B.

Solution: clearly $A \cup B = \{3,4,5,6,8,10\}$

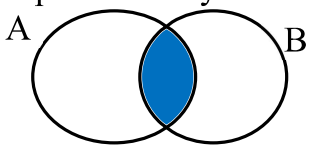
Example 2: If $A = \{a, e, i, o\}$ and $A = \{a, e, i\}$, then find A ∪ B

Solution: $A \cup B = \{a, e, i, o\}$

1.3.2 The intersection of sets

Definition 1.7 The intersection of two sets A and B, denoted by A ∩ B and read "A intersection B", is the set of all elements which belong to both A and B.

Using Venn diagram, A ∩ B is represented by the shaded region



Example 3: If $A = \{3,4,5,6\}$ and $B = \{4,6,8,10\}$, then find A ∩ B.

Solution: $A \cap B = \{4,6\}$

Example 4: If $A = \{a, e, i, o\}$ and $A = \{a, e, i\}$, then find $A \cap B$.

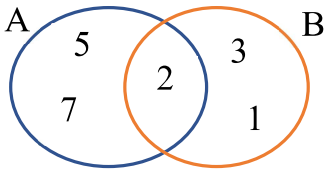
Solution: $A \cap B = \{a, e, i\}$

Example 5: If $A = \{4,7,9\}$ and $A = \{a, e, i\}$, then find $A \cap B$.

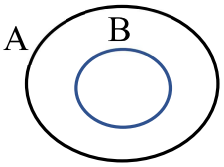
Solution: $A \cap B = \{\ \ \}$

Exercise 1.3.1

1. Given $A = \{1,3,4,6\}, B = \{1,4,3\}, C = \{\ \ \}$ and $D = \{1,2,3,4,5,6\}$,find
   a. $A \cap B$ d. $A \cup B$ g. $(A \cap B) \cup C$
   b. $A \cap C$ e. $C \cup B$ h. $(B \cup C) \cap D$
   c. $A \cap D$ f. $B \cup D$ i. $(A \cap B) \cup (B \cap D)$
2. The relation between set A and B is shown in Venn diagram below



   a. List all elements of set A c. find $A \cap B$
   b. List all elements of set A d. find $A \cup B$
3. In the following Venn diagram below shade $A \cap B$.
4. If $A \subset B$, then
   a. $A \cap B =$ _____
   b. $A \cup B =$ _____



Summary for unit one

- Set is a collection of well-defined objects.
- The individual objects in a set are called members or elements of the set.
- A set which do not have an element is called empty set.
- Two sets A and B are said to be equal, if they have exactly the same elements.
- A set A is said to be a subset of set B, written $A \subseteq B$ if every element of A is also an element of B.
- A set A is said to be a proper subset of set B, written $A \subset B$, if and only if A is a subset of B and B has at least one more element.
- the union of two sets A and B, denoted by $A \cup B$, is the set of all elements which belong to A or B or both.
- The intersection of two sets A and B, denoted by $A \cap B$, is the set of all elements which belongs to both A and B.

Review Exercise for unit 1

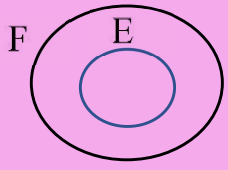
I. State whether the following is true or false.

a. All equivalent sets are equal sets.
b. A set is a subset of itself.
c. $\{m, n, o\} \subseteq \{m, n\}$
d. $2 \in \{0,4,12\}$

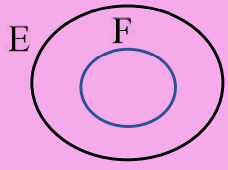
II. Choose the correct answer

1. Which one of the following is **not** a proper subset of the set $\{e, f\}$?
   A. $\{\ \}$ B. $\{e\}$ C. $\{f\}$ D. $\{e, f\}$
2. Which one of the following is equivalent to the set $\{a, 3\}$?
   A. $\{\ \}$ B. $\{2,3\}$ C. $\{a\}$ D. $\{3\}$
3. If $A \subseteq B$, then which one of the following is necessarily true?
   A. $A \cap B = B$ B. $A \cap B = A$ C. $A \cup B = A$ D. $A \subset B$
4. If $A = \{a, b\}, B = \{c, d\}$,then $A \cap B$ is equal to:
   A. $\{\ \}$ B. $\{a, b, c, d\}$ C. $\{c, d\}$ D. $\{d\}$
5. Which one of the following is the Venn diagram representation of $E \subset F$?

A.



B.



Unit 2
Integers

Learning Outcomes:

At the end of this unit, you should able to:

- *understand the concept of integers.*
- *represent integers on a number line.*
- *identify the commutative, associative and distributive properties of operation of integers.*
- *perform the operations addition and subtraction on integers.*
- *apply integers in the real-life situation.*

Introduction

In the previous grades you had already learnt about whole numbers and natural numbers. These numbers together with negative numbers form another set of numbers called integers. In these unit you will learn about integers and the four fundamental operations on integers.

2.1. Revision on whole and natural numbers

By the end of these section you should be able to:

- *describe whole and natural numbers.*
- *identify the relation between whole and natural numbers.*

Activity 2.1.1

Discuss with your friends/partners

1. List down the first three smallest natural numbers
2. List down the first five smallest whole numbers
3. What is the smallest and greatest natural number
4. What is smallest and the greatest whole number
5. Write : a) The set of natural numbers
   b) The set of whole numbers

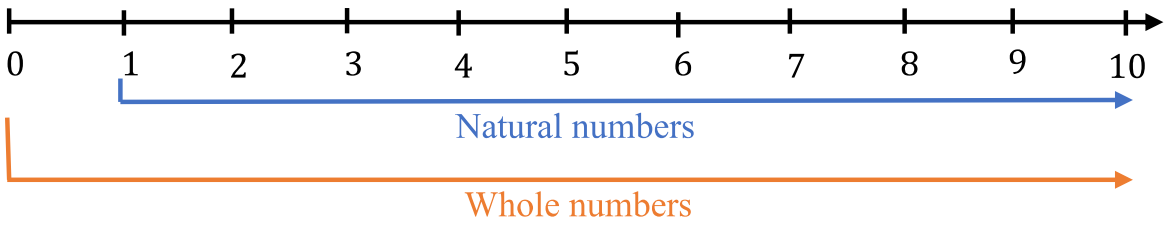
Definition 2.1: **Natural number** is a collection or set of numbers denoted by $\mathbb{N}$ and is described by: $\mathbb{N} = \{1, 2, 3, \ldots\}$

Natural numbers are also called counting numbers.

Definition 2.2: **Whole number** is a number containing natural numbers and zero. Whole number is denoted by $\mathbb{W}$ and is described by:

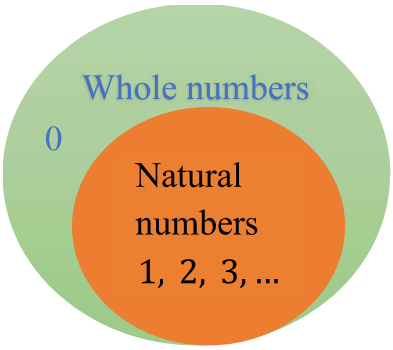
$$\mathbb{W} = \{0, 1, 2, 3, \ldots\}$$

Relationship between natural numbers and whole numbers is described below



Relationship between $\mathbb{N}$ and $\mathbb{W}$

Using Venn diagram, the relationship between natural and whole numbers is shown below.



Venn diagram relationship between $\mathbb{N}$ and $\mathbb{W}$

From above diagrams you can see that,

- All-natural numbers are whole numbers.
- But all whole numbers are not natural numbers.
- A collection of whole numbers without zero is natural numbers.
- A collection of natural numbers and zero is whole numbers.

Revision of operations on natural numbers and whole numbers

Example 1. Find the following sum

a. $452 + 323$
b. $3234 + 598$
c. $423 + 5672$

Solution:

a. $\begin{array}{r} 452 \\ +323 \\ \hline = 775 \end{array}$ b. $\begin{array}{r} 3234 \\ +598 \\ \hline = 3832 \end{array}$ c. $\begin{array}{r} 5672 \\ +432 \\ \hline = 6104 \end{array}$

Example 2. Find the following difference.

a. $453 - 223$
b. $7456 - 352$
c. $456 - 456$

Solution:

a. $\begin{array}{r} 453 \\ -223 \\ \hline = 230 \end{array}$ b. $\begin{array}{r} 7456 \\ -352 \\ \hline = 7104 \end{array}$ c. $\begin{array}{r} 456 \\ -456 \\ \hline = 0 \end{array}$

Example 3. Find the product of the following numbers.

a. $46 \times 44 = 2024$
b. $72 \times 78 = 5616$
c. $123 \times 342 = 42066$

Example 4. Find the quotient of the following

a. $455 \div 5 = 91$
b. $4824 \div 12 = 402$

Note:

- The sum of any two natural numbers is always natural number.
- The sum of any two whole numbers is always whole number.
- The product of any two natural numbers is always natural number.
- The product of any two whole numbers is always whole number
- The quotient of any two natural numbers is not always natural number.
- The difference of any two whole numbers is not always whole number.
- The difference of any two natural numbers is not always natural number.

Exercise 2.1.1

1. Calculate the following.

a. $345 + 892$	d. $23 + (654 - 345)$
b. $342 + 435$	e. $5427 \div 9$
c. $6543 - 234$	f. 342×345

2.2. Introduction to integer

By the end of these section you should be able to:

- *apply real-life applications of integers in terms of temperature, altitude and money to express positive and negative numbers*
- *define the set of integers*
- *indicate integers on the number line.*
- *describe the relations, among natural numbers, whole numbers and integers* $\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z}$

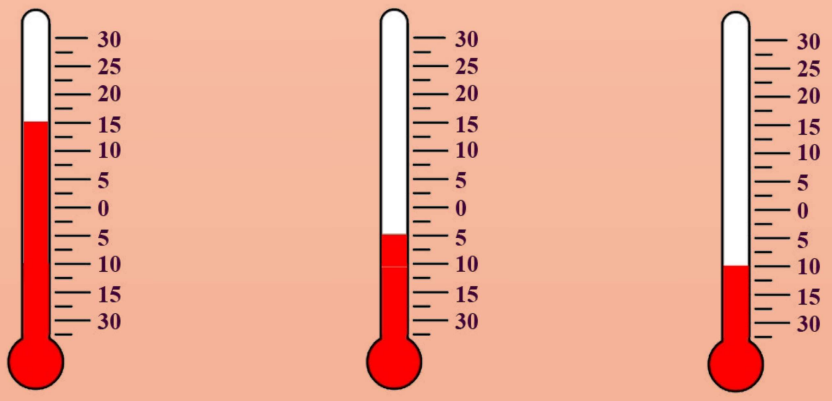
Activity 2.2.1

Discuss with your friends/partners

Anders Celsius the Swedish astronomer who lived between 1701 and 1744 A.D. He devised a way of measuring temperature which was adjusted and improved after his death.

The table below shows that the temperature recorded on three consecutive days for Debre Birhan city

Day	Temperature in degree Celsius
Monday	$15\ ^0$c above zero
Tuesday	5^0c below zero
Wednesday	10^0c below zero



Monday Tuesday Wednesday

Thermometer

1. What was the temperature on each day using the sign plus or minus?
   a. How far did the temperature from zero in each day as shown in table and figure above?
   b. What was the coldest temperature?
   c. What was the hottest temperature?
2. What is the number that comes after zero?
3. What is the number that comes before zero?

Note: Very cold temperatures below zero can be described by number with **minus** sign which are called **negative numbers.**

The temperatures above zero can be described by a number with **plus** sign which are called **positive numbers**.

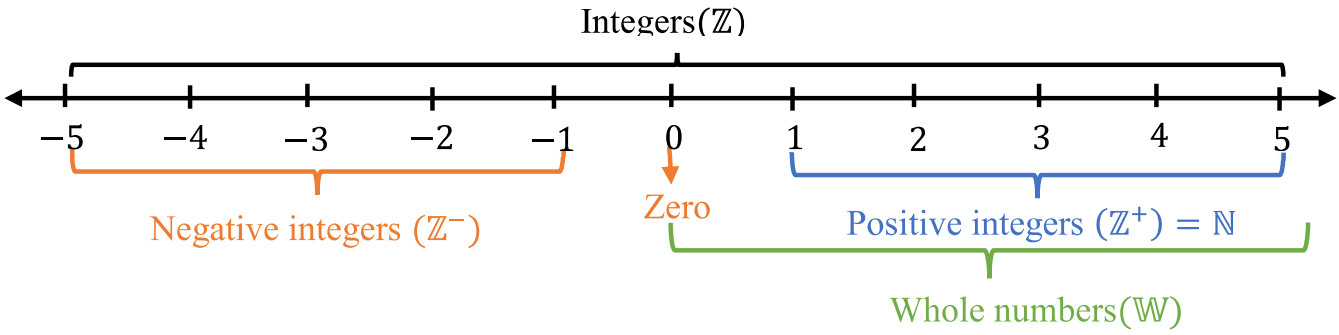
Example 1: The temperature 5^0c below zero can be written as -5^0c and

The temperature 5^0c above zero can be written as $+5^0c$ or 5^0c

Note: positive numbers can be written without plus sign. For instance, $+2$ simply written as 2 .

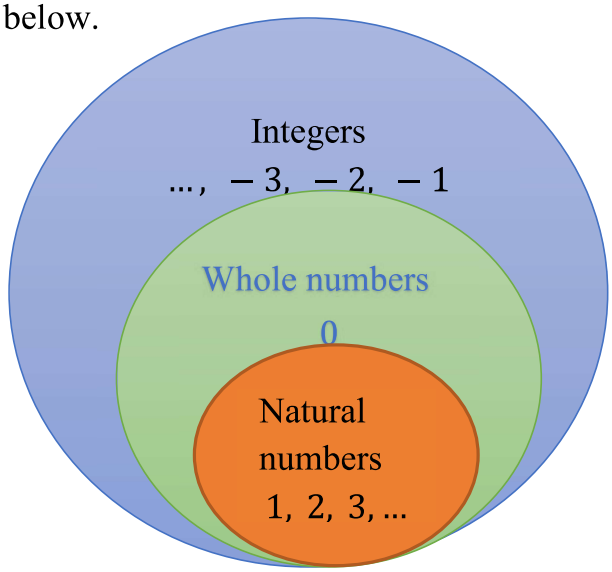
Definition 2.3: An integer is a set of numbers consisting of whole numbers and negative numbers. The set of integers is denoted by

$$\mathbb{Z} = \{\ldots 4, -3, -2, -1, 0, 1, 2, 3, 4, \ldots\}$$



Relationship between ℕ , 𝕎 and ℤ

Using Venn diagram, the relationship between natural numbers, whole numbers and integers is shown below.



Relationship between ℕ , 𝕎 and ℤ

Note:

- Negative integer is denoted by $\mathbb{Z}^-$ and positive integer is denoted by $\mathbb{Z}^+$.
- The set of positive integers is the same as the set of natural numbers.$(i.e.\ \mathbb{Z}^+ = \mathbb{N})$
- Zero is an integer which is neither positive nor negative.
- Decimals and fractions are not integers.
- Integers, whole numbers and natural numbers are related as $\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z}$.
- The number -4 is read as negative four and 4 is read as positive four or four.

Example 2:

a. List five Positive integers

b. List five negative integers

c. Give example of integer which is neither positive nor negative

d. List four examples of numbers which are not integers

Solution:

a. $9, 54, 380, 6284, 89123$ are Positive integers.

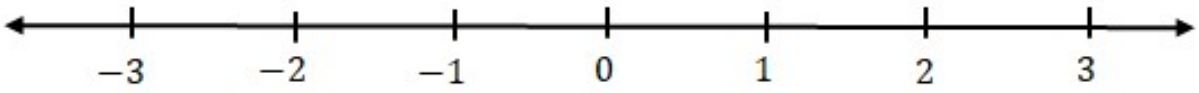
b. $-9, -54, -125\ -647, -32564$, are negative integers.

c. 0

d. $2.5, \frac{3}{4}, 1.08, \frac{10}{7}$ are not integers.

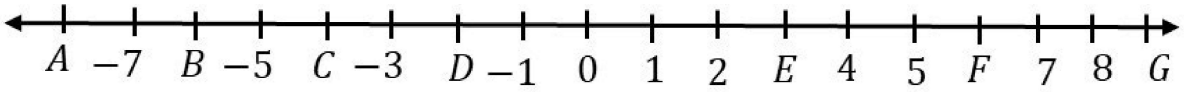
Example 3: Represent all integers from -3 to 3 on number line.

Solution: First draw horizontal line and mark the numbers on the number line

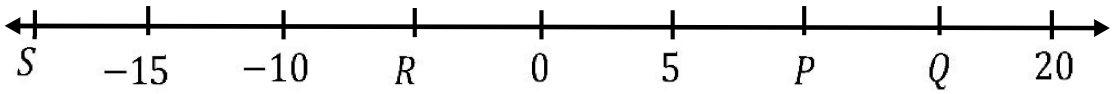


Exercise 2.2.1

1. Give the numbers represented by the letters on the number line below



2. Give the numbers represented by the letters on the number line below



3. Represent the numbers 1, -1, 5, -5, 2 and -2 on a number line.

4. Write down four numbers which are not integer.

5. Write the following integers in word

   a. -3 b. 21 c. -132 d. 99

6. write the following numbers as numeral

   a. Positive four
   b. Negative twenty
   c. negative fifty-three
   d. one hundred one

7. Represent the following fact by using a numeral and $+ \; or \; -$ signs.

   a. A loss of Birr 100.
   b. A rise of 10^0c temperature.
   c. A walk of 7 km forward
   d. Five minutes late.
   e. 6^0c below zero.
   f. 21^0c above zero.

8. Write true for correct statement and false for wrong statement.

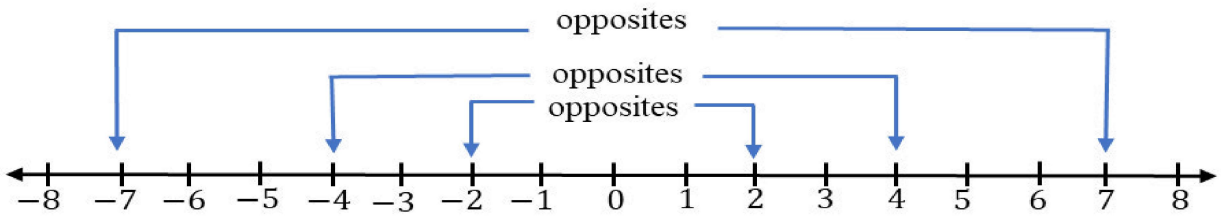
   a. $-2 \in \mathbb{Z}$ d. $\mathbb{N} = \mathbb{Z}^+$ g. $\mathbb{Z}^+ \subseteq \mathbb{N}$ j. $\mathbb{W} \cup \mathbb{N} = \mathbb{N}$
   b. $2.5 \in \mathbb{Z}$ e. $\mathbb{N} \subset \mathbb{W}$ h. $\mathbb{Z}^- \subset \mathbb{Z}$ k. $\mathbb{Z}^+ \cup \mathbb{Z}^- = \mathbb{Z}$
   c. $0 \in \mathbb{Z}^+$ f. $\mathbb{W} \subseteq \mathbb{Z}$ i. $\mathbb{W} \cap \mathbb{N} = \mathbb{N}$ l. $\mathbb{Z}^+ \cup \{0\} = \mathbb{W}$

9. What is the relationship between natural number and positive integer?
10. Explain the relationship between whole number, natural number and integer.
11. From the numbers listed below select integers.

    a. $2\frac{1}{3}$ d. 1.02 g. 0.25
    b. 0 e. 12 h. 455
    c. $\frac{2}{5}$ f. -294

Opposite numbers

Definition 2.4: Opposite numbers are numbers that are at the same distance from zero in opposite direction and the opposite of zero is zero.



Using the above number line, the opposite of some integers is listed in the table below

Number	**-10**	**-9**	**-8**	**-7**	**-6**	**0**	**2**	**3**	**4**	**5**
Opposite number	10	9	8	7	6	0	-2	-3	-4	-5

Example 4: Find the opposite of the following numbers

a.12

b. -9

c. 413

d. -564

Solution:

a. The opposite of 12 is -12
b. The opposite of 413 is-413
c. The opposite of 0 is 0
d. The opposite of -564 is 564

Note: For any integer $a \neq 0$

i. The opposite of a is $-a$
ii. For integer $a, -(-a) = a$
iii. the opposite of positive number is negative number
iv. the opposite of negative number is positive number
v. the opposite of zero is zero.

Exercise 2.2.2

1. Find the opposite of each integer given below.
   a. 70 b. -23 c. -170 d. 0

2. Find the opposite of the following

   a. Gain of Birr 120

   b. 15^0c above zero

   c. A walk of 5 km backward

3. Ahmed saw on weather report that Hawassa's temperature was 9 ^{0}C above zero.
   a. Write Hawassa's temperature as an integer.
   b. If the temperature in Bale is opposite of Hawassa's temperature. What is the temperature in Bale?

2.3. Comparing and ordering integers

By the end of these section you should be able to:

- *compare and order integers using a number line*
- *determine the predecessor and successor of a given integer.*

Activity 2.3.1

Discuss with your friends/partners

1. On October, the lowest temperature of three different cities was recorded as follows

City	Temperature in ^0c
Debire Birhan	4^0c below zero
Addis Ababa	4^0c above zero
Adama	10^0c above zero

   a. Which city has the lowest(minimum) temperature?
   b. Which city has the highest temperature?
   c. Indicate the above temperatures on number line and arrange the temperatures from the smallest to largest
2. Indicate each of the following pairs of numbers on number line and circle the number which is found to the right of the other.
   a. 0 and 6 b. 0 and -6 c. -6 and -4 d. 6 and 4
3. Circle the bigger number from the following pairs of numbers.
   a. 0 and 6 b. 0 and -6 c. -6 and -4 d. 6 and 4

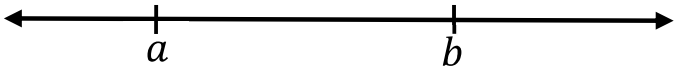
To compare integers, draw number line and indicate the numbers on number line then,

- The number which is to the right of the other is bigger number.
- The number which is to the left of the other is smaller number.

We use the symbols " $>$ "," $<$ "," $=$ " to compare numbers.

- $a > b$ means a is greater than b
- $a < b$ means a is less than b
- $a = b$, means a is equal to b

Consider the following number line

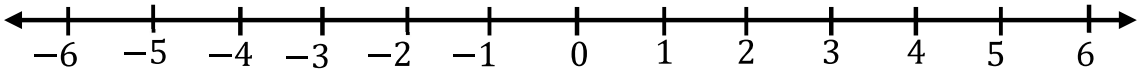


- a is to the left of b, hence a is smaller than b $(i.e.\, a < b)$
- b is to the right of a, hence b is bigger than a $(i.e.\, b > a)$

Example 1. For each pair of numbers below select the number which is to the right of the other

a. $3\ and\ 6$ b. $-3\ and\ -6$

Solution: First draw a number line and mark the numbers

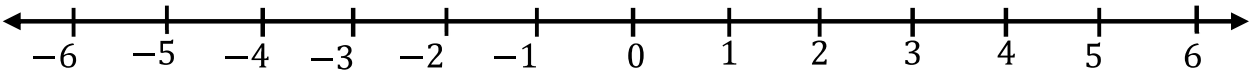


a. 6 is to the right 3 b. -3 is to the right of -6

Example 2. Insert " $>$ " *or* " $<$ " to express the corresponding relationship between the following pairs of integers

a. -4 ___ 0 b. 0 ___ -3

Solution: use a number line to compare numbers



a. -4 is to the left of 0, hence -4 is smaller than 0, symbolically, $-4 < 0$.

b. 0 is to the right of -3, hence 0 is greater than -3, symbolically $0 > -3$.

Note:

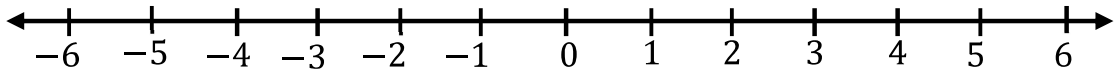
- All negative integers are to the left of 0, hence every negative integer is less than 0
- 0 is to the right of any negative integer, hence 0 is greater than any negative integer.

Example 3. Insert " $>$ " *or* " $<$ " to express the corresponding relationship between the following pairs of integers

a. $-4 ___ 2$ c. $4 ___ -6$

b. $5 ___ -6$ d. $-2 ___ 4$

Solution: use a number line to compare numbers



a) -4 is to the left of 2, hence -4 is less than 2 symbolically$(-4 < 2)$.

b) 5 is to the right of -6, hence 5 is greater than -6 , symbolically$(5 > -6)$.

c) 4 is to the right of -6, hence 4 is greater than -6, symbolically$(4 > -6)$.

d) -2 is to the left of 4, hence -2 is less than 4, symbolically $(-2 < 4)$.

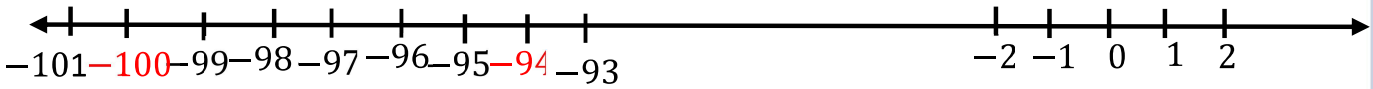
Note:

- All negative integers are to the left of positive integers; hence every negative integer is less than any positive integer.
- All positive integers are to the right of any negative integer; hence every positive integer is greater than any negative integer.

Example 4. Compare the following integers using the sign " $>$ "" $=$ " *or* " $<$ "

a. $-94 ___ -100$

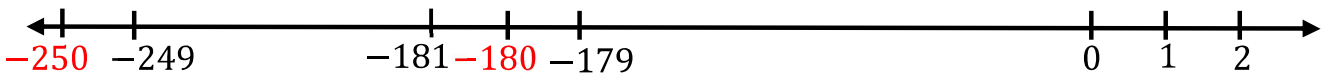
Solution: To compare the numbers draw part of number line



-94 is to the right of -100, hence -94 is bigger $(-94 > -100)$

b. $-250 ___ -180$

Solution: draw part of number line



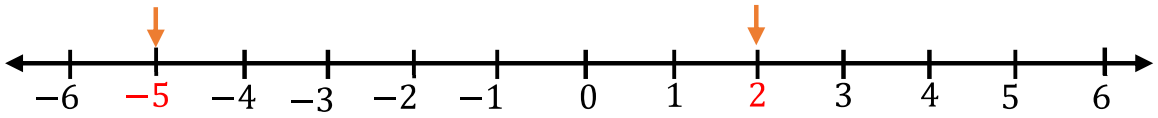
-250 is to the left of -180, hence -250 is smaller $(-250 < -180)$

c. -40 ___ -40

This two numbers are the same numbers; hence they are equal $(-40 = -40)$

Example 5. Name all integers which lie between -5 *and* 2

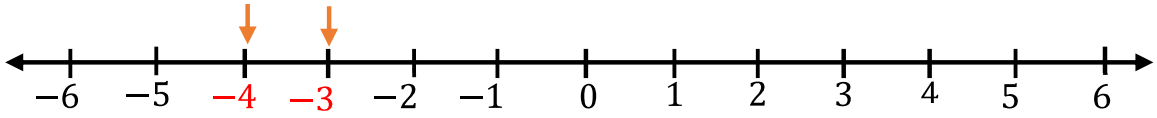
Solution: Draw a number line



The integers that lie between -5 *and* 2 are $-4, -3, -2, -1, 0$ *and* 1

Example 6. List all integers which lie between -4 *and* -3

Solution: Draw a number line



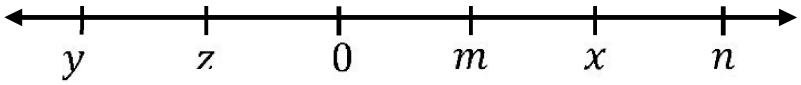
There is no integer between -4 *and* -3

Exercise 2.3.1

1. Insert " $>$ "" $=$ " *or* " $<$ " to express the corresponding relationship between the following pairs of integers
   a. -150 ___ 0
   b. 0 ___ -300
   c. -1200 ___ -74
   d. -34 ___ 34
   e. 65 ___ -20
   f. -340 ___ -341
   g. -23 ___ -23
   h. -8 ___ 2
   i. -120 ___ -80
   j. -360 ___ 185
   k. 0 ___ 13
   l. 14 ___ 0
2. From each pair of numbers below circle the number which is to the right of the other.
   a. 3 *and* 6
   b. -3 *and* -6
   c. 10 *and* 5
   d. -1005 *and* -105
   e. -87 *and* -99
   f. -360 *and* -1
   g. 35 *and* -45
   h. 0 *and* -12
3. List all integers which lie between the following pair of numbers
   a. -2 *and* 3
   b. -1 *and* 1
   c. -98 *and* -95
   d. -10 *and* -9

4. The five integers $x, y, z, n, and\ m$ are represented on the number line below

Using " $<$ "or" $>$ " fill in the blank space



a. z _____ x b. m _____ x c. z _____ n d. 0 _____ z

5. Answer the following questions

a. What is the greatest negative integer?

b. What is the smallest positive integer?

c. What is the greatest integer?

d. What is the smallest integer?

Successor and predecessor of integers

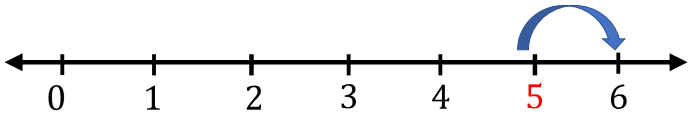
Successor of an integer is an integer that comes after the given integer

Example 7: Find the successor of the given integers

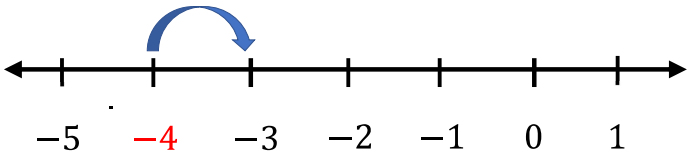
a. 5 c. -4 d. -99 e. 99

Solution:

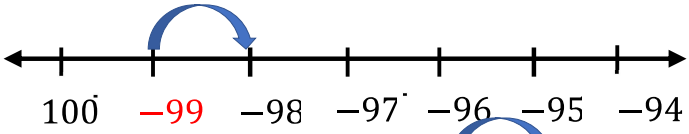
a. The successor of 5 is 6.



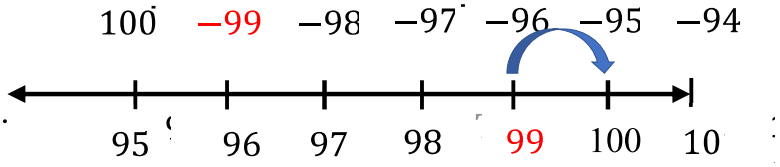
b. The successor of -4 is -3.



c. The successor of -99 is -98.



d. The successor of 99 is 100.



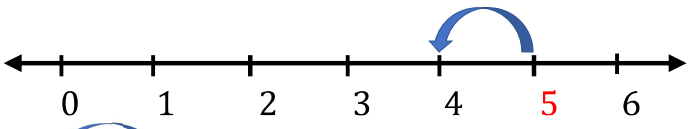
Predecessor of an integer is an integer that comes before the given integer.

Example 8: Find the predecessor of the given integers

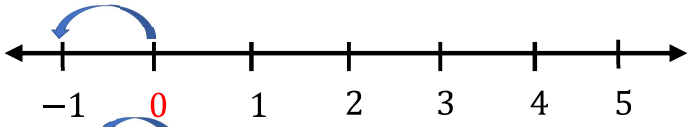
a. 5 b. 0 c. -4 d. -99 e. 99

Solution:

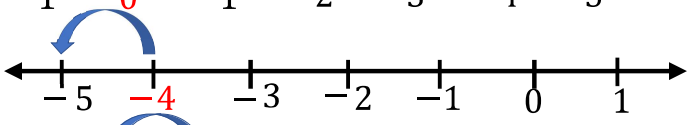
a. The predecessor of 5 is 4.



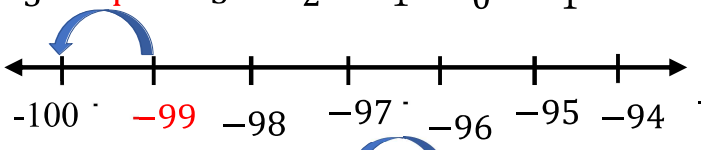
b. The predecessor of 0 is −1.



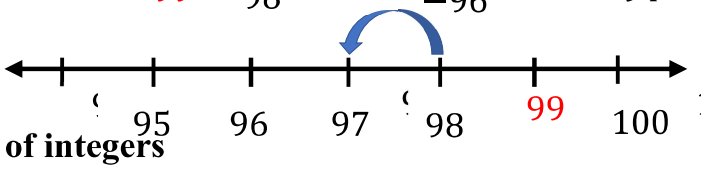
c. The predecessor of −4 is −5.



d. The predecessor of −99 is −100.



e. The predecessor of 99 is 98.



Ascending and Descending order of integers

Ascending order means arranging the given numbers in increasing order.

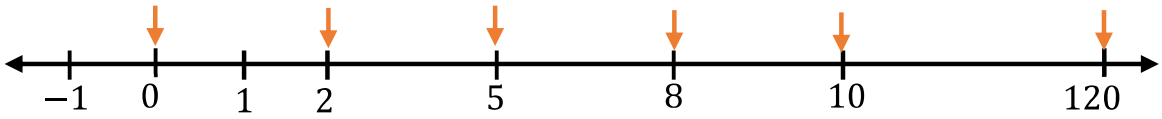
(i.e. arranging from the smallest to the largest number)

Example 9: Arrange the following set of numbers in ascending order.

a. 10, 2, 5, 0, 8, 120
b. −40, −8, 0, −12, −10

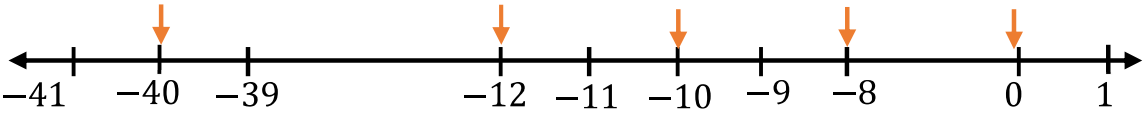
Solution:

a.



0, 2, 5, 8, 10, 120

b.



−40, −12, −10, −8, 0

Descending order means arranging the given numbers in decreasing order.

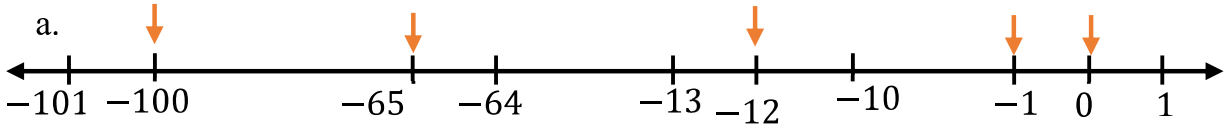
(i.e. arranging from the largest to the smallest)

Example 10: Arrange the following set of numbers in descending order

a. 0, −65, −100, −12, −1
b. 55, −75, 0, 15, −35

Solution:

a.



$0, -1, -12, -65, -100$

b.

$55, 15, 0, -35, -75$

Exercise 2.3.2

1. Find the successor of the following integers
   a. 20 b. -16 c. 999 d. -999
2. Find the predecessor of the following integers
   a. 1000 b. -77 c. -999 d. -1000
3. Arrange the following numbers in ascending order
   a. $0, -78, 17, -24, 71$
   b. $-230, -157, -400, -350, 50$
4. Arrange the following numbers in descending order
   a. $-9, -99, 69, 59, -89$
   b. $200, -300, -757, -445, 400$
5. Abebe, Almaz, Hailu and Derartu played basketballs. The results are shown in the table below

	First play	Second play	Final result
Abebe	Loss 5 basket balls	Won 7 basket balls	Won 2 baskets (+2)
Almaz	Won 5 basket balls	Loss 5 basket balls	
Hailu	Loss 3 basket balls	Loss 2 basket balls	
Derartu	Won 1 basket balls	Won 2 basket balls	

   a. Complete the final result of the players in the above table
   b. Who was the winner?
   c. Who was the loser?
   d. Arrange the final result of the players in descending order.

Project work:

Register the daily temperature of Addis Ababa, Adama, Hawassa, Debire Birhan, Gambela, Semera, and Bahir Dar from Ethiopian mass media in a given day and answer the following:

a. Graph the temperature of each city on number line
b. Which city was coldest?
c. Which city was warmest?

2.4. Addition and subtraction of Integers

By the end of these section you should be able to:

- *find the sum of integers.*
- *find the difference between two integers.*

1. Hussen, Almaz, Hailu and Derartu are selling different articles. Their profit or loss is shown in the table below

	First sell	Second sell	Total profit/ loss
Hussen	Loss Birr 15	Profit Birr 17	
Almaz	Profit Birr 15	Loss Birr 15	
Hailu	Loss Birr 13	Loss Birr 20	
Derartu	Profit Birr 10	Profit Birr 12	

2. Find the sum of the following using number line

   a. $4 + (-8)$ b. $-2 + 5$ c. $-3 + (-4)$ d. $-4 + 0$

3. Find the sum of the following integers

   a. $3 + 2$ b. $-20 + 0$ c. $-3 + 2$ d. $4 + (-3)$ e. $-10 + 10$

Rules: For addition of integers on number line

1. Start the arrow from the first addend.
2. Move the arrow to the right with the same magnitude as the second addend, if the sign of the second addend is positive.
3. Move the arrow to the left with the same magnitude as the second addend, if the sign of the second addend is negative.
4. The sum of the integers is at the point where the arrow ends.

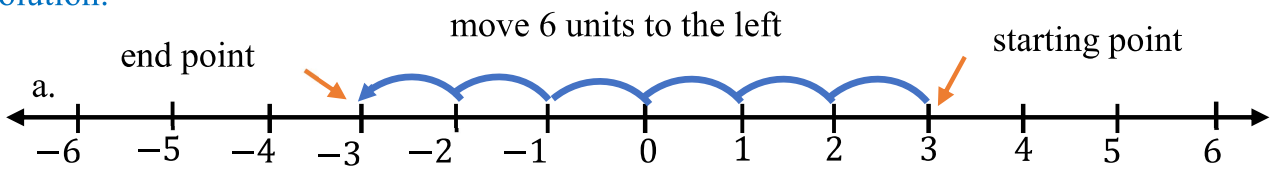
In $a + b = c$, a and b are called addends and c is called sum.

Example 1: Find the sum of the following integers using number line

a. $3 + (-6)$ b. $-3 + (-6)$ c. $-3 + 6$

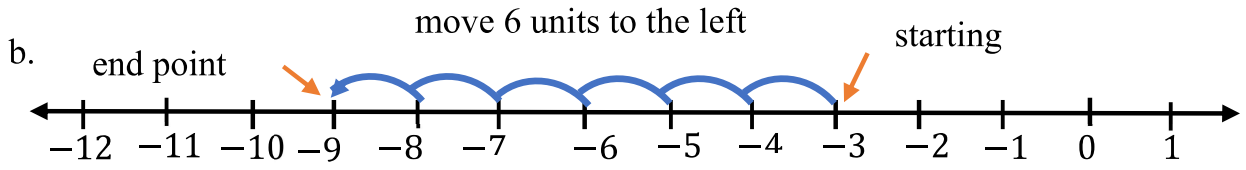
Solution:

a.



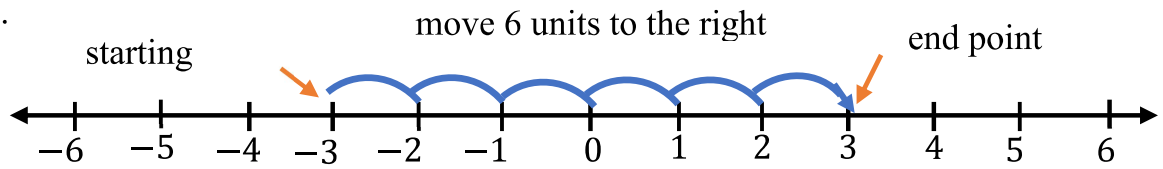
$3 + (-6) = -3$

b.



$-3 + (-6) = -9$

c.



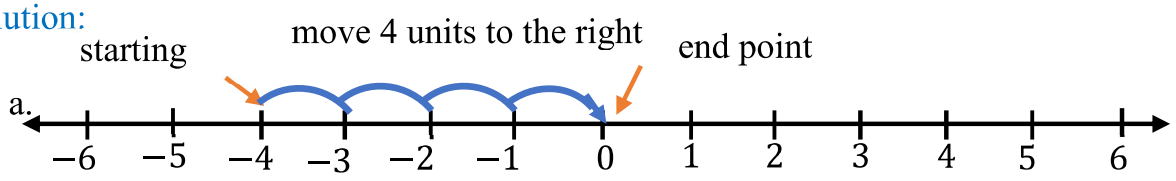
$-3 + 6 = 3$

Example 2: Add the following numbers

a. -4+4 b. 6+(-6) c. -1+1 d. 2+(-2)

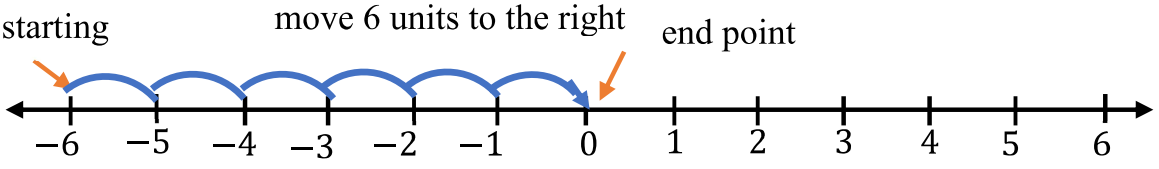
Solution:

a.



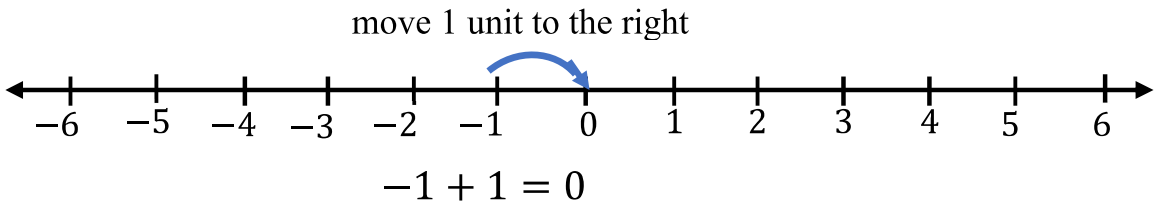
$-4 + 4 = 0$

b.



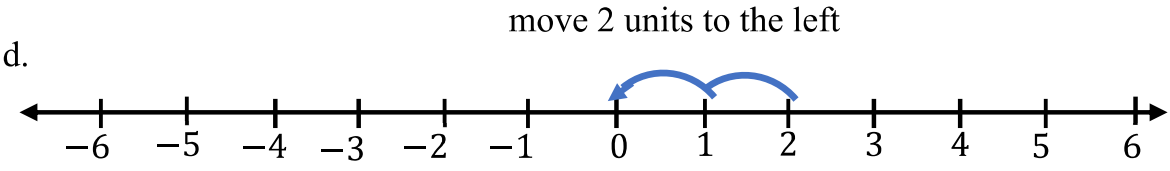
$-6 + 6 = 0$

c.



$$-1 + 1 = 0$$

d.



$$2 + (-2)$$

Note: The sum of any two opposite integers is zero.

Addition of positive and negative integers without using number line

To find the sum of positive and negative integers, follow the following steps:

Step 1: Take the sign of a number with greater magnitude.

Step 2: Find the difference of magnitude of the numbers.

Step 3: put the sign in front of the difference.

Example 3: Find the sum of the following integers

a. $22 + (-43)$

Solution: $43 > 22$, the sign of 43 is $(-)$,hence the sum is $(-)$

$$22 + (-43) = -(43 - 22)$$
$$= -21$$

b. $625 + (-214)$

Solution: $625 > 214$, 625 is $(+)$,hence the sum is $(+)$

$$625 + (-214) = +(625 - 214)$$
$$= 411$$

c. $-1300 + 560$

Solution: $1300 > 560$, 1300 is $(-)$,hence the sum is $(-)$

$$-1300 + 560 = -(1300 - 560) \quad = -740$$

Addition of two negative integers

To find the sum of two negative integers:

Step 1: The sign of the sum is always negative.

Step 2: Find the sum of magnitude of the numbers.

Step 3: put the negative sign in front of the sum

Example 4: find the sum of the following

a. $-15 + (-35) = -(15 + 35) = -50$
b. $-36 + (-40) = -(36 + 40) = -76$
c. $-120 + (-76) = -(120 + 76) = -196$

Example 5: A person saved Birr 200 and then he spent Birr 150. How much money does he have now?

Solution:

Saved 200 can be represented by + 200

Spent 150 can be represented by -150

We get: $200 + (-150) = +(200 - 150)$

$= 50$

Therefore, he has left Birr 50.

Example 6: A submarine at 21m below sea level suddenly moves up about 6m. At what depth is the submarine located now?

Solution:

$21m$ below sea level can be represented by -21

Moves up about $6m$ can be represented by $+6$, then

We get $-21 + 6 = -(21 - 6)$

$= -15$

The submarine is 15m below sea level.

Property of addition of Integers

I. The sum of two integers is an integer

Example:

a. $2 + 5 = 7$, 7 is integer b. $4 + (-6) = -2$, -2 is integer

II. Addition has commutative property: for any two integers m and n

$$m + n = n + m$$

Example: $5 + (-3)$ $\qquad$ $-3 + 5$

$= +(5 - 3)$ $\qquad$ $= +(5 - 3)$

$= \mathbf{2}$ $\qquad$ $= \mathbf{2}$

Therefore, $5 + (-3) = -3 + 5$

III. Associative property: for any three integers x, y and z

$$x + (y + z) = (x + y) + z$$

Example: $3 + (4 + (-5))$ $\qquad$ $(3 + 4) + (-5)$

$= 3 + (-1)$ $\qquad$ $= 7 + (-5)$

$= \mathbf{2}$ $\qquad$ $= \mathbf{2}$

Therefore, $3 + (4 + (-5)) = (3 + 4) + (-5)$

IV. Property of zero:

For any integer $x, x + 0 = x = 0 + x$

V. Property of opposites:

For any integer x, $x + (-x) = 0$

Exercise 2.4.1

1. Find the sum of the following integers?

   a. $250 + (-250)$ c. $160 + (-160)$

   b. $-170 + 170$ d. $-15 + 15$

2. Find the sum of the following using number line.

   a. $-3 + 7$ c. $6 + (-3)$ e. $5 + (-2)$

   b. $-4 + (-5)$ d. $-10 + 10$ f. $-7 + 4$

3. Find the sum of the following integers

   a. $23 + (-10)$ d. 230+(-63) g. $12 + (-23)$

   b. $-30 + 40$ e. -325+234 h. 2250+(-457)

   c. -560+(-50) +(210) f. -50+(-45) +62 i. -20+(-10)

4. You are Birr 5 in debt. You borrow Birr 12 more. What is the total amount of your debt?
5. An airplane takes off and then climbs 2500 meter. After 20 minutes, the airplane descends 150 meters. What is the airplane's current height?

Subtraction of Integers

Activity 2.4.2

1. Write the following differences in the form of sum, the first is done for you

   a. $3 - 6 = 3 + (-6)$

   b. $-3 - 5 =$

   c. $4 - 2 =$

   d. $-5 - (-6) =$

2. Find the following differences by expressing in the form of sum, the first is done for you

   a. $5 - 6 = 5 + (-6) = -1$

   b. $-2 - 2 =$

   c. $4 - 2 =$

   d. $-7 - (-6) =$

Subtracting an integer b from a is the same as adding opposite of b to a

i.e. for any integers a and b, $a - b = a + (-b)$

Example 1: Find the difference of the following integer by expressing in the form of sum.

a. $5 - 2$ b. $4 - 7$ c. $2 - 6$

Solution:

a. $5 - 2$, subtracting 2 means adding the opposite of 2

$$5 - 2 = 5 + (-2)$$
$$= 3$$

b. $4 - 7$, subtracting 7 means adding the opposite of 7

$$4 - 7 = 4 + (-7)$$
$$= -(7 - 4)$$
$$= -3$$

c. $2 - 6$, subtracting 6 means adding the opposite of 6

$$2 - 6 = 2 + (-6)$$
$$= -(6 - 2)$$
$$= -4$$

Example 2: Find the difference of the following integer by expressing in the form of sum.

a. $25 - (-30)$ b. $-12 - (-18)$ c. $-55 - (-20)$ d. $8 - (-9)$

Solution:

a. $25 - (-30)$, subtracting -30 means adding the opposite of -30

$$25 - (-30) = 25 + 30$$
$$= 55$$

b. $-12 - (-18)$, subtracting -18 means adding the opposite of -18

$$-12 - (-18) = -12 + 18$$
$$= +(18 - 12)$$
$$= 6$$

c. $-55 - (-20)$, subtracting -20 means adding the opposite of -20

$$-55 - (-20) = -55 + 20$$

Note: For any two-integer a and b

$a - (-b) = a + b$

$= -(55 - 20)$

$= -35$

d. $8 - (-9)$, subtracting -9 means adding the opposite of -9

$8 - (-9) = 8 + 9 = 17$

Example 3:

Find the difference of the following integer by expressing in the form of sum.

a. $-21 - 12$ b. $-248 - 125$

Note: For any two-integer a and b
$-a - b = -(a + b)$

Solution:

a. $-21 - 12$, subtracting 12 means adding the opposite of 12

$\mathbf{-21 - 12 = -21 + (-12)}$

$\mathbf{= -(21 + 12)}$

$= -33$

b. $-248 - 125$, subtracting 125 means adding the opposite of 125

$-248 - 125 = -248 + (-125)$

$= -(248 + 125)$

$= -373$

Exercise 2.4.2

1. express the following differences in the form of sum
   a. $3 - (-10)$ c. $9 - 12$ e. $-178 - 57$
   b. $-5 - (-6)$ d. $25 - 17$ f. $-365 - 1000$
2. Find the following differences.
   a. $0 - 3$ c. $-75 - 15$ e. $465 - (-778)$
   b. $7 - (-8)$ d. $-1789 - (-1000)$ f. $25 - 75$
3. The melting point of dry ice is $-109^{\circ}F$ .The boiling point of dry ice is $109^{\circ}F$ then, how many degrees is the boiling point above the melting point?

2.5. Multiplication and Division of Integer numbers

By the end of these section you should be able to:

- *solve problems on multiplication of integers.*
- *identify the commutative and associative property of multiplication.*
- *identify the distributive property of multiplication over addition.*
- *divide integers whose quotient as expressed in decimals (2 decimal places)*

Multiplication of Integer numbers

Activity 2.5.1

1. Express each of the following product as sum and find the product.
   a. 5×2 c. -5×4
   b. 9×3 d. -6×2
2. Express each sum as product.
   a. $0 + 0 + 0 + 0$ c. $-2 + (-2) + (-2)$
   b. $3 + 3 + 3 + 3 + 3$ d. $-10 + (-10) + (-10) + (-10) + (-10) + (-10)$
3. Multiply the following.
   a. 2×4 c. 3×51 e. $4 \times (-25)$
   b. 0×121 d. -10×2

Example1 Find the product of the following

a. 3×5 b. 11×6

Note: The product of two positive integers is positive integer

Solution:

a. $3 \times 5 = 5 + 5 + 5 = 15$

b. $11 \times 6 = 11 + 11 + 11 + 11 + 11 + 11 = 66$

Example 2 Find the product of the following numbers expressing in the form of Sum

a. $4 \times (-5)$ b. -1×6 c. -365×2

Solution:

a. $4 \times (-5) = -5 + (-5) + (-5) + (-5)$

$= -20$

b. $-1 \times 6 = -1 + (-1) + (-1) + (-1) + (-1) + (-1)$

$= -6$

c. $-365 \times 2 = -365 + (-365) = -730$

Example 3 Multiply the following

a. $4 \times (-3)$

b. b. $12 \times (-11)$

c. c. -10×267

Note: From above examples, you can see that the product of positive and negative integer is negative integer.

Solution a. $4 \times (-3) \rightarrow$ The result is negative $(-)$

$\Rightarrow 4 \times 3 = 12$

so, $4 \times (-3) = -12$

b. $12 \times (-11) \rightarrow$ The result is negative $(-)$

$\Rightarrow 12 \times 11 = 132$

so, $12 \times (-11) = -132$

c. $-10 \times 267 \rightarrow$ The result is negative $(-)$

$\Rightarrow 10 \times 267 = 2670$

so, $-10 \times 267 = -2670$

Note: The product of two negative integers is positive integer

Example 4: find the product of the following

a. $-2 \times 2 = -4$
b. $-2 \times 1 = -2$
c. $-2 \times 0 = 0$
d. $-2 \times (-1) = ?$

In the above example, the product pattern is $-4, -2, 0, \ldots$, hence $-2 \times (-1) = 2$

Example 5: Multiply

a. $-3 \times (-5)$

b. $-21 \times (-1)$

c. $-6 \times (-13)$

d. $-11 \times (-8)$

Solution:

a. $-3 \times (-5) \rightarrow$ The result is positive $(+)$

$\Rightarrow 3 \times 5 = 15$

$so, -3 \times (-5) = 15$

b. $-21 \times (-1) \rightarrow$ The result is positive $(+)$

$\Rightarrow 21 \times 1 = 21$

$so, -21 \times (-1) = 21$

c. $-6 \times (-13) \rightarrow$ The result is positive $(+)$

$\Rightarrow 6 \times 13 = 78$

$so, -6 \times (-13) = 78$

d. $-11 \times (-8) \rightarrow$ The result is positive $(+)$

$\Rightarrow 11 \times 8 = 88$

$so, -11 \times (-8) = 88$

Summary of product of integers

Two factors	Sign of Product	Example
Both are positives	Positive	$3 \times 5 = 15$
Opposite sign $(+ \times -)$	Negative	$-3 \times 5 = -15$
Both are negatives	Positive	$-3 \times (-5) = 15$

Exercise 2.5.1

1. Multiply

a. 6×12

b. 13×15

c. -7×13

d. $16 \times (-25)$

e. $-17 \times (-103)$

f. $-170 \times (-345)$

Properties of multiplication of integers

Activity 2.5.2

Which of the following statements are true and which are false

a. $2 \times 4 = 4 \times 2$

b. $21 \times (-3) = -3 \times 21$

c. $5 \times (3 \times 2) = (5 \times 3) \times 2$

d. $-5 \times (3 + 7) = (-5 \times 3) + (-5 + 2)$

1. ***Commutative properties of multiplication***

For any two integers a and b,

$a \times b = b \times a$ ... commutative property of multiplication

Example 1: $-2 \times 5 = 5 \times (-2)$

$-10 = -10$

$-12 \times (-20) = -20 \times (-12)$

$240 = 240$

2. ***Associative properties of multiplication***

For any three integer numbers a, b and c, then

$a \times (b \times c) = (a \times b) \times c$ ... Associative property

Example 2: $-2 \times (5 \times 6) = (-2 \times 5) \times 6$

$-2 \times 30 = -10 \times 6$ (by multiplying numbers in the bracket)

$-60 = -60$

3. ***Distributive properties of multiplication over addition***

Let a, b, & c be any integers, then

$a \times (b + c) = (a \times b) + (a \times c)$ ... distributive property

Example 3: $-6 \times (5 + 3) = (-6 \times 5) + (-6 \times 3)$

$-6 \times (8) = (-30) + (-18)$ (applying operations in the bracket)

$-48 = -48$

4. ***Properties of zero and 1 on multiplication***

Example 4 Find the product

a. 2×0 $b. -122 \times 0$

Solution a. $2 \times 0 = 0$

b. $-122 \times 0 = 0$

Example 5

$a.\ 9 \times 1 = 9$

b. $1 \times (-291) = -291$

Note The product of any integer and zero is zero.

Let 'a' be any integer, then

$a \times 0 = 0$ or $0 \times a = 0$

Note: The product of any integer and 1 is the number itself.

Let 'a' be any integer then

$a \times 1 = a$ or $1 \times a = a$

Exercise 2.5.2

1. State the properties for the following
   a. $2 \times (-3 + 54) = (2 \times (-3)) + (2 \times 54)$
   b. $-525 \times 955 = 955 \times (-525)$
   c. $2 \times (-13 \times 5) = (2 \times (-13)) \times 5$
   d. $5 \times 1 = 5$
   e. $0 \times (-8) = 0$
2. Simplify each of the following pairs and compare their result
   a. $3\times (-2\times 3)$ and $(3\times (-2)) \times 3)$
   b. 259×0 *and* 0×259
   c. $-6 \times (5 + 2)$ *and* $(-6 \times 5) + (-6 \times 2)$
   d. $-5 \times (-3 + (-8))$ *and* $(-5 \times (-3)) + (-5 \times (-8))$

3. Copy and complete the following table below

a	b	c	$a \times b$	$b \times a$	$a \times (b \times c)$	$(a \times b) \times c$
-7	4	-3				
9	-2	5				
-6	-8	-12				

Multiplication of three or more integer

Activity 2.5.3

1. Multiply the following numbers and notice the sign of the product.
   a. $-4 \times 5 \times 3 \times 1$
   b. $-4 \times 5 \times (-3) \times 1$
   c. $-4 \times (-5) \times (-3) \times 1$
   d. $-4 \times (-5) \times (-3) \times (-1)$
2. Multiply the following numbers
   a. $-4 \times 5 \times 0 \times 1$
   b. $-4 \times 5 \times (-3) \times 1 \times 125 \times 0$

   What do you conclude from a and b?

The following properties are helpful in simplifying products with three or more factors:

- The product of an even number of negative factors is positive.
- The product of an odd number of negative factors is negative.
- Product of integers with at least one factor 0 is zero.
- If you multiply an integer a by -1, then you get the opposite of a ($-1 \times a = -a$).

➢ When you multiply a number by a variable, you can omit the multiplication sign and keep the number in front of the variable $(i.e. 3 \times y = 3y)$.

Example 1: determine the sign of product

a. $-1 \times 5 \times (-6)$

b. $-265 \times (-81) \times 35 \times (-68)$

c. $-5 \times (-2) \times 3 \times (-8) \times (-4) \times 7$

solution: a. The number of negative factors is 2 (even), then the product is positive

b. The number of negative factors is 3 (odd), then the product is Negative

c. The number of negative factors is 4 (even)then the product is positive

Example 2 Find the product of the following

a. $-3 \times 5 \times (-4)$

b. $-5 \times (-5) \times 3 \times (-8)$

c. $-5 \times (-2) \times 3 \times (-8) \times (-4) \times 0$

Solution:

a. The number of negative factors is 2 (even), then the product is positive

$$(-3 \times 5) \times (-4) = -15 \times (-4) = 60$$

b. The number of negative factors is 3 (odd), then the product is Negative

$$\Rightarrow -5 \times (-5) \times 3 \times (-8) = (-5 \times (-5)) \times (3 \times (-8))$$

$$= 25 \times (-24)$$

$$= 600$$

c. There is one factor 0 in the product, hence the product is 0

$$-5 \times (-2) \times 3 \times (-8) \times (-4) \times 0 = 0$$

Exercise 2.5.3

1. Find the sign of the product
   a. $12 \times (-50) \times (-61)$
   b. $(-125) \times (-3) \times (-52)$
   c. $-1 \times (-2) \times (-3) \times (-4) \times (-5)$
   d. $3 \times (-6) \times (-9) \times (-12)$
2. Find the product of the following
   a. $9 \times (-3) \times (-6)$
   b. $(-12) \times (-13) \times (-5)$
   c. $(-1) \times (-2) \times (-3) \times (-4) \times (-5)$
   d. $(3) \times (-6) \times (-9) \times (-12)$
   e. $15625 \times (-2) \times 0 \times (-15625) \times 98$

Division of integers

Activity 2.5.4

1. Check the quotient of the following using product, the first is done for you
   a. $20 \div 4 = 5$, hence $20 = 5 \times 4$
   b. $-36 \div 9 = -4$
   c. $84 \div (-7) = -12$
   d. $-8 \div (-2) = 4$
2. Referring the above activity
   a. The quotient of negative and positive numbers is $(Positive/negative)$
   b. The quotient of positive and negative numbers is$(Positive/negative)$
   c. The quotient of negative and negative numbers is$(Positive/negative)$

Division is an inverse operation of multiplication

For example, divide 12 by 3 means find a number that gives product 12 when it is multiplied by 3. The number is 4 and $12 \div 3 = 4$, because $4 \times 3 = 12$

In the division $12 \div 3 = 4$, 12 is dividend, 3 is divisor and 4 is quotient.

Example 1: Find the quotient by converting in to products

Note:

For any numbers $a, b, and\ c\ where\ b \neq 0$,

$a \div b = c$, if and only if $a = b \times c$

- $a \div b$ read as a divided by b
- $a \div b$ is also denoted by $\frac{a}{b}$ or a/b

a. $8 \div 2$ b. $12 \div (-4)$ c. $-10 \div (-5)$ d. $-8 \div 4$

Solution:

a. $8 \div 2 = ___$, means $8 = 2 \times ?$, the number is 4, because $8 = 2 \times 4$

Hence, $8 \div 2 = 4$

b. $12 \div (-4) = ___$, means $12 = -4 \times ?$, the number is -3, because

$12 = -4 \times -3$

Hence, $12 \div (-4) = -3$

c. $-10 \div (-5) = ___$, means $-10 = -5 \times ?$, the number is 2, because

$-10 = -5 \times 2$,

Hence, $-10 \div (-5) = 2$

d. $-8 \div 4 = ___$, means $-8 = 4 \times ?$, the number is -2, because

$-8 = 4 \times (-2)$

Hence, $-8 \div 4 = -2$

From above example,

- The quotient of two positive integers is positive
- The quotient of two negative integers is positive
- The quotient positive and negative integers is negative

To divide integers without converting in to multiplication follow the following procedure:

1. If the sign of dividend and divisor are the same:
   - The sign of the quotient is $(+)$
   - Find the magnitude of the quotient
   - Put the sign in front of the quotient
2. If the sign of dividend and divisor is different:
   - The sign of the quotient is $(-)$
   - Find the magnitude of the quotient
   - Put the sign in front of the quotient

Example 2: Find the quotient of the following.

a. $-368 \div 8$
b. $\frac{-264}{12}$
c. $175 \div (-7)$
d. $\frac{8211}{-12}$

Solution:

a. $-368 \div 8$ →The result is negative $(-)$

$$368 \div 8 = 46$$

$$\Rightarrow -368 \div 8 = -46$$

b. $\frac{-264}{12}$ →The result is negative $(-)$

$$264 \div 12 = 22$$

$$\Rightarrow \frac{-264}{12} = -22$$

c. $175 \div (-7$ →The result is negative $(-)$

$$175 \div 7 = 25$$

$$\Rightarrow 175 \div (-7) = -25$$

d. $\frac{8211}{-21}$ →The result is negative $(-)$

$$8211 \div 21 = 391$$

$$\Rightarrow \frac{8211}{-21} = -391$$

Example 3: Find the quotient of the following.

a. $-1 \div 4$
b. $-24 \div (-5)$
c. $-5 \div 4$
d. $-150 \div (-20)$

Solution

a. $-1 \div 4$ →The result is negative $(-)$

$$1 \div 4 = 0.25$$

$$\Rightarrow -1 \div 4 = -0.25$$

b. $-24 \div (-5)$ → The result is positive $(+)$

$$24 \div 5 = 4.8$$

$$-24 \div (-5) = 4.8$$

c. $-5 \div 4$ →The result is negative $(-)$

$$5 \div 4 = 1.25$$

$$\Rightarrow -5 \div 4 = -1.25$$

d. $-150 \div (-20)$ →The result is positive $(+)$

$$150 \div 20 = 7.5$$

$$\Rightarrow -150 \div (-20) = 7.5$$

Example 4 find the quotient of the following

a. $-387 \div (-9)$
b. $\frac{-1984}{-16}$

Solution a. $-387 \div (-9)$ →The result is positive $(+)$

$$387 \div 9 = 43$$

$$\Rightarrow -387 \div (-9) = 43$$

b. $\frac{-1984}{-16}$ →The result is positive $(+)$

$$1984 \div 16 = 124 \quad \Rightarrow \frac{-1984}{-16} = 124$$

Division of integers is summarized in the following table

Dividend and divisor	Sign of quotient	Example
Both positive	Positive	$15 \div 3 = 5$ or $\frac{15}{3} = 3$
Opposite sign$(+\&-)$	Negative	$15 \div (-3) = -5$ or $\frac{15}{(-3)} = -5$
Both negative	positive	$-15 \div (-3) = 5$ or $\frac{-15}{-3} = 5$

Properties of division of integers

Properties	Examples
1.The quotient of any two integers is not necessarily an integer	$20 \div 8 = 2.5 = \frac{5}{2}$
2. A non-zero integer divided by itself is 1	$20 \div 20 = 1$
3.An integer divided by zero is not defined. i.e Any integer cannot be divided by zero	$\frac{a}{0}$ *is not defined* $(undefined)$
4.Zero divided by any non-zero integer is zero.	$0 \div 20 = 0$
5. An integer divided by 1 is the number itself.	$20 \div 1 = 20$

Exercise 2.5.4

1. Identify dividend, devisor and quotient of the following
   a. $-96 \div 12 = -8$ b. $\frac{-56420}{-124} = 455$
2. Find the quotient by converting in to products
   a. $18 \div 3$ b. $24 \div (-4)$ c. $-40 \div (-5)$ d. $-44 \div 4$
3. Divide
   a. $-130 \div 5$
   b. $0 \div (-4965)$
   c. $(-1048) \div (-1048)$
   d. $\frac{-2574}{33}$
   e. $768 \div 0$
   f. $\frac{-17622}{-99}$
   g. $0 \div 0$

4. Fill the blank space
   a. $56 \div$ ------ $= 7$
   b. $-1092 \div$ ------- $=14$
   c. -------- $\div (-18) = 46$
   d. -------- $\div (55) = -101$
5. Divide the following
   a. $-12 \div 5$
   b. $-37 \div (-2)$
   c. $-26 \div (-8)$
   d. $45 \div (-4)$

2.6. Even and Odd integers

By the end of these section you should be able to:

- *describe even and odd integers*
- *identify even and odd integers*

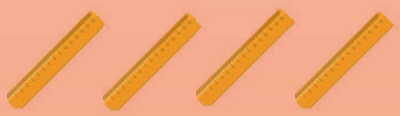
Activity 2.6.1

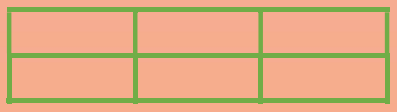
Discuss with your friends/partners

1. Group the following objects by two

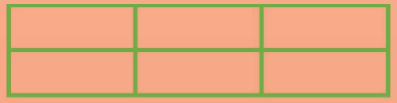
a.

b.

c. 



d.



2. From activity 1 a and c, explain what you observe?
3. From activity 1 b and d, explain what you observe?
4. What do we call numbers that can be grouped by two (divisible by two) without leaving remainder?
5. What do we call numbers that can be grouped by two leaving one remainder?
6. Identify the following numbers as even or odd

   6, 5, 4,3

   Even integers = _____

   Odd integers = _____

Even integer is an integer that can be divisible by 2 without leaving remainder.

Example 1: The following are single digit even integers

$$-8, -6, -4, -2, 0, 2, 4, 6, and\ 8$$

Odd integer is an integer that cannot be divisible by 2.

When you divide any odd integer by 2, its remainder is 1.

Example 2: The following are single digit odd integers

$$-9, -7, -5, -3, -1, 1, 3, 5, 7, and\ 9$$

Many digits even and odd integers

Example 3: Identify the following numbers as even and odd integers.

$$20, -16, 31, -23, 14, -15, 345, -252$$

For many digits number:

If the unit digit is $0, 2, 4, 6, or\ 8$, then the number is even integer, otherwise it is odd.

Solution: The unit digit of even integer is $0, 2, 4, 6, or\ 8$, hence $20, -16, 14\ and\ -252$ are even integers. The rest $31, -23, -15, and\ 345$ are odd integers

Example 4: List three consecutive even integers next to 31.

Solution: 32, 34, and 36

Example 5: List the next three consecutive even integers greater than -6.

Solution: $-4, -2, and\ 0$

Exercise 2.6.1

1. Identify the following integers as even or odd.

   $-36, 55, 0, -71, 103, -180$
2. Identify the following integers as even or odd.

   $-3675, 5568, -7109, 1034, -1807$
3. List the next three consecutive even integers greater than 143
4. List the next three consecutive odd integers greater than -54
5. List an even integer which is neither positive nor negative
6. What is the smallest positive even integer?
7. What is the greatest odd negative integer?
8. What is the greatest three-digit positive integer?
9. What is the smallest two-digit negative integer?

Sum of even and odd integers

Example 1: Add the following integers

a. $-40 + 36$ b. $24 + (-18)$ c. $-4 + 4$ d. $246 + 344$

Solution:

a. $-40 + 36 = -(40 - 36) = -4$
b. $24 + (-18) = +(24 - 18) = 6$
c. $-4 + 4 = 0$
d. $246 + 344 = 590$

From above example, the sum of two even integers is always _____(even/odd)

Example 2: add the following integers

a. $31 + 25$ b. $-45 + 23$ c. $11 + (-33)$ d. $-21 + (-15)$

Solution:

a. $31 + 25 = 56$
b. $-45 + 23 = -(45 - 23) = -22$
c. $11 + (-33) = -(33 - 11) = -22$
d. $-21 + (-15) = -(21 + 15) = -36$

From above example, the sum of two odd integers is always _____(even/odd)

Example 3: add the following integers

b. $8 + 19$ b. $-4 + (-23)$ c. $17 + (-32)$ d. $-21 + 30$

Solution:

a. $8 + 19 = 27$
b. $-4 + (-23) = -(4 + 23) = -27$
c. $17 + (-32) = -(32 - 17) = -15$
d. $-21 + 30 = +(30 - 21) = 9$

From above example, the sum of odd and even integer is always ____(even/odd)

Difference of even and odd integers

Example 4: subtract the following integers

a. $8 - 6$ b. $12 - 20$ c. $-10 - 10$ d. $-6 - (-30)$

Solution:

a. $8 - 6 = 2$
b. $12 - 20 = 12 + (-20) = -(20 - 12) = -8$
c. $-10 - 10 = -10 + (-10) = -(10 + 10) = -20$
d. $-6 - (-30) = -6 + 30 = +(30 - 6) = 24$

From the above example, we can conclude that $even - even =$ _____

Example 5: subtract the following integers

a. $7 - 5$ b. $11 - 23$ c. $-15 - 13$ d. $-9 - (-11)$

Solution:

a. $7 - 5 = 2$
b. $11 - 23 = 11 + (-23) = -(23 - 11) = -12$
c. $-15 - 13 = -15 + (-13) = -(15 + 13) = -28$
d. $-9 - (-11) = -9 + 11 = +(11 - 9) = 2$

From the above example, we can conclude that $odd - odd =$ _____

Example 6: Subtract the following integers.

a. $12 - 5$ b. $31 - 50$ c. $-18 - 15$ d. $-22 - (-9)$

Solution:

a. $12 - 5 = 7$
b. $31 - 50 = 31 + (-50) = -(50 - 31) = -19$
c. $-18 - 15 = -18 + (-15) = -(18 + 15) = -33$
d. $-22 - (-9) = -22 + 9 = -(22 - 9) = -13$

From the above example, we can conclude that

$odd - even =$ _____

$even - odd =$ _____

Product of even and odd integers

Example 1: Multiply the following integers

a. 12×4 b. $0 \times (-6)$ c. -6×4 d. -2×10 e. $-8 \times (-4)$

Solution:

a. $12 \times 4 = 48$
b. $0 \times (-6) = 0$
c. $-6 \times 4 = -24$
d. $-2 \times 10 = -20$
e. $-8 \times (-4) = 32$

From the above example, we can conclude that

$even \times even =$ _____

Example 2: Multiply the following integers

a. 2×3 b. $0 \times (-5)$ c. -7×4 d. -3×8 e. $-9 \times (-4)$

Solution:

a. $2 \times 3 = 6$ c. $-7 \times 4 = -28$

b. $0 \times (-5) = 0$ d. $-3 \times 8 = -24$ e. $-9 \times (-4) = 36$

From the above example, we can conclude that

$$odd \times even = ____$$
$$even \times odd = ____$$

Example 3: Multiply the following integers

b. 5×3 b. $1 \times (-5)$ c. -7×11 d. -13×3 e. $-9 \times (-5)$

Solution:

a. $5 \times 3 = 15$ c. $-7 \times 11 = -77$

b. $1 \times (-5) = -5$ d. $-13 \times 3 = -39$ e. $-9 \times (-5) = 45$

From the above example, we can conclude that

$$odd \times odd = ____$$

Exercise 2.6.2

1. Fill in the blank space with the correct answer
   a. $even + even = ____$
   b. $even - odd = ____$
   c. $odd \times even = ____$
   d. $even + odd + even = ____$
   e. $odd - even - odd + odd = ____$
   f. $odd \times odd \times even = ____$
   g. $even \times odd + even = ____$
   h. The difference between any two consecutive even integers is equal to ____.
   i. The difference between any two consecutive odd integers is equal to ____.

Summary for unit two

- Natural number is a set of numbers denoted by $\mathbb{N}$ and is described by $\mathbb{N} = \{1, 2, 3, \dots\}$
- Whole numbers are numbers contains natural number and zero denoted by $\mathbb{W}$ and is described by $\mathbb{W} = \{0, 1, 2, 3, \dots\}$
- An integer is a set of numbers consisting of whole numbers and negative numbers. The set of integers is denoted by

  $$\mathbb{Z} = \{\dots -4, -3, -2, -1, 0, 1, 2, 3, 4, \dots\}$$
- Every negative integer is less than any positive integer.
- Every positive integer is greater than any negative integer.
- Successor of an integer is an integer that comes after the given integer
- Predecessor of an integer is an integer that comes before the given integer.
- The sum of any two integers is always an integer.
- The difference of any two integers is always an integer.
- The sum of two opposite integers is zero.
- The product of positive and negative integer is negative integer.
- The product of an even number of negative factors is positive.
- The product of an odd number of negative factors is negative.
- The product of any two integer is always an integer.
- Any negative integer divided by negative integer is always positive
- Even integer is an integer that can be divisible by 2 without leaving remainder.
- Odd integer is an integer that cannot be divisible by 2.

Review Exercise for unit 2

1. Represent the following integers on number line.
   a. $-12, 3, 6, 8, -5, -9$
   b. $-11, 6, -4, -8$
   c. $-2, 3, 4, -3, -10$
2. Show the relation between $\mathbb{N}, \mathbb{W}$ *and* $\mathbb{Z}$.
3. List down 3 examples of non-integers.
4. Find the opposite of the following integers?
   a. -80
   b. 34
   c. -34
   d. 1002
   e. 835
   f. 543
   g. -1020
   h. -90
   i. -768
5. List down all integers that lie between the following pair of integers.
   a. -70 *and* -60
   b. -34 *and* -34
   c. -3 *and* -4
   d. -12 *and* -10
   e. 0 *and* 7
   f. -12 *and* 0
6. Compare the following pair of integers using the sign $<, =$, and $>$.
   a. -65____0
   b. 34____-678
   c. -34____-304
   d. 1002____-1002
   e. -835____-835
   f. 543____-56
   g. -1020____220
   h. -90____-65
   i. -768____34
7. If $-5 > x$, then list down four possible values of x, where x is an integer.
8. Arrange the following integers in descending and ascending order.

   $$-1, -5, 1, -51, -69, 101, -101, 21, -81$$

9. Find the following sum.
   a. $0 + (-59)$
   b. $26 + (-52)$
   c. $-95 + 124$
   d. $-545 + 511$
   e. $-215 + (-349)$
   f. $121 + (-290)$
   g. $-1020 + 20$
   h. $-90 + (-489) + 315$
   i. $-768 + 2001$

10. Find the following difference.

a. $0 - 592$

b. $75 - 105$

c. $-89 - 482$

d. $513 - (-109)$

e. $-1099 - (-412)$

f. $299 - 1251 - (-191)$

11. Multiply the following.

a. $10 \times (-24)$

b. $-45 \times (-12)$

c. $-9 \times (-24)$

d. $-4 \times (-25) \times 14(-1)$

e. $-9 \times (-55) \times 998 \times 0 \times 32$

f. $-1 \times (-1) \times 1 \times (-1) \times (-1) \times (-1) \times (-1)$

12. Name the property of the following for any integers m, n, &p.

a. $m \times n = n \times m$

b. $m \times (n + p) = (m \times n) + (m \times p)$

c. $m \times (n \times p) = (m \times n) \times p$

d. $0 \times m = 0$

e. $1 \times m = m$

13. Find the quotient of the following.

a. $1080 \div 8$

b. $-214 \div (-214)$

c. $-19225 \div (-1)$

d. $0 \div 492$

e. $-492 \div 0$

f. $-7632 \div 36$

g. $1 \div 20$

h. $-54 \div 4$

i. $-110 \div (-20)$

14. Write four consecutive negative even integers.

15. Write all odd integers between -106 and -96

16. Decide whether the following statements are Even or odd

a. The sum of any two even integers.

b. The difference of any two odd integers.

c. The product of any two even integers.

d. The product of any three odd integers.

e. The sum of even and odd integers.

f. The difference of odd and even integers.

g. The product of odd and even

17. Find three integers whose sum is zero.

18. list examples of a pair of integers whose sum is zero.

19. Abel was playing a two-round game in which he could gain or loss points. During the first round he lost 30 points. During the second round gained 13 points. What was his net(total) score at the end of the game?

Unit 3
Linear equations

Learning Outcomes:

At the end of this unit, you should be able to:

- *identify variables, terms and variables in algebraic expressions*
- *simplify algebraic expressions*
- *develop their skills on rearranging and solving linear equations.*
- *apply the rules of transformation of linear equations for solving problems*
- *draw a line through the origin whose equation is given.*
- *apply real-life situations in solving linear equations*

Introduction

In this unit you will learn the importance of variables in mathematics, apply variables in solving linear equation, like linear equation with one variable involving brackets and linear equation involving fractions.

The unit has four sections. The first section deals with the use of variables in formula and the second section deals with solving linear equation, the third section deals with cartesian coordinate system and the last section deals about application of linear equations. Here the student will see how to use variables in formula and how to solve linear equation with one variable involving brackets and fractions.

3.1. Algebraic terms and expressions

By the end of these section you should be able to:

- *describe algebraic terms and expressions*
- *simplify algebraic expressions with and without brackets*

3.1.1. Use of Variables in Formula

Activity 3.1.1

1. Identify the variables

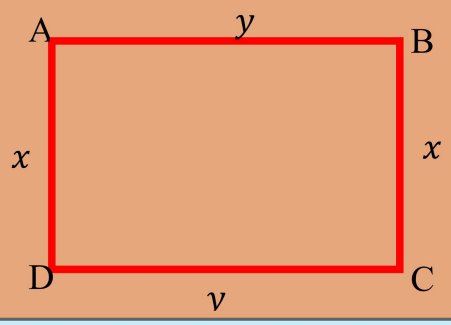
   $23, x, 4, y, z, 0.5, 3x, \frac{3}{4}, a, 3b, \frac{1}{2}$

2. What is variable?
3. Express the following using variables.
   a. A number x plus 4.
   b. The difference of 5 and y, where y is greater than 5.
   c. The product of 3 and x.
   d. I start with x, add 8 and then, double the result.
4. From the above figure

a. what is the length of AB

b. what is the length of BC

c. Find the perimeter of the figure in terms of x and y.



Definition 3.1: a variable is any letter or a symbol that represent some unknown number or value. Such as x, y, z, n, m ...

Example1: In $x + 2$, x is variable

Example2: Describe the following using variables

a. Twice a number.
b. Three times a number.
c. Two more than a number.
d. Half a number.
e. Two less than a number
f. The product of a number and 10.
g. Product of ten and sum of two numbers.

Solution:

a. Let the number is x, then twice the number $= 2x$
b. Let the number is z, then three times a number $= 3z$
c. Let the number y. then two more than a number $= y + 2$
d. Let the number is n, then half of the number $= \frac{n}{2}$
e. Let the number is m, then two less than a number $= m - 2$
f. Let the number is k, then product of a number and $10 = 10k$
g. Let the two numbers are x and y ,then product of ten and sum of two numbers $= 10(x + y)$

Example 3: If you are 13 years old now. How old will/were you

a. after x years? b. after z years? c. y years ago?

Solution:

a. $13 + x$ b. $13 + z$ c. $13 - y$

Exercise 3.1.1

1. Identify the variables in each of the following.
   a. $2 + 4x$
   b. $5x + x$
   c. $2y + b$
   d. $\frac{x}{2} - p$
2. Describe the following using variables.
   a. Four times a number.
   b. One third of a number.
   c. Ten more than a number.
   d. The sum of two different numbers.
   e. The product of a number and 20.
   f. Five less than a number.
   g. Ten more than twice a number
   h. Six less than twice a number.
3. Express the following in words.
   a. $x - 1$
   b. $7x$
   c. $3 + 4x$
   d. $\frac{x}{3} - 1$
4. Find the value of y in the relation $y = 2x + 3$,when
   a. $x = 1$ b. $x = 5$ c. $x = -4$ d. $x = \frac{2}{3}$

3.1.2. Variables, Terms and Expressions

Activity 3.1.2

1. Which of the following are terms and which are not terms?
   a. $x + y$ b. $2x + 3$ c. $2x$ d. $-3x$ e. $\frac{3}{y}$ f. 4
2. Find the numerical coefficient of the following terms.
   a. $-3x$ b. y c. $-xy^2$ d. $\frac{xy}{4}$
3. Identify whether each pair of the following are like terms or unlike terms?
   a. $5x^2 \text{ and } 3y^2$ b. $3x \text{ and } x$ c. $2xy \text{ and } yx$ d. $8x^2y \text{ and } 4xy^2$

Definition 3.2: a constant (a number), a variable, product or quotient of a number and variable is called a term.

Example 1:

- 4 is a term.
- xy is a term.
- $12xy - 4$ is not a term.
- $\frac{3x}{y}$ is a term.
- $x - y$ is not a term.

Definition 3.3 In the product of a number and variable, the numerical factor of the term is called numerical coefficient.

Example 2:

a. The numerical coefficient of $6ab$ is 6.
b. The numerical coefficient of $-3xy$ is -3.
c. The numerical coefficient of x is 1.
d. The numerical coefficient of $-\frac{2ab}{3}$ is $-\frac{2}{3}$.

Definition 3.4: Like terms are terms whose variable and exponent of variable are exactly the same but they may differ in their numerical coefficients. Terms which are not like terms are called unlike terms.

A term which do not contain a variable is called constant term. All constant terms are like terms.

$3x$ is the same as $3x^1$ ← Exponent

$5x^2$ ← Exponent

Example 3:

- 2 and -5 are like terms, because they are constant terms.
- $\frac{3}{5}a^2b^2$ *and* $-\frac{2}{3}b^2a^2$ are like terms.
- $-20abc$ and acb are like terms.
- $3xyz$ *and* $x^2y^2z^2$ are unlike terms.
- $-2a^3b^3c^3$ *and* $-2a^3b^2c^3$ are unlike terms.
- $3xy$ *and* $3yx$ are like terms.

Exercise 3.1.2

1. Identify whether each pair of the following are like or unlike terms.
   a. $3x$ and $-5x$
   b. $20xy$ and ab
   c. $-80ab$ and $15a^2b^2$
   d. $5ab$ and $\frac{1}{3}ab$
   e. xy^2 *and* y^2x
   f. $3x^2$*and* $2x^3$
   g. xyz^2 *and* z^2xy
   h. $\frac{3}{x}$ *and* $\frac{5}{x}$
2. Which of the following are terms and which are not term?
   a. $5x - 4y$ b. $8x^2$ c. $\frac{2x-y}{5}$ d. $-9abc$ e. $\frac{8y}{x}$ f. 0
3. In each of the following expressions, determine the numerical coefficient.
   a. $2bm$
   b. $5yx^2$
   c. $-4x$
   d. $\frac{3x}{4}$
   e. $\frac{y}{2}$
   f. $-z$

Definition 3.5: Algebraic expressions are formed by using numbers, letters and the operation of addition, subtraction, multiplication and division.

Example 4: $12, 2x-3, a+10, 3x, \frac{3}{4}a, 3x, 3x+2a-5$, etc. are algebraic expressions.

Note: The terms of an algebraic expression are parts of the expression that are connected by plus or minus signs.

Example 5: In the expression $3x+2a-5$, the terms are $3x, 2a$ *and* -5. And the constant term is -5.

Definition 3.6: An algebraic expression in algebra which contains one term is called a monomial.

Example 6: $5x, 12ab, 3xyz, 0.2ab$ *…are monomials.*

Definition 3.7: An algebraic expression in algebra which contains two terms is called a binomial.

Example 7: $5x+3y, 12-ab, 3x+yz, x-b$ *…are binomials.*

Definition 3.8: An algebraic expression in algebra which contains three terms is called a trinomial.

Example 8: $2x+3y-3, 12+a-b, x-y-z, ab+6+z$ *are trinomials.*

Simplifying Algebraic Expressions

To simplify any algebraic expression, follow the following basic steps

- Remove brackets using distributive property.
- collecting like terms and
- add or subtract like terms.

Note:

1. When you add or subtract like terms, add or subtract their numerical coefficients.

   Example: $-3x + 5x = (-3 + 5)x = 2x$

2. Unlike terms cannot be added or subtracted

   Example: $5x + 2y$ cannot be added.

3. When you remove brackets using distributive property be careful of signs.

   Example:

   a. $-2(x - 4) = -2x - (-2) \times 4 = -2x + 8$

   b. $-3(-2x - 1) = -3(-2x) - (-3)(1) = 6x + 3$

4. When you multiply a number with variable, you can write the number as coefficient without multiplication sign

   Example:

   a. $4 \times y = 4y$

   b. $5 \times (3y) = 15y$

Example 9: Simplify the following algebraic expression

$a(b + c) = ab + ac$

a. $3x + 2y - 4x + 5y$

b. $3(x + 4) - 5(x - 2)$

c. $20 + 4(x + 3y) - 4x - 8y$

d. $4(a + 2b) - 3(2a - b) + 6a - 7b$

e. $3(2a + 3) - 2(a + 8)$

Solution:

a. $3x + 2y - 4x + 5y$

$= 3x - 4x + 2y + 5y$ ... ... collecting like terms

$= -x + 7y$ ..... adding/subtracting coefficient of like terms

b. $3(x + 4) - 5(x - 2)$

$= 3x + 12 - 5x + 10$ ..... distributive

$= 3x - 5x + 12 + 10$ ..... collecting like terms

$= (3 - 5)x + 22$ ..... combining constants

$= -2x + 22$ ..... simplifying

c. $20 + 4(x + 3y) - 4x - 8y$

$= 20 + 4x + 12y - 4x - 8y$ .....distributive

$= 4x - 4x + 12y - 8y + 20$ ..... collecting like terms

$= (4 - 4)x + (12 - 8)y + 20$ ..... combining constants

$= 4y + 20$ ....simplifying

d. $4(a + 2b) - 3(2a - b) + 6a - 7b$

$= 4a + 8b - 6a + 3b + 6a - 7b$ ..... distributive

$= 4a - 6a + 6a + 8b + 3b - 7b$ ..... collecting like terms

$= (4 - 6 + 6)a + (8 + 3 - 7)b$ .....combining constants

$= 4a + 4b$ ..... Simplify

e. $3(2a + 3) - 2(a + 8)$

$= 6a + 9 - 2a - 16$ ....distributive

$= 6a - 2a + 9 - 16$ .....collecting like terms

$= (6 - 2)a + (9 - 16)$ ..... combine constants

$= 4a - 7$ ..... simplify

Exercise 3.1.3

1. Categorize the following expressions as monomial, binomial or trinomial and determine the constant terms in each expression.

   a. 24 d. $5x - 3y - 4$ g. $z - y - 25x$

   b. $5x$ e. $20w + w^2$ h. y

   c. $-21t - 2x$ f. $3x - 2y + 1$ i. $2b - 4a$

2. Simplify the following expressions.

a. $3x + 35y - 8x$
b. $15 + 21b - 32b + 3$
c. $3(3x - 6y) + x + 18y - 4$
d. $3z - 12 - 11y + x + y - 48z - 3$
e. $-3(-a + 6) + x - (z + 4) - 2x$
f. $xy + ab - cd + 2xy - ab + dc$
g. $3x^2 + 4x + 6 - x^2 - 3x - 3$
h. $5 + 2y + 3y^2 - 8y - 6 + 2y^2$
i. $2x^2 - 3x + 8 + x^2 + 4x + 4$
j. $5(2 - x^2) - 3(x^2 - 4y)$

3.2. Solving linear equations

By the end of these section you should be able to:

- *define a linear equation.*
- *solve linear equations involving brackets.*
- *solve linear equations involving fractions.*

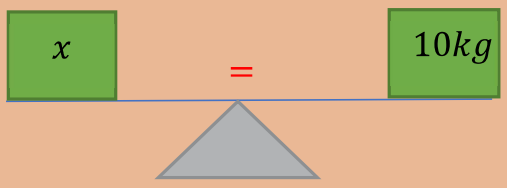
3.2.1. Linear Equations Involving Brackets

Activity 3.2.1

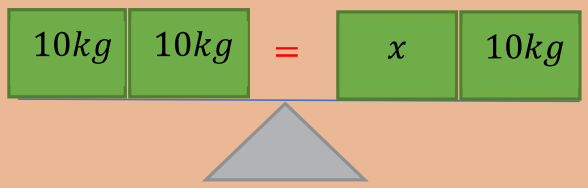
Discuss with your friends

1. Connect the following a pair of expressions by using '=' sign.
   a. $2x$ *and* 2 b. $3x - 1$ *and* 6 c. $y + 4$ *and* $5 - y$

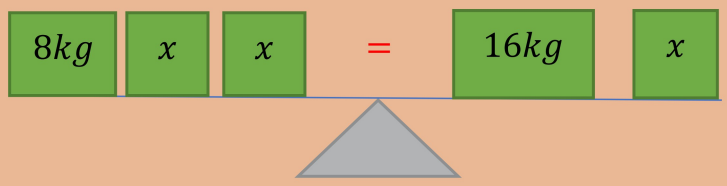
2. Using the following balance, find the unknown variables.
   a.



   c.



b.



3. What replacement of x will make the following equality true?
   a. $x-1=1$ c. $4x=8$
   b. $x+13=0$ d. $2x=0$
4. Determine whether each the following are equations or not.
   a. $2+x$ c. $x>2$
   b. $x=2$ d. $2x-1=0$
5. What is equation?

Definition 3.9: Two different algebraic expressions connected by equal $(=)$ sign is called equation.

Definition 3.10: A Linear equation in one variable x is an equation which can be written in standard form $ax+b=0$,where a and b are constant numbers with a$\neq 0$.

Note: an equation of a single variable in which the highest exponent of the variable is one is a linear equation.

Example 1: which of the following equations are linear and which of them are not.

a. $2x=2$ c. $2x^2+3=2x-1$
b. $x+3=2x-1$ d. $x=2x^3-6$

Solution:

- a and b are linear equation, because the highest exponent of variable is one.
- c and d are not linear equation, because the highest exponent of the variable is not one.

Note: All equations have two sides with respect to the equal sign. Left hand side and (L.H.S) and right-hand side (R.H.S)

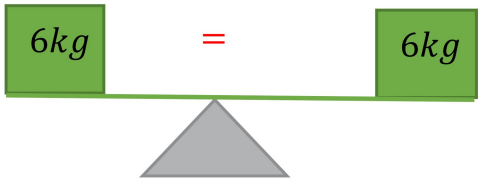
Example 2: In the equation $3x + 5 = x - 1$

L.H.S is $3x + 5$ and R.H.S is $x - 1$

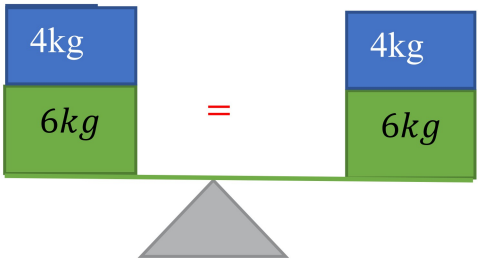
Rules of transforming equations to simpler form

Consider the following balances,

a.



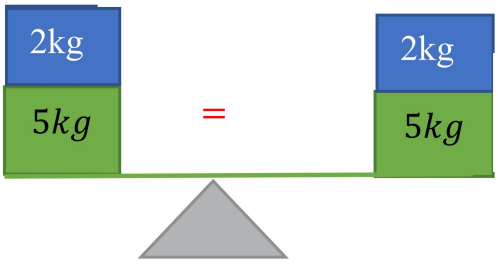
adding 4 kg to both sides



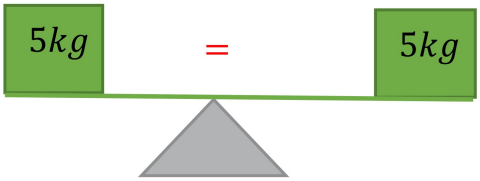
Rule 1: Adding the same number to both sides of equation will not affect the equation.

i.e. $if\ a = b, then\ a + c = b + c, where\ a, b, and\ c\ are\ any\ constant\ number$

b.



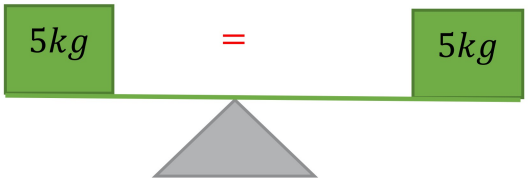
subtracting 2 kg from both sides



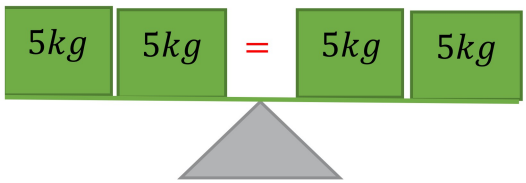
Rule 2: Subtracting the same number from both sides of equation will not affect the equation.

i.e. $if\ a = b, then\ a - c = b - c, where\ a, b\ and\ c\ are\ any\ constant\ number.$

c.

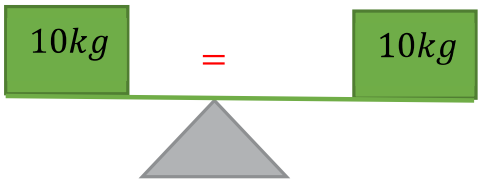


multiplying both sides by 2

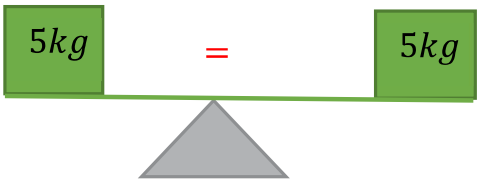


Rule 3: Multiplying both sides of equation with the same none zero number will not affect the equation. $If\ a = b, then\ a \times c = b \times c, where\ a, b, c\ any\ constant\ number\ and\ c \neq 0$

d.



dividing both sides by 2



Rule 4: Dividing both sides of equation with the same none zero number will not affect the equation.

$If\ a = b, then\ \frac{a}{c} = \frac{b}{c}, where\ a, b, and\ c\ are\ any\ constant\ number\ and\ c \neq 0$

Example 3: Find the value of x by using rules of transformation

a. $x - 10 = 6$ b. $10 - x = 6$ c. $x + 4 = -6$

Solution:

a. $x - 10 + 10 = 6 + 10$ ……adding 10 in both sides

$\underline{x = 16}$

b. $10 - 10 - x = 6 - 10$ ……. subtracting 10 from both sides

$-x = 6 - 10$

$\frac{-x}{-1} = \frac{-4}{-1}$ …….. dividing both sides by -1

$\underline{x = 4}$

c. $x + 4 - 4 = -6 - 4$ …… subtracting 4 from both sides

$\underline{x = -10}$

Sum of opposite numbers is zero

$-4 + 4 = 0$

$10 + (-10) = 0$

Example 4: Find the value of y by using rules of transformation

a. $8y = 72$ b. $\frac{y}{5} = -2$

Solution:

a. $8y = 72$, to find the value of y divide both sides by 8

$\frac{8y}{8} = \frac{72}{8}$

$\underline{y = 9}$

b. $\frac{y}{5} = -2$, to find the value of y multiply both sides by 5

$5 \times \frac{y}{5} = -2 \times 5$

$y = -10$

Exercise 3.2.1

1. Decide each of the following are either linear equation or not.
   a. $2y - 2$ c. $2x^2=4$ e. $3x = 0$
   b. $x - 1 = 5$ d. $x - 3 + 4x = 2x - 2$ f. $x > 0$
2. By what transformation the second equation is obtained from the first?
   a. $x + 1 = 2$ b. $-8x = 12$ c. $2x - 4 = 6$ d. $\frac{2x}{3} = 5$
   $x = 1$ $x = -1.5$ $2x = 10$ $2x = 15$
3. Find the value of unknown variable by using rules of transformation
   a. $-8x = 12$ b. $\frac{2x}{5} = 10$ c. $20 - x = 15$ d. $2x - 4 = 16$

Solving equations containing variable on both sides of the equation

Consider the equation $x - 5 = 10$, in solving this equation you add 5 on both sides and get $x = 10 + 5$, this is the same as shifting -5 to the other side of the equation and it becomes $+5$. i.e.

$x - 5 = 10 \quad +5$, so $x = 10 + 5 = 15$

Shifting -5 to the other side, becomes $+5$

Note: when you shift a variable(term) or number from one side of the equation to the other side its sign changes.

For example, in the equation $3x - 4 = 2x - 1$, shifting $2x$ to the left side you get

$-2x$

$3x - 4 = +2x - 1$ so, it becomes $3x - 2x - 4 = -1$

In solving equation containing variable on both sides, collect the terms containing variables on one side and numbers on the other side and do not forget changing the signs when you shift a term from one side to the other.

Example 5: solve the following equations and check the result

a. $2x = x + 4$

b. $2x - 8 = 4$

c. $5x - 10 = 6 - 3x$

d. $x + 5 - 4x = 3 + 2x - 6 - 4x$

Solution:

a. $2x = x + 4$

$2x - x = 4$ ($shifting\ \ x\ to\ the\ left\ it\ becomes\ -x$)

$\underline{\underline{x = 4}}$

Check: $2x = x + 4$, substitute 4 in the place of x

$2 \times 4 = 4 + 4$

$8 = 8$, (True)

Hence, $\underline{\underline{x = 4}}$ is the solution.

b. $2x - 8 = 4$

$2x = 4 + 8$ (shifting -8 to the right becomes$+8$)

$2x = 12$

$\frac{2x}{2} = \frac{12}{2}$ ($divide\ both\ sides\ by\ 2$)

$\underline{\underline{x = 6}}$

Check: $2x - 8 = 4$, substitute 6 in place of x

$(2 \times 6) - 8 = 4$

$12 - 8 = 4$

$4 = 4\ (True)$

Hence, $\underline{\underline{x = 6}}$ is the solution

c. $5x - 10 = 6 - 3x$

$5x + 3x = 6 + 10$ ……. $Transform\ the\ variable\ to\ the\ left\ side\ \&$ $the\ constants\ to\ the\ right$

$8x = 16$

$\frac{8x}{8} = \frac{16}{8}$ $(divide\ both\ sides\ by\ 8)$

$\underline{\underline{x = 2}}$

Check: $5x - 10 = 6 - 3x$, substitute 2 in place of x

$(5\times 2) - 10 = 6 - (3 \times 2)$

$0 = 0 \quad (True)$

Hence, $\underline{\underline{x = 2}}$ is the solution

d. $x + 5 - 4x = 3 + 2x - 6 - 4x$

$x - 4x + 5 = 3 - 6 + 2x - 4x$ .... (collect and simplify like terms on the same side) $-3x + 5 = -3 - 2x$

$-3x + 2x = -3 - 5$

$-x = -8$

$\frac{-x}{-1} = \frac{-8}{-1}$ $(divide\ both\ sides\ by -1)$

$\underline{\underline{x = 8}}$

Check: $x + 5 - 4x = 3 + 2x - 6 - 4x$

$8 + 5 - (4 \times 8) = 3 + (2 \times 8) - 6 - (4 \times 8)$

$13 - 32 = 3 + 16 - 6 - 32$

$-19 = -19(True)$, Hence, $\underline{\underline{x = 8}}$ is the solution.

Exercise 3.2.2

1. Solve the following equations and check the result.
   a. $3x - 9 = 4x + 5$
   b. $5x - 3 - 4x = 13$
   c. $12x + 7 = 5 - 3x + 17$
   d. $-x + 3 = x + 9$
   e. $3x - 9 = -x + 19$
   f. $8x + 5 = 2x + 17$
   g. $10 = 3x - 5 + 2x$
   h. $x - 5 = 2x + 13 + x$

Solving Linear equations involving brackets

Activity 3.2.2

Discuss with your friends

1. Simplify the following by using distributive property.
   a. $3(x+2)$
   b. $-(-2x+9)$
   c. $4(x-1)+2(1-2x)$
   d. $4(x-1)-2(x-9)$
2. Solve the following linear equations.
   a. $3(x+2)=0$
   b. $-(-2x+9)=1$
   c. $4(x-1)=2(1-2x)$
   d. $4(x-1)=2(x-9)$

To solve linear equations involving brackets

1. Remove the brackets using distributive property.
2. Collect the terms containing variables on one side and constants on the other side, do not forget to change sign while shifting the terms.

Note: For any numbers a, b and c

1. $a+(b+C)=a+b+c$
2. $a-(b+C)=a-b-c$
3. $a-(b-C)=a-b+c$
4. $a(b+c)=ab+ac$
5. $a(b-c)=ab-ac$

Example 6: Remove the bracket and simplify the following expressions

a. $3-(2x-5)$
b. $2y+(4-y)$
c. $3m-2(m-3)$
d. $-5n+3(4-2n)$

Solution:

a. $3-(2x-5)=3-2x+5$
$=3+5-2x$
$=8-2x$
$=-2x+8$

Note:

1. $a-b=-b+a$
2. $8-2x=-2x+8$

b. $2y + (4 - y) = 2y + 4 - y$

$= 2y - y + 4$

$= y + 4$

c. $3m - 2(m - 3) = 3m - 2m + 6$

$= m + 6$

d. $-5n + 3(4 - 2n) = -5n + 12 - 6n$

$= -5n - 6n + 12$

$= -11n + 12$

Example 7: solve each of the following equations.

a. $4(2x + 3) = 36$
b. $6(5x - 7) = 4(3x + 7)$
c. $-10y - (-2y + 3) = 21$
d. $4(x - 1) + 3(x + 2) = 5(x - 4)$

Solution:

a. $4(2x + 3) = 36$

$4 \times 2x + 4 \times 3 = 36$ ….. (Removing the bracket)

$8x + 12 = 36$

$8x = 36 - 12$ …… (shifting +12 in to right side, becomes -12)

$\frac{8x}{8} = \frac{24}{8}$ …….. (Divide both sides by 8)

$\underline{\underline{x = 3}}$

b. $6(5x - 7) = 4(3x + 7)$

$30x - 42 = 12x + 28$ ... ... (Removing brackets)

$30x - 12x = 28 + 42$ …….. (Shifting variables to the left and constants to the right)

$\frac{18x}{18} = \frac{70}{18}$ …….. (divide both sides by 18)

$x = \frac{35}{9}$

c. $-10y - (-2y + 3) = 21$

$-10y + 2y - 3 = 21$ …… (Removing the brackets)

$-8y - 3 = 21$

$-8y = 21 + 3$

$\frac{-8y}{-8} = \frac{24}{-8}$ …….. (Divide both sides by -8)

$\underline{\underline{y = -3}}$

d. $4(x-1)+3(x+2)=5(x-4)$

$4x-4+3x+6=5x-20$ ..... (Removing the brackets)

$4x+3x-4+6=5x-20$ ..... (Collecting like terms in one side)

$7x+2=5x-20$ ..... (Simplify in one side)

$7x-5x=-20-2$

$\frac{2x}{2}=\frac{-22}{2}$ ..... (Divide both sides by 2)

$\underline{x=-11}$

Exercise 3.2.3

1. Solve each of the following equations
   a. $(x-1)=11$
   b. $6(8y+3)=4(7y+5)$
   c. $8(2y-6)=5(3y-7)$
   d. $3-2(2x+1)=x-14$
   e. $7-(x+1)=9-(2x-1)$
   f. $5(1-2a)-3(4-4a)=0$
   g. $3x+7+3(x-1)=2(2x+6)$
   h. $5(2x+1)+3(3x-4)=4(5x-6)$
2. Solve for x in terms of m and n.
   a. $m(x-1)=0, m\neq 0$
   b. $m(x+n)=mn\ , m\neq 0$
   c. $n(x-2)=m+x\ , n\neq 1$
   d. $x(n-1)=4-x, n\neq 0$

3.2.2. Solving linear equation involving fractions

Activity 3.2.3

1. Find the LCM of the following numbers
   a. 2 *and* 3
   b. 4 and 6
   c. 3, 4 *and* 6
   d. 4 *and* 8
2. Multiply both side of the equation by the LCM of the denominator, simplify and Solve the equation
   a. $\frac{x}{2}=\frac{5}{2}$
   b. $\frac{3}{4}=\frac{x}{8}$
   c. $\frac{x}{3}+5=7$
   d. $\frac{3}{4}+\frac{x}{2}=\frac{x}{3}$

Example 8: Solve the following linear equations

a. $\frac{x+2}{4} = \frac{5}{3}$

b. $\frac{x}{8} + \frac{7}{4} = \frac{5}{6}$

c. $\frac{x+1}{3} + \frac{x-1}{10} = 12$

d. $\frac{2x+3}{3} + \frac{x-1}{2} + \frac{x+1}{4} = 8$

Solution:

a. $\frac{x+2}{4} = \frac{5}{3}$

The LCM of 4 and 3 is 12,

therefore multiply both sides by 12

$12(\frac{x+2}{4}) = 12(\frac{5}{3})$

$3(x+2) = 4(5)$ .....simplifying

$3x + 6 = 20$ .....removing brackets

$3x = 20 - 6$ ..... collecting like terms

$3x = 14$ .....simplifying

$\frac{3x}{3} = \frac{14}{3}$..... dividing both side by 3

$x = \frac{14}{3}$

b. $\frac{x}{8} + \frac{7}{4} = \frac{5}{6}$

The LCM of 8, 4 and 6 of is 24

$24(\frac{x}{8} + \frac{7}{4}) = 24(\frac{5}{6})$..... multiplying both sides by 24

$24\left(\frac{x}{8}\right) + 24(\frac{7}{4}) = 24 \times \frac{5}{6}$ ..... distributive property

$3x + 42 = 20$ .....simplifying

$3x = 20 - 42$ .....collecting like term

$3x = -22$ ..... simplifying

$\frac{3x}{3} = \frac{-22}{3}$..... dividing both side by 3

$x = \frac{-22}{3}$

Note: To solve linear equations involving fractions, follow the following steps:

Step1. Find the list common multiple (LCM) of denominators.

Step 2. Multiply both sides of the equation by LCM.

Step 3. Solve the linear equation by shifting the terms containing variable to one side and constants to the other side.

c. $\frac{x+1}{3}+\frac{x-1}{10}=12$

The LCM of 3 and 10 is 30, therefore multiplying both sides by 30

$30(\frac{x+1}{3}+\frac{x-1}{10})=30(12)$

$30\left(\frac{x+1}{3}\right)+30(\frac{x-1}{10})=30\times 12$……distributive property

$10(x+1)+3(x-1)=360$ ……simplifying

$10x+10+3x-3=360$ ……removing brackets

$10x+3x=360-10+3$ ……collecting like terms

$13x=353$ ……simplifying

$\frac{13x}{13}=\frac{353}{13}$……dividing both side by 13

$x=\frac{353}{13}$

Then, the solution is $x=\frac{353}{13}$

d. $\frac{2x+3}{3}+\frac{x-1}{2}+\frac{x+1}{4}=8$

The LCM of 3, 2 and 4 is 12

$12(\frac{2x+3}{3}+\frac{x-1}{2}+\frac{x+1}{4})=12\times 8$ multiplying both sides by 12

$12(\frac{2x+3}{3})+12(\frac{x-1}{2})+12(\frac{x+1}{4})=12\times 8$ ……distributive property

$4(2x+3)+6(x-1)+3(x+1)=96$ ……simplifying

$8x+12+6x-6+3x+3=96$ ……removing brackets

$8x+6x+3x=96-12+6-3$……collecting like terms

$17x=87$ ……simplifying

$\frac{17x}{17}=\frac{87}{17}$…… dividing both side by 17

$x=\frac{87}{17}$

Definition 3.11: Equivalent equations are two or more equations that have the same solution.

Example 9: show that $2x - 1 = 11$ and $x + 12 = 3x$ are equivalent equations.

Solution:

$2x - 1 = 11$ and $x + 12 = 3x$

$2x = 11 + 1$ and $x - 3x = -12$

$2x = 12$ and $-2x = -12$

$x = 6$ and $x = 6$

Hence, $2x - 1 = 11$ and $x + 12 = 3x$ are equivalent equations.

Example 10: show that $2(x - 5) = x + 6$ *and* $2x - 4 = 10$ are not equivalent

Solution:

$2(x - 5) = x + 6$ *and* $2x - 4 = 10$

$2x - 10 = x + 6$ *and* $2x - 4 = 10$

$2x - x = 6 + 10$ *and* $2x = 10 + 4$

$x = 16$ *and* $x = 7$

Therefore, $2(x - 5) = x + 6$ *and* $2x - 4 = 10$ do not have the same solution, so they are not equivalent.

Exercise 3.2.4

1. Solve the following

a. $\frac{5x}{13} + \frac{5x}{26} = 1$

b. $\frac{3x}{7} + \frac{35x}{8} = 10$

c. $\frac{2x}{5} - \frac{2}{3} = \frac{x}{2} + 6$

d. $\frac{-3}{5} + \frac{x}{10} = \frac{-1}{5} - \frac{x}{5}$

e. $\frac{9w}{7} - 6 = \frac{5w}{7} + 12$

f. $\frac{7}{10}x + \frac{3}{2} = \frac{3}{5}x + 2$

g. $\frac{1}{2}(4x + 6) = \frac{1}{3}(9x - 24)$

h. $\frac{n+1}{2} + \frac{n+2}{3} + \frac{n+4}{4} = 3$

i. $\frac{1}{3}(x + 6) - \frac{1}{2}(3x - 4) = 5$

2. Which of the following pairs of equations are equivalent?

a. $x = 0$ *and* $2x - 1 = -1$ c. $5x = 10$ *and* $14 = 2x + 4$

b. $2x - 6 = 2$ *and* $-x = -5$ d. $x = 0$ *and* $324x + 55 = 55$

3.3. Cartesian coordinate system

By the end of these sections you should be able to:

- *describe the Cartesian coordinate system*
- *draw the four quadrants of the Cartesian plane and mark the origin,* x *−axis and* y *− axis*
- *plot points on the Cartesian coordinate plane given their coordinates*
- *draw graph of linear equations like* $y = mx$*, in a Cartesian coordinate plane using table values manually and computer applications.*

3.3.1. The Four Quadrants of the Cartesian coordinate Plane

Activity 3.3.1

Discuss with your friends

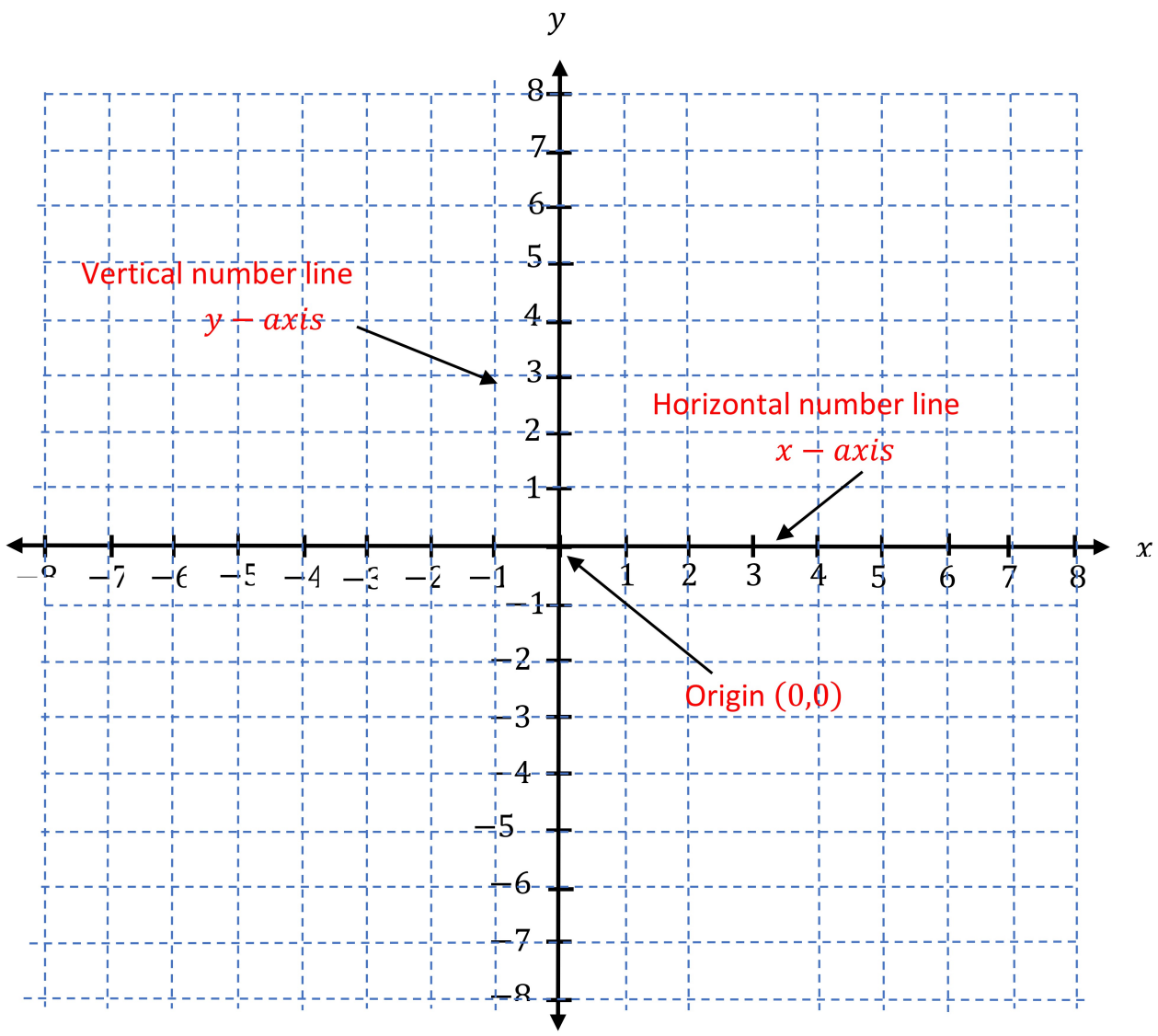
1. Draw horizontal number line and mark numbers.
2. Draw vertical number line perpendicular to horizontal number line that pass through zero.
3. Mark the numbers above and below zero on the vertical line.
4. The horizontal number line is called _____.
5. The vertical number line is called_____.
6. The two intersecting number lines (vertical and horizontal) set up a plane is called_____.

Definition 3.12: The two perpendicularly intersecting horizontal and vertical number lines together set up a plane is called Cartesian coordinate plane.

- The horizontal number line is called *x*-axis.

- The vertical number line is called *y*-axis.
- The point where, *x*-axis and the *y*-axis intersect is called origin.

Coordinate plane or Cartesian Plane



Locating points on the coordinate plane

- Points on Cartesian plane is described by two numbers (a, b) that are called coordinates.
- The first number a, is the horizontal position of the point from the origin. It is called $x - coordinate$ (abscissa).

- The second number b is the vertical position of the point from the origin. It is called $y-coordinate$ (ordinate).

To locate a point (a, b) on a coordinate plane follow the following steps.

Step 1: Move a unit in the horizontal direction from the origin.

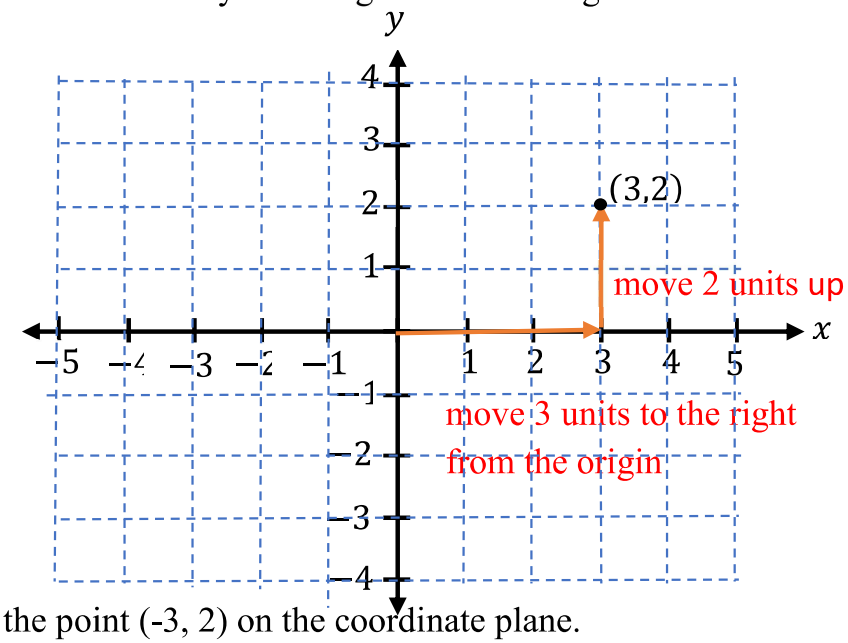
- Move a unit to the right if a is positive.
- Move a unit to the left if a is negative.

Step 2: Then, move b unit in the vertical direction

- Move b unit up if b is positive.
- Move b unit down if b is negative.

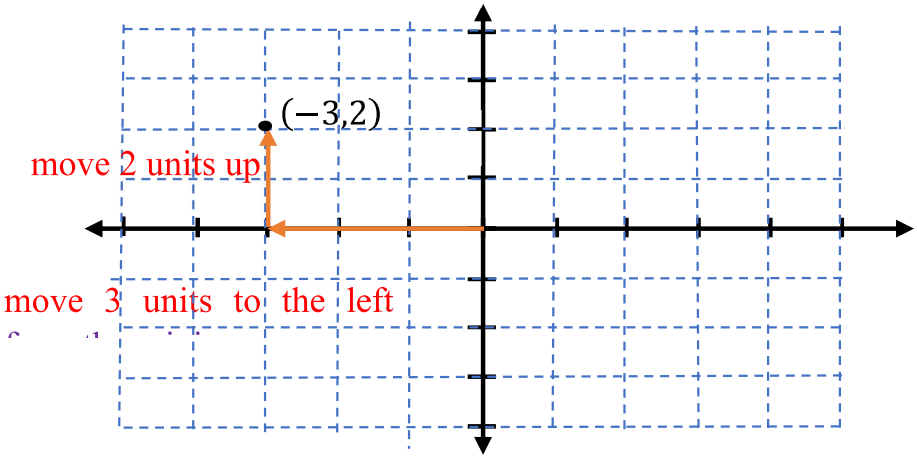
Example 1: Locate the point (3, 2) on the coordinate plane.

Solution: Move 3 units horizontally to the right from the origin and then 2 units up.



Example 2: Locate the point (-3, 2) on the coordinate plane.

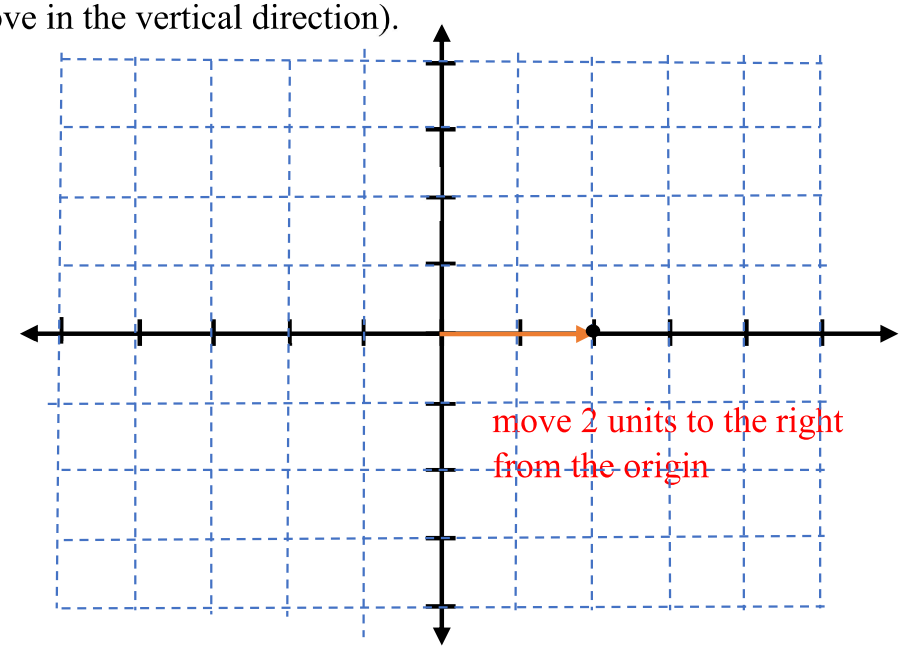
Solution: Move 3 units horizontally to the left from the origin and then 2 units up.



Example 3: Locate the point (2, 0) on the coordinate plane.

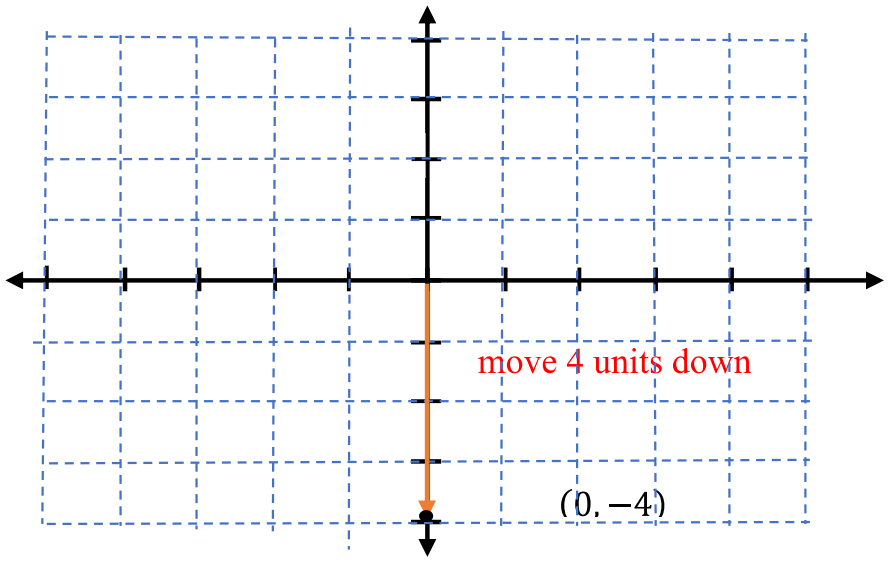
Solution: Move 2 units to the right from the origin and then 0 unit in the vertical direction

(no move in the vertical direction).



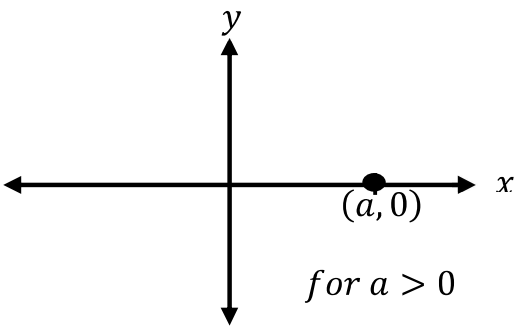
Example 4: Locate the point (0, -4) on the coordinate plane.

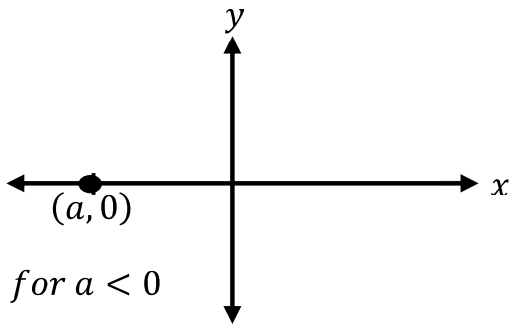
Solution: Start from origin and no move on horizontal, then move 4 units down in vertical direction.



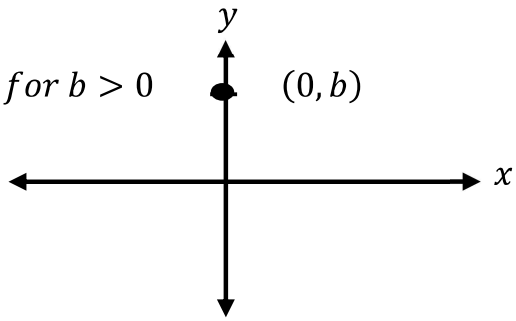
Note:

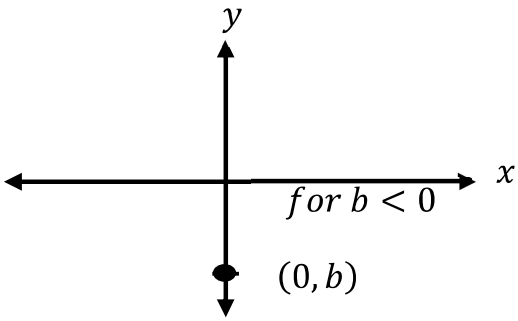
1. The coordinate of all points on the $x-axis$ has the form $(a, 0)$





2. The coordinate of all points on the $y-axis$ has the form $(0, b)$.



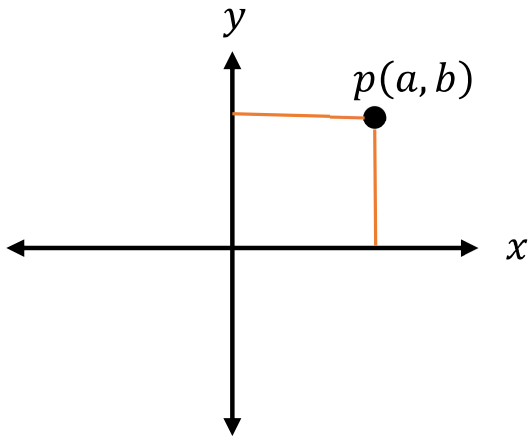


Determining the coordinate of a given point.

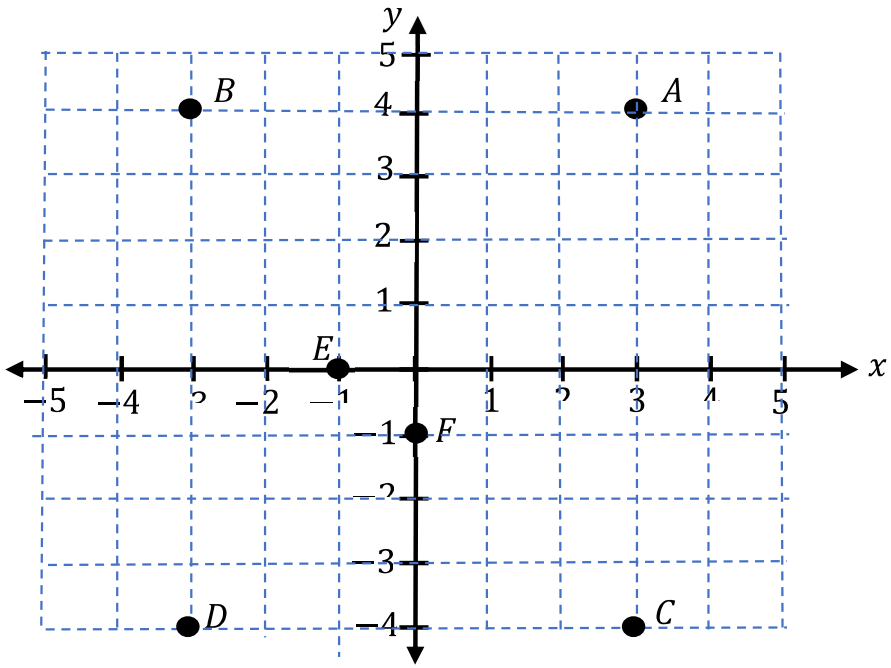
To determine the coordinate of a point p.

- First draw vertical line passing through the point p crossing $x-axis$. Let it crosses at $'a'$,then the x coordinate of p is a

- Then, draw horizontal line passing through p crossing $y-axis$. Let it crosses at $'b'$,then the y coordinate of p is b
- So, the coordinate of p is (a, b).



Example 5: Determine the coordinates of the points indicated below



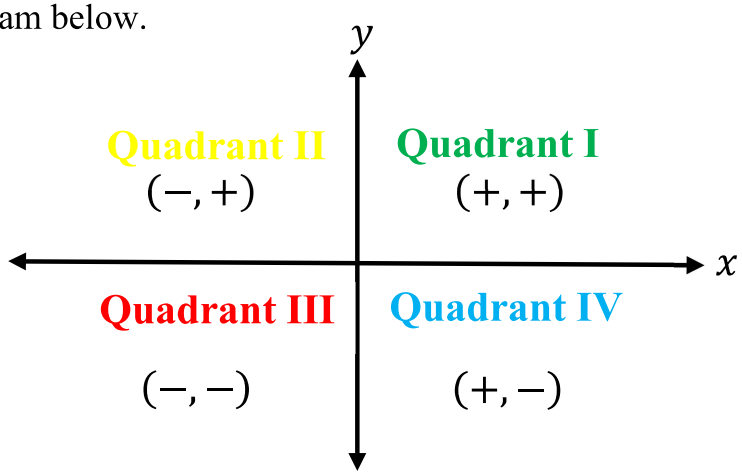
Solution: To determine coordinate of A, draw vertical line passing through A. it crosses *x -axis* at 3, hence its *x* coordinate is 3. Then draw horizontal line passing through A, it crosses *y – axis* at 4, hence its y coordinate is 4.

Therefore, the coordinate of A is $(3,4)$. In similar method you can determine the coordinate of the rest points. B (-3, 4), C (3, -4), D (-3, -4), E (-1, 0), F (0, -1).

Quadrants

The x-axis and y-axis divide the Cartesian plane into 4 regions known as quadrants.

The four quadrants (1st quadrant, 2nd quadrant, 3rd quadrant, and 4th quadrant) are shown in the diagram below.



- Quadrant 1 contains positive x values and positive y values.
- Quadrant 2 contains negative x values and positive y values.
- Quadrant 3 contains negative x values and negative y values.
- Quadrant 4 contains positive x values and negative y values.

Example 6: In which quadrant or axis, the following points lie?

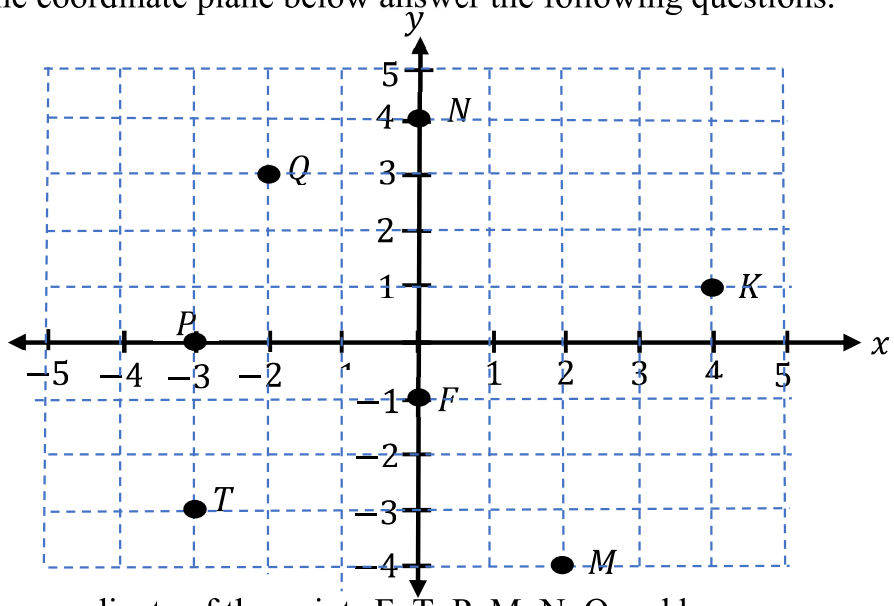
a. $(4, 1)$
b. $(-4, 1)$
c. $(0, 9)$
d. $(-3, -3)$
e. $(-3, 0)$
f. $(1. -3)$

Solution:

a. both x and y values are positive then $(4, 1)$ lies on 1st quadrant.
b. x is negative and y is positive then $(-4, 1)$ lies on 2nd quadrant.
c. x coordinate is 0, then $(0, 9)$ lie on *y-axis*.
d. both x and y values are negative, then $(-3, -3)$ lie on 3rd quadrant.
e. y coordinate is 0 then $(-3, 0)$ lie on *x-axis*.
f. x is positive and y is negative then $(1, -3)$ lie on 4th quadrant.

Exercise 3.3.1

1. locate the following points on the same Cartesian coordinate plane.
   a. M (-1, 4)
   b. N (4, 6)
   c. O (4, -1)
   d. P (-3, -2)
   e. Q (-5, 0)
   f. R (0. -3)
   g. S (-2, 1.5)
   h. T $(\frac{3}{2}, -5)$
   i, U (0, 0)
2. Based on the coordinate plane below answer the following questions.

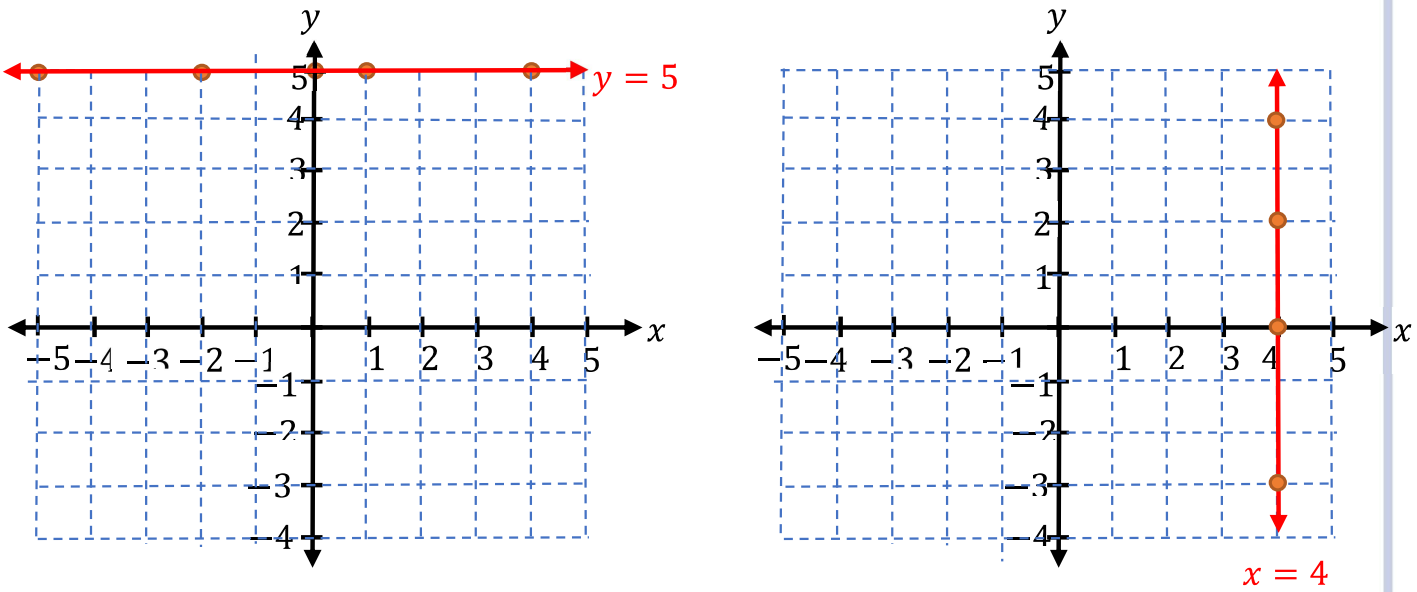


   a. Write the coordinate of the points F, T, P, M, N, Q and k.
   b. Which points has the coordinate (-2, 3).
3. In which quadrant or axis does the following coordinate points lie?
   a. $(5, 212)$
   b. $(-3,27)$
   c. $(0, -16)$
   d. $(-45, -4)$
   e. $(10, 0)$
   f. $(12. -30)$
   g. $(\frac{5}{2}, 0)$
   h. $(35, -12.5)$
   i. $(-\frac{7}{4}, -\frac{13}{5})$

3.3.2. Coordinates and graph of linear equations

Activity 3.3.2

1. Write the coordinate of the points that lie on the lines $x = 4$ and $y = 5$ indicated below.



2. Plot the following points on coordinate plane and connect them using straight line. $(-2, 0),\ (-2, -3),\ (-2, 3)$.
3. Plot the following points on coordinate plane and connect them using straight line. $(0, -2), (-3, -2), (3, -2)$.
4. What is the equation of the line in activity 2?
5. What is the equation of the line in activity 3?

Graph of an equation of the form $= a$, where $'a'$ is constant

To draw the graph of a line $x = a$, follow the following steps:

Step 1: Prepare table of values relating x *and* y. In the case of $x = a$, for any different possible values of y, the value of x is constant. It is $x = a$.

Step 2: plot the coordinate of the points in step 1.

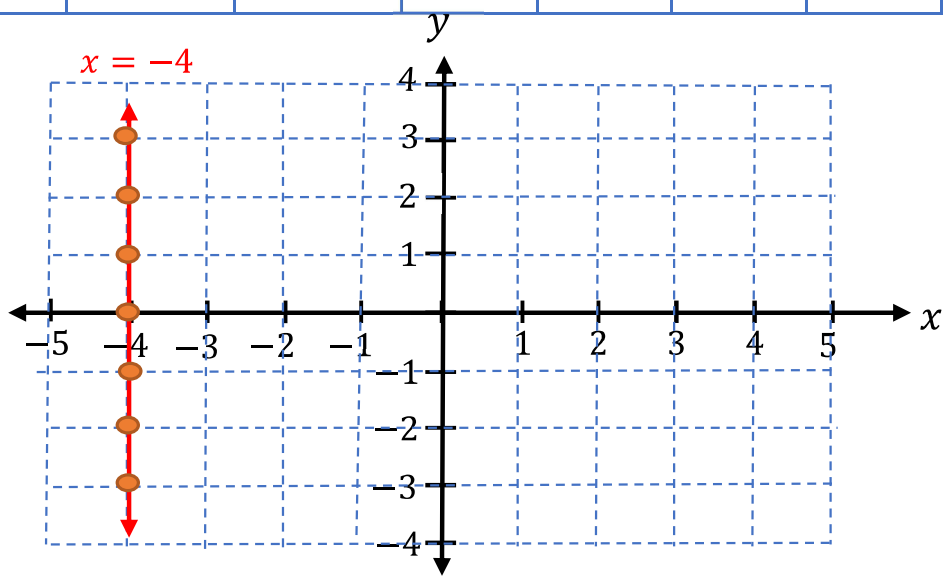
Step 3: Join all points in step 2 using straight line.

Example 7: Draw the graph of the following equation of lines.

a. $x = -4$ b. $x + 3 = 0$

Solution: a. $x = -4$, for any possible values of y, the value of x is -4

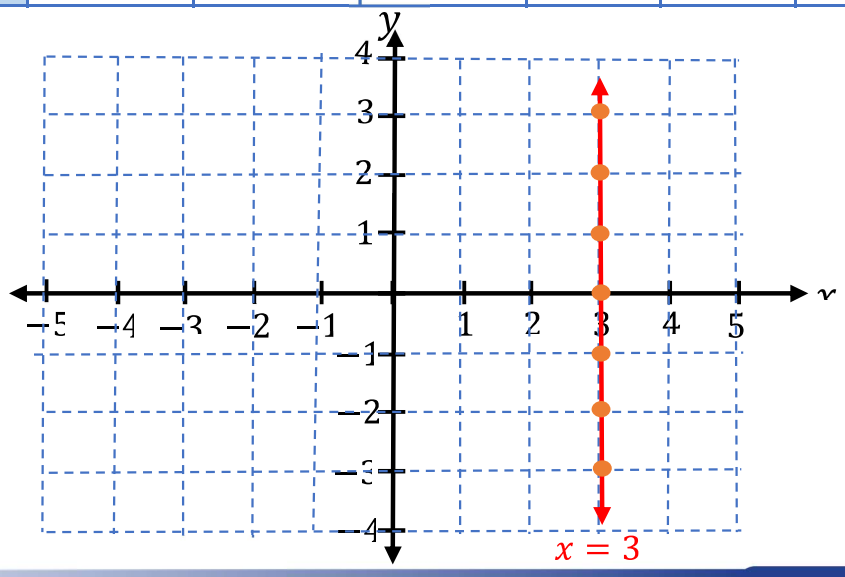
x	-4	-4	-4	-4	-4	-4	-4
y	-3	-2	-1	0	1	2	3
(x, y)	$(-4, -3)$	$(-4, -2)$	$(-4, -1)$	$(-4,0)$	$(-4,1)$	$(-4,2)$	$(-4,3)$



b. $x - 3 = 0$

$x = 3$, for any possible values of y, the value of x is 3

x	3	3	3	3	3	3	3
y	-3	-2	-1	0	1	2	3
(x, y)	$(3, -3)$	$(3, -2)$	$(3, -1)$	$(3,0)$	$(3,1)$	$(3,2)$	$(3,3)$



Note:

- The graph of the equation of the line $x = a$,(a is constant) is a line parallel to y-axis at a distance of a unit from the origin.
- If a is positive, then the line lies to the right of the y-axis.
- If a is negative, then the line lies to the left of y-axis.
- If $a = 0$, then the line lies on y-axis.

Graph of an equation of the form $= b$, where $'b'$ is constant

To draw the graph of a line $y = b$, follow the following steps:

Step 1: Prepare table of values relating x *and* y. In the case of $y = b$, for any different possible values of x, the value of y is constant. It is $y = b$.

Step 2: plot the coordinate of the points in step 1.

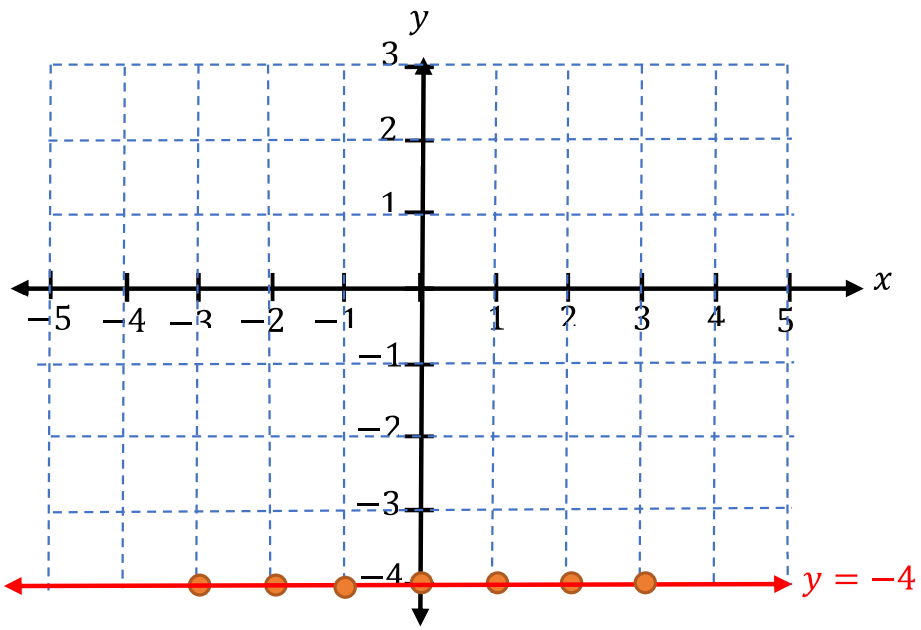
Step 3: Join all points in step 2 using straight line.

Example 8: Draw the graph of the following equation of lines.

a. $y = -4$ b. $y = 3$

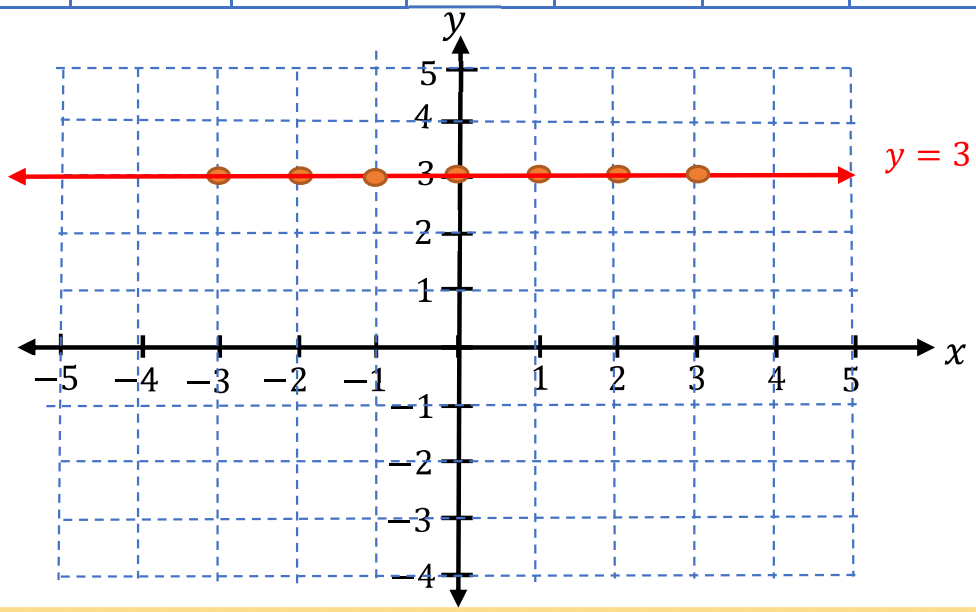
Solution: a. $y = -4$

x	-3	-2	-1	0	1	2	3
y	-4	-4	-4	-4	-4	-4	-4
(x, y)	$(-3, -4)$	$(-2, -4)$	$(-1, -4)$	$(0, -4)$	$(1, -4)$	$(2, -4)$	$(3, -4)$



b. $y = 3$

x	-3	-2	-1	0	1	2	3
y	3	3	3	3	3	3	3
(x, y)	$(-3,3)$	$(-2,3)$	$(-1,3)$	$(0,3)$	$(1,3)$	$(2,3)$	$(3,3)$



Note:

- The graph of the equation of the line $y = b$, (b is constant) is a line parallel to x-axis at a distance of b unit from the origin.
- If b is positive, then the line lies above the x-axis.
- If b is negative, then the line lies below the x-axis.
- If $b = 0$,then the line lies on x-axis.

Exercise 3.3.2

1. Draw the graph of the following equations on the same coordinate plane.

   a. $x = 7$ b. $y = 5$ c. $x = \frac{1}{2}$ d. $y = 0$

2. Which of the following points lie on a line y=1

   a. $(-1, 1)$ d. $(0, 1)$ g. $(2, 2)$
   b. $(1, 1)$ e. (-3.1) h. $(-12, 1)$
   c. $(0, 0)$ f. $(1, 2)$ i. $(1, 15)$

3. A point (m, 5), (m, 6) and (m, -2) lie on the line $x = -8$, the find the value of m?

4. Draw the graph of a line that passes through the following points.
   a. $(2,3),(1,3),(-2,3)$ c. $(0,0),(0,1),(0,-2)$
   b. $(-1,0),(-1,-1),(-1,4)$
5. List any three points that lie on the line $x = -2$
6. List any three points that lie on the line $y = -3$

Graph of the equation of the form $y = mx$ where m is constant, and $m \neq 0$

Activity 2.3.3

1. If $y = x$, then complete the given table below.

x	-3	-2	-1	0	1	2	3
y							
(x, y)							

2. Draw the graph of the line $y = x$ using table values above.

Steps: To draw the graph of $y = mx$

Step 1: Choose some values of x and find corresponding value of y.

Step 2: plot the coordinate of the points in step 1.

Step 3: Join all points in step 2 using straight line.

Example 9: Draw the graph

a. $y = 2x$ b. $y = -2x$

Solution:

a. $y = 2x$

Choose some values of x, and determine the corresponding values of y using the equation $y = 2x$

choose the values of $x = -3, -2, -1, 0, 1, 2, 3$

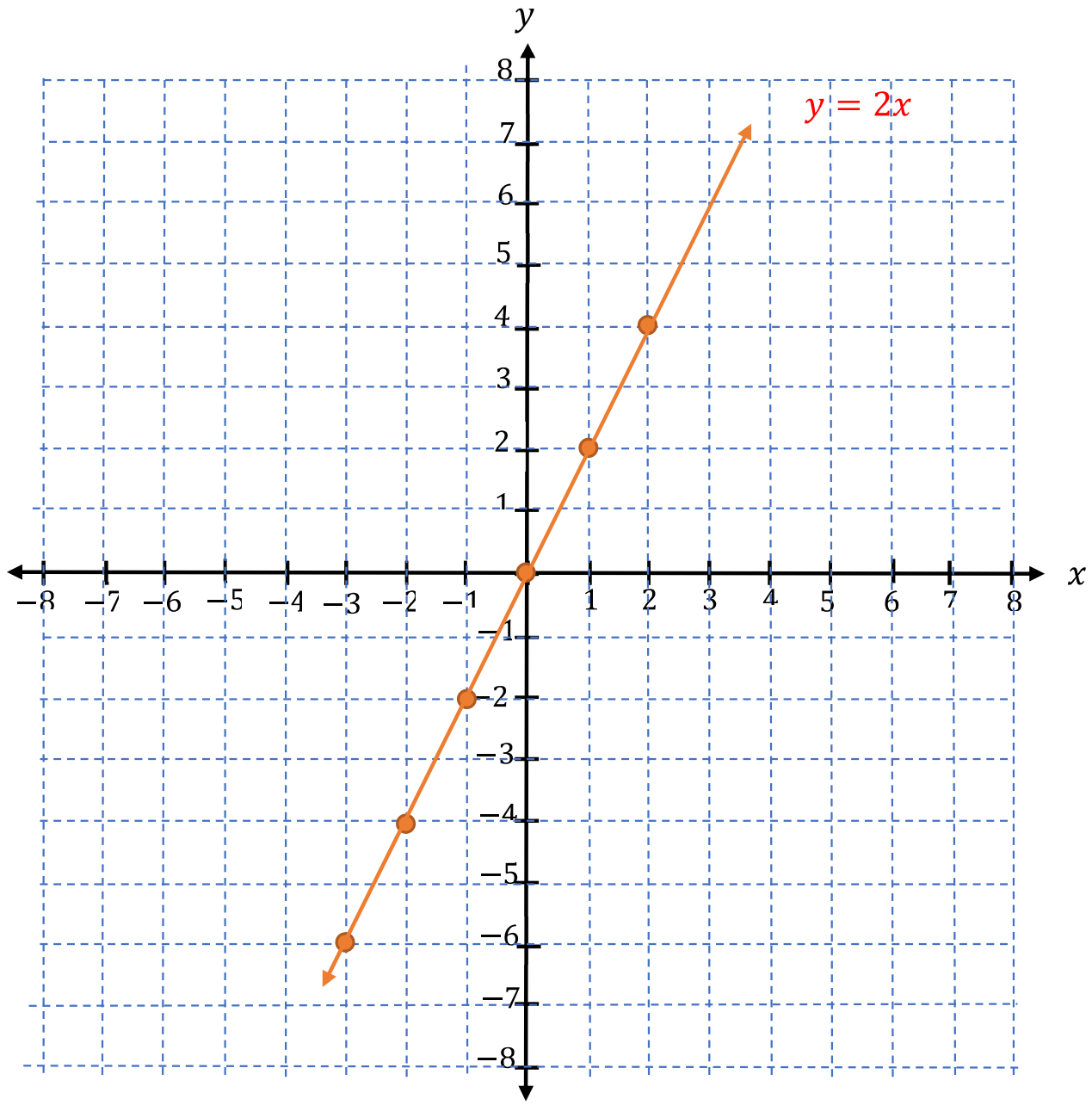
when, $x = -3, y = 2(-3) = -6$, *when* $x = -2, y = 2(-2) = -4$

when, $x = -1, y = 2(-1) = -2$, *when* $x = 0, y = 2(0) = 0$

when, $x = 1, y = 2(1) = 2$, *when* $x = 2, y = 2(2) = 4$

when , $x = 3, y = 2(3) = 6$

x	-3	-2	-1	0	1	2	3
y	-6	-4	-2	0	2	4	6
(x, y)	$(-3,-6)$	$(-2,-4)$	$(-1,-2)$	$(0,0)$	$(1,2)$	$(2,4)$	$(3,6)$



b. $y = -2x$

Choose some values of x, and determine the corresponding values of y using the equation

$y = -2x$

Choose the value $x = -3, -2, -1, 0, 1, 2, 3$

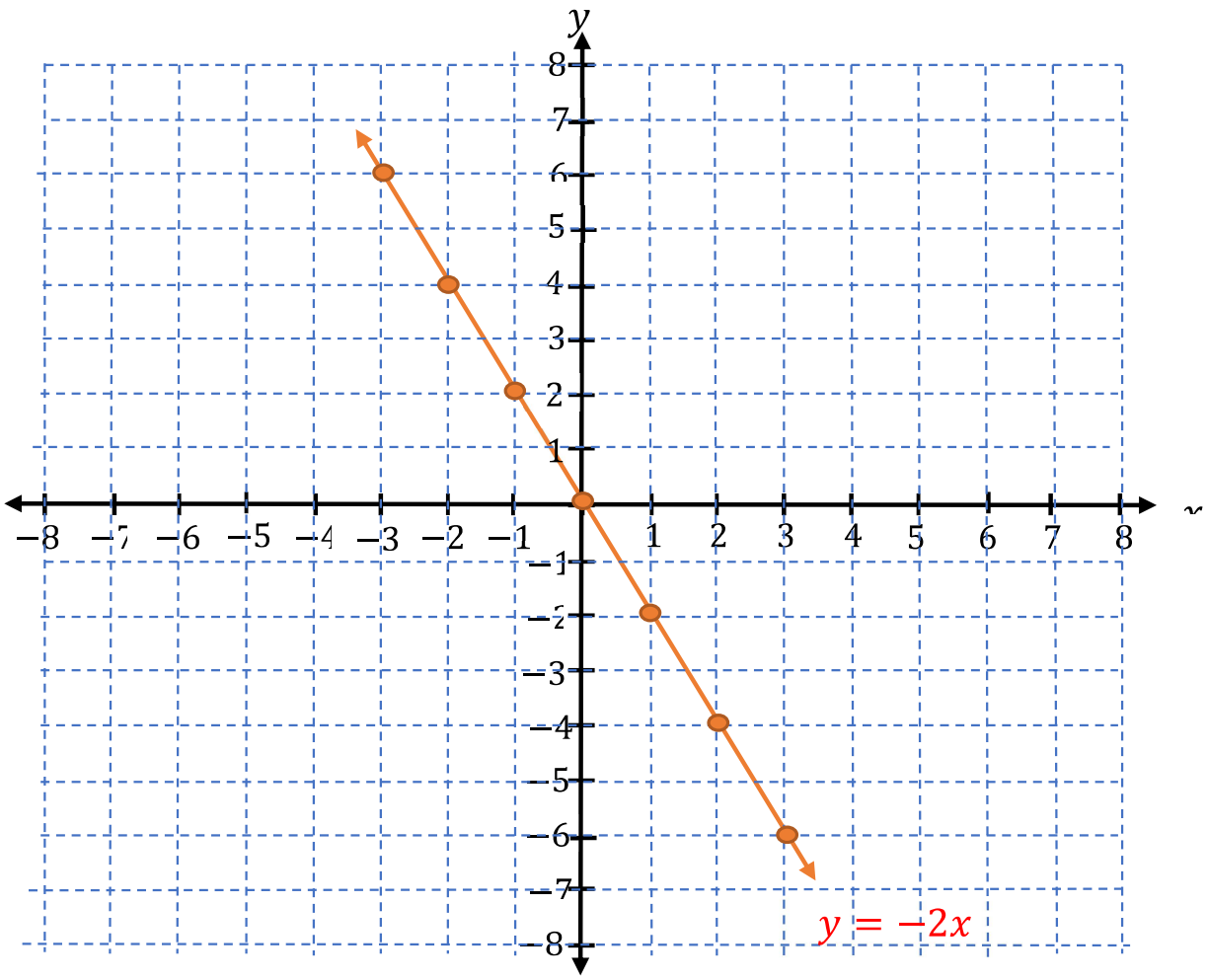
when, $x = -3, y = -2(-3) = 6$, *when* $x = -2, y = -2(-2) = 4$

when, $x = -1, y = -2(-1) = 2$, *when* $x = 0, y = -2(0) = 0$

when, $x = 1, y = -2(1) = -2$,*when* $x = 2, y = -2(2) = -4$

when, $x = 3, y = -2(3) = -6$

x	-3	-2	-1	0	1	2	3
y	6	4	2	0	-2	-4	-6
(x, y)	$(-3,6)$	$(-2,4)$	$(-1,2)$	$(0,0)$	$(1,-2)$	$(2,-4)$	$(3,-6)$



Note:

- The graph of $y = mx$ passes through 1st and 3rd quadrant when $m > 0$.
- The graph of $y = mx$ passes through 2nd and 4th quadrant when $m < 0$.
- For any values of m the graph of $y = mx$ passes through the origin.
- In the equation of the line y=mx, m is called **slope** of a line.

Exercise 3.3.3

1. If $y = 4x$ then complete the table below and draw the graph of $y = 4x$

x	-3	-2	-1	0	1	2	3
y							
(x, y)							

2. Which of the following equation of lines passes through 1st and 3rd quadrant
   a. $y = 7x$ c. $y - 12x = 0$ e. $y + 6x = 0$
   b. $y = -10x$ d. $y - x = 0$ f. $y = -\frac{3}{2}x$
3. Draw the graph of the following equations.
   a. $y = 5x$ c. $y - 3x = 0$ e. $2y + 6x = 0$
   b. $y = -5x$ d. $y + x = 0$ f. $y = \frac{3}{2}x$
4. If a point $(2, 8)$ lies on a line $y = mx$, then find the value of m.

3.4. Applications

Activity 3.4.1

Discuss with your friends

1. Translate each of the following phrases in to mathematical equations.
   a. A number increased by 5 is 11.
   b. A number decreased by 5 is 11.
   c. The product of a number and 5 is 30.

d. The sum of a number and 7 is 29.
e. Half of a number is 20.
f. 10 less than a number is 12.
g. 5 more than four times a number is 30.

Solving word problems

To solve word problems, follow the following important steps.

1. Read the problem carefully
2. Select variables for unknown quantities
3. Write a mathematical equation
4. Solve the equation
5. Interpret the result and write the final answer in words.
6. Check the answer

Example 1: A number increased by 10 is 23. What is the number?

Solution: Let y be the unknown number

Mathematical equation $\Rightarrow y + 10 = 23$

solve the mathematical equation $y + 10 = 23$

$y = 23 - 10$

$y = 13 \leftarrow$ solution

Hence, the unknown number is 13.

Example 2: 26 less than three times a number is 46. What is the number?

Solution: Let x be the unknown number

mathematical equation $\Rightarrow 3x - 26 = 46$

solve the mathematical equation $3x - 26 = 46$

$3x = 46 + 26$

$3x = 72$ $x = 24 \leftarrow$ solution

Hence, the unknown number is 24.

Example 3: A number decreased by 11 is 35. What is the number?

Solution: Let x be the unknown number

mathematical equation $\Rightarrow x - 11 = 35$

solve the mathematical equation $x - 11 = 35$

$x = 35 + 11$

$x = 46 \leftarrow$ solution

Hence, the unknown number is 46.

Example 4: The sum of one seventh of a number and 15 is 19. What is the number?

Solution: Let x be the unknown number

mathematical equation $\Rightarrow \frac{x}{7} + 15 = 19$

solve the mathematical equation $\frac{x}{7} + 15 = 19$

$\frac{x}{7} = 19 - 15$

$\frac{x}{7} = 4$ $x = 28 \leftarrow$ solution

Hence the unknown number is 28.

Exercise 3.4.1

1. A number increased by 7 gives 20, what is the number?
2. A number decreased by 3 gives 13. What is a number?
3. The difference between a positive number x and 3 is 21. What is a number?
4. 15 more than twice a number is 37. What is a number?
5. A number is increased by 7, twice the result is 41. Find the number.
6. If the sum of $\frac{1}{4}$ of a number and 15 is 13. What is a number?
7. 8 less than three times a number is 55. What is a number?
8. A number is doubled and the result is increased by 8. If the final result is 36. What is a number?

Example 5: (Application involving consecutive integers)

Find three consecutive integers whose sum is 156.

Solution: Let x be the smaller integer, then

The three consecutive integers $are\ x, x+1$, and $x+2$

mathematical equation $\Rightarrow x+(x+1)+(x+2)=156$

solve the mathematical equation $x+(x+1)+(x+2)=156$

$$x+x+1+x+2=156$$

$$3x+3=156$$

$$3x=156-3$$

$$3x=153$$

$$x=51 \leftarrow \text{smallest number}$$

Hence the unknown integers are 51, 52 and 53.

Example 6:(applications involving ages)

The sum of the ages of a man and his wife is 96 years. The man is 6 years older than his wife. How old is his wife?

Solution:

- let his wife's age is y
- The man's age will be $y+6$ (he is 6 years older)
- The sum of their age is 96, so

$$y+(y+6)=96$$

$$2y+6=96$$

$$2y=96-6$$

$$2y=90$$

$$y=45 \quad \ldots\ldots\ldots \quad \text{his wife's age}$$

Hence, the age of his wife is 45 years.

Example 7:

A farmer has sheep and hens. The sheep and hens together have 100 heads and 356 legs. How many sheep and hens does the farmer have?

Solution: Since the total heads is 100, the sheep and hens together are 100 in number.

Let the farmer has x sheep. And the number hens in this case is $100 - x$

A sheep has 4 legs, so x sheep have $4x$ legs.

A hen has 2 legs, so $(100 - x)$ hens have $2(100 - x)$ legs

The total number of legs is 356, so

$$number\ of\ legs\ of\ sheep + number\ of\ legs\ of\ hens = 356$$

$$4x + 2(100 - x) = 356$$

$$4x + 200 - 2x = 356$$

$$4x - 2x = 356 - 200$$

$$2x = 156$$

$$x = 78$$

Therefore, the farmer has $x = 78\ \ sheep$ and $100 - x = 100 - 78 = 22\ hens$

Exercise 3.4.2

1. The sum of three consecutive integers is 345. What are the numbers?
2. The sum of two consecutive even integers is 170. What are the integers?
3. The sum of two consecutive odd integers is 144.What are the numbers?
4. The sum of the ages of a man and his wife is 83. The man is 3 years older than his wife. How old is a man and his wife?
5. In a class there are 45 students. The number of girls is 1.5 times the number of boys. How many boys and how many girls are there in the class?

6. There are 26 girls and 22 boys in class room. Every student contributes equal amount of Birr for the renaissance dam (Hidassie Gidib). If Birr 2400 is collected from all students, then what amount of Birr contributed by
   a. Each student? b. boys? c. girls?
7. A government has 87 hospitals. If the government plan to build 2 hospitals in a year. How many years will it takes so that the total number of hospitals be 109.
8. In a triangle ABC, angle A is twice as large as angle B and angle B is 20 more than angle C. What is the measure of each angle? (hint sum of interior angle of a triangle is 180^0.)

Summary for unit 3

- A variable is any letter or a symbol that represent some unknown number or value.
- a constant (a number), a variable, product or quotient of a number and variable is called a term.
- In the product of a number and variable, the numerical factor of the term is called numerical coefficient.
- like terms are terms whose variable and exponent of variable are exactly the same but they may differ in their numerical coefficients. Terms which are not like terms are called unlike terms.
- Algebraic expressions are formed by using numbers, letters and the operation of addition, subtraction, multiplication and division.
- Two different algebraic expressions connected by equal $(=)$ sign is called equation.
- A Linear equation in one variable x is an equation which can be written in standard form $ax + b = 0$,where a and b are constant numbers with a$\neq 0$.
- To transform linear equations to simpler form, you can apply the following rules.

$i.$ if $a = b$, then $a + c = b + c$, where $a, b, and\ c$ are any constant numbers.

$ii.$ if $a = b$, then $a - c = b - c$, where $a, b, and\ c$ are any constant numbers.

$iii.$If $a = b$, then $a \times c = b \times c$, where $a, b\ and\ c$ are any constant number $and\ c \neq 0$.

$iv.$ If $a = b, then\ \frac{a}{c} = \frac{b}{c}$, where $a, b\ and\ c$ any constant number $and\ c \neq 0$.

- To solve linear equations involving brackets, remember the following key properties. For any numbers a, b and c

$$a + (b + C) = a + b + c$$
$$a - (b + C) = a - b - c$$
$$a - (b - C) = a - b + c$$
$$a(b + c) = ab + ac$$
$$a(b - c) = ab - ac$$

- Points on cartesian plane is described by two numbers (a, b) that are called coordinates.
  i. The first number a, is the horizontal position of the point from the origin. It is called $x - coordinate$ (abscissa).
  ii. The second number b is the vertical position of the point from the origin. It is called $y - coordinate$ (ordinate).
- The graph of the equation of the line $x = a$, a is constant is a line parallel to y-axis at a distance of a unit from the origin.
- The graph of the equation of the line y=b, b is constant is a line parallel to x-axis at a distance of b unit from the origin.
- The graph of $y = mx$ passes through 1^{st} and 3^{rd} quadrant when $m > 0$
- The graph of $y = mx$ passes through 2^{nd} and 4^{th} quadrant when $m < 0$
- In the equation of the line y=mx, m is called **slope** of a line.
- For any values of m the graph of $y = mx$ passes through the origin.
- In solving word problems follow the following steps
  i. Read the problem carefully
  ii. Select variables for unknown quantities
  iii. Write a mathematical equation
  iv. Solve the equation
  v. Interpret the result and write the final answer in words.
  vi. Check the answer

Review Exercise for unit 3

I. Write True for the correct statements and False for the incorrect statements.

1. For any integers a, b and $c, a(b+c) = ab + ac$.
2. An algebraic expression $2x^2 + 2xy$ is trinomial algebraic expression.
3. $3a^2b$ and $3ab^2$ are like terms.
4. If $a = b$, then $a - b = 0$.
5. The equation of the line $x + y = 0$ is vertical line.
6. 4 is the slope of a line $y - 4x = 0$.
7. The numerical coefficient of $3xy^2$ is 2.
8. If $c = d$, then $a + c = d + a$.

II. Choose the correct answer from the given alternatives.

9. Which one of the following pair of terms are like terms?

   a. $3x^2$ and $3x$ c. $2ab^2$ and $2a^2b$

   b. $-x^2$ and $-x$ d. $-4xy$ and $9yx$

10. Which of the following is equivalent to the equation $-3(2x-1) = 2x + 12$?

    a. $-6x + 3 = 2x + 12$ c. $x = -1$

    b. $-8x = 8$ d. $-6x = 2x - 4$

11. The solution of $\frac{x}{2} - x = \frac{3x}{8} - 11$ is:

    a. $\frac{7}{2}$ b. $\frac{7}{11}$ c. $\frac{88}{7}$ d. $-\frac{11}{7}$

12. Which of the following value of x satisfies the equation $2x = 3x - 1$?

    a. -1 b. 1 c. 0 d. 2

13. Which of the following is the slope of the line $y - 3x = 0$?

    a. -3 b. 3 c. $\frac{1}{3}$ d. has no slope

14. The sum of the age of a girl and her brother is 34. The girl is 8 years older than her brother, how old is her brother?

    a. 21 b. 13 c. 11 d. 23

15. Which one of the following equations is equation of horizontal line?

a. $y = 4x$ b. $y + 1 = 0$ c. $x - 2 = 0$ d. $x = 0$

III. Work out problems

16. Write the following word phrases in to algebraic expression.

a. Six less than a number.

b. Nine more than twice of a number.

c. A number increased by five.

d. 35 more than half of a number.

17. Simplify the following algebraic expressions

a. $-5x + 4 - 21x - 21$

b. $5x - (3x - 1) - 6(7 - x)$

c. $2(6y - x) - 10y + 2(y + x)$

18. Solve each of the following linear equations.

a. $4x + 36 = 86 - 8x$

b. $12x - 8 + 2x - 17 = 3x - 4 - 8x + 74$

c. $2(6y - 18) - 102 = 78 - 18(y + 2)$

d. $7(x + 26 + 2x) = 5(x + 7)$

e. $\frac{2x+7}{3} - \frac{x-9}{2} = \frac{5}{2}$

f. $\frac{x+3}{6} - \frac{x-5}{4} = \frac{3}{8}x$

g. $\frac{x-2}{3} + \frac{x+3}{8} = \frac{5x}{6} - 6$

19. Draw the graph of each of the following equation of the line on the same Cartesian coordinate plane

a. $y + 4x = 0$ b. $y + 1 = 0$ c. $x = 0$ d. $y = x$

20. Find the equation of the line passes through the origin with slope 4.

21. What is the slope of a line $y = \frac{x}{2}$

22. 16 less than four times a number is 12, what is the number?

23. The sum of one fifth of a number and 12 is 30. what is the number?

24. I think a number. If I subtract 8 from it and multiply the difference by 3, the result is 21. What is a number?

25. Bezawit is 8 years older than her sister. How old is Bezawit if the sum of their ages is 32.

Unit 4
Ratio, proportion and percentage

Learning Outcomes:

At the end of this unit, students will able to:

- *understand the notions of ratio and proportions.*
- *solve problems involving ratio and proportion*
- *describe a percentage*
- *solve problems involving percentages.*
- *relate fractions, decimals and percentages to real life situations*
- *apply the concept of percentage in solving real life problems*

Introduction

In this unit you will learn the basic mathematical concepts that will be applied in business, like profit, loss, simple and compound interest.

The unit has three sections. The first section deals with the concept of ratio and proportion. The second section deals with percentages and the third section is about application of ratio, proportion and percentage. Here you will see how to calculate loss, profit, simple interest, compound interest, income tax, VAT, and turnover tax.

4.1 Ratio and proportion

By the end of these section you should be able to:

- *Explain the notation of ratio*
- *Visualizes the ratio of 2 given numbers.*
- *expresses ratios in their simplest forms.*
- *finds the missing term in a pair of equivalent ratios.*
- *Solve problems on proportion*

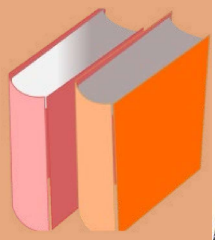
4.1.1 Ratio

Activity 4.1.1

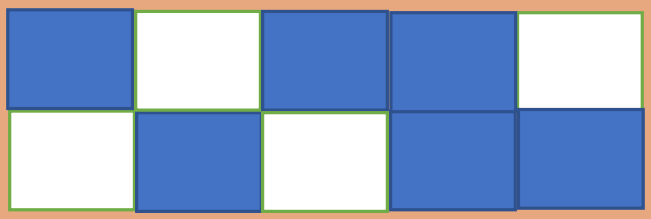
Discuss with your friends

1. Write a simple ratio for each of the following
   a. Cats to hens

   b. What is the ratio books to pens to pencils?



2. Consider the following figure



   a. What is the ratio of shaded part to unshaded part?
   b. What is the ratio of shaded part to the whole part?
3. Can you define a ratio based on the above activities in your own words?
4. What is the ratio of girls to boys in your class?

Definition 4.1: The method of comparing two or more quantities of the same kind and in the same units by division is known as a ratio.

- The symbol to denote the ratio is:
- The ratio of one quantity a to another quantity b is usually denoted by $a:b$.Where, a is called antecedent and b is called consequent.
- The ratio of a to b can be written as any one of the following forms:

  $i.\ a:b \rightarrow$ read as a ratio b or a to b

  $ii.\ \frac{a}{b} \rightarrow$ read as a over b

  $iii.\ a \div b \rightarrow$, a divided by b, provided that $b \neq 0$

- In most cases the ratio $a:b$ is written in simplified form, where a and b are natural numbers.
- To write the ratio of two quantities, the quantities must have the same units of measurement.
- Quantities having different units of measurement cannot be compared in the form of ratio.
- Ratio of quantities have no unit (unitless).

Example 1: In a class there are 18 boys and 24 girls.

a. What is the ratio of boys to girls

b. What is the ratio of girls to boys

c. What is the ratio of girls to the total number of students in the class?

d. What is the ratio of boys to the total number of students in the class?

Solution: *Total number of students*$= number\ of\ boys + number\ of\ girls$

$$= 18 + 24 = 42$$

a. Ratio of boys to girls $= \frac{number\ of\ boys}{number\ of\ girls}$

$$= \frac{18}{24} = \frac{18\div6}{24\div6} = \frac{3}{4}\ or\ 3:4$$

b. Ratio of girls to boys $= \frac{number\ of\ girls}{number\ of\ boys}$

$$= \frac{24}{18} = \frac{24\div6}{24\div6} = \frac{4}{3}\ or\ 4:3$$

c. Ratio of girls to total number of students $= \frac{number\ of\ girls}{total\ number\ of\ students}$

$$= \frac{24}{42} = \frac{24\div6}{42\div6} = \frac{4}{7}\ or\ 4:7$$

d. Ratio of boys to total number of students $= \frac{number\ of\ boys}{total\ number\ of\ students}$

$$= \frac{18}{42} = \frac{18\div6}{42\div6} = \frac{3}{7} or\ 3:7$$

Example 2: Write down the ratio of the first number to the second, in the simplest form.

a. 4 to 8 b. 54 to 128 c. 48 to 6 d. 540 to 48

Solution:

a. Ratio of 4 to 8 $= \frac{4\div4}{8\div4} = \frac{1}{2} or\ 1:2$

b. Ratio of 54 to 128 $= \frac{54\div2}{128\div2} = \frac{27}{64} or\ 27:64$

c. Ratio of 48 to 6 $= \frac{48\div6}{6\div6} = \frac{8}{1}\ or\ 8:1$

d. Ratio of 540 to 48 $= \frac{540\div2}{48\div2} = \frac{270\div2}{24\div2} = \frac{135\div3}{12\div3} = \frac{45}{4} or\ 45:4$

Example 3: Find the ratio of the following and express simplest form of fraction

a. $\frac{3}{4}$ to $\frac{5}{3}$

b. $\frac{2}{13}$ to $\frac{15}{16}$

c. 3.2 to 1.2

d. $\frac{7}{9}$ to 0.7

Solution:

$$\frac{a/b}{c/d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$$

a. Ratio of $\frac{3}{4}$ to $\frac{5}{3} = \frac{\frac{3}{4}}{\frac{5}{3}} = \frac{3}{4} \times \frac{3}{5} = \frac{9}{20} = 9:20$

b. Ratio of $\frac{2}{13}$ to $\frac{15}{16} = \frac{\frac{2}{13}}{\frac{15}{16}} = \frac{2}{13} \times \frac{16}{15} = \frac{32}{195} = 32:195$

c. Ratio of 3.2 to 1.2 $= \frac{3.2}{1.2} = \frac{32}{10} \times \frac{10}{12} = \frac{32 \div 4}{12 \div 4} = \frac{8}{3} = 8:3$

d. Ratio of $\frac{7}{9}$ to 0.7 $= \frac{\frac{7}{9}}{0.7} = \frac{\frac{7}{9}}{\frac{7}{10}} = \frac{7}{9} \times \frac{10}{7} = \frac{10}{9} = 10:9$

Example 4: Find the ratio of the following quantities

a. 65 cents to 5 Birr
b. 12 meters to 200cm
c. 700 grams to 6kg
d. 600 seconds to 8 minutes

Solution: First change the quantities in to the same units

a. 1 Birr$= 100cents, then\ 5\ Birr\ = 500cents$

The ratio of 65cents to 5 Birr = Ratio of 65 cents to 500cents $= \frac{65\ cents}{500\ cents}$

$$= \frac{65 \div 5}{500 \div 5} = \frac{13}{100} = 13:100$$

b. $1m = 100cm, then\ 12m = 1200cm$

The ratio of 12m to 200cm= Ratio of $1200cm$ to $200cm = \frac{1200\ cm}{200\ cm}$

$$= \frac{12 \div 2}{2 \div 2} = \frac{6}{1} = 6:1$$

c. $1kg = 1000grams\ , then\ 6kg = 6000grams$

The ratio of 700gram to 6kg = Ratio of 700 grams to 6000grams

$$= \frac{700\ gram}{6000\ gram}$$

$$= 7:60$$

d. $1minute = 60seconds, then\ 8minute = 480\ seconds$

The ratio of 600 seconds to 8minutes = Ratio of 600 seconds to 480 seconds

$$= \frac{600\ second}{480\ second}$$

$$= \frac{60 \div 12}{48 \div 12} = \frac{5}{4} = 5:4$$

Exercise 4.1.1

1. Write down the ratio of the first number to the second one in the simplest form

a. $48\ and\ 80$

b. $360\ and\ 72$

c. $4.8\ and\ 9.6$

d. $13.6\ and\ 10.2$

e. $\frac{2}{21}\ and\ \frac{8}{21}$

f. $2\frac{11}{12}\ and\ 1\frac{2}{3}$

2. Express the following ratios as in their lowest term:

a. 4 Birr to 16 cents

b. 2 liters to 2250 milliliters

c. 5 days to 100 hours

d. 3 hours and 30minutes to 180 minutes

e. 3.5 kg to 6500 grams

f. 3 hectares to 3000 square meters

Note:

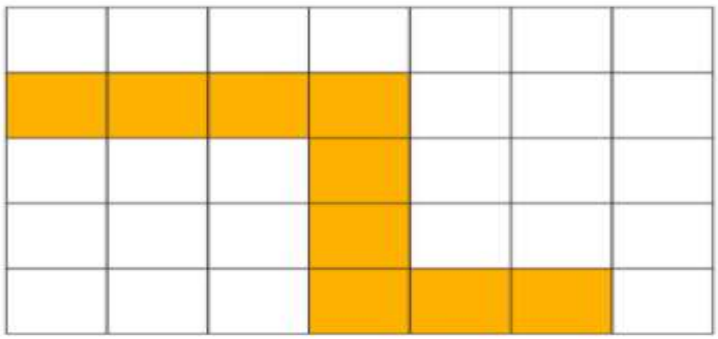
1 liter = 1000 milliliters

1 hectare = 10,000 square meters

1 day = 24 hours

1kg = 1000g

3. Using the figure below answer the following questions.



a. What is the ratio of shaded part to unshaded part?

b. What is the ratio of shaded part to the whole part?

c. What is the ratio of unshaded part to the whole part?

Dividing the given quantity in the given ratio

Examples:

1. Divide Birr 300 with ratio 3: 2

Solution: let the first part is $3x$ and the second part is $2x$, then

$$3x + 2x = 300$$

$$\frac{5x}{5} = \frac{300}{5}$$

$$x = 60$$

Therefore, the first part is $3x = 3 \times 60 = 180\ Birr$ and the second part is

$$2x = 2 \times 60 = 120\ Birr.$$

2. The ratio of two numbers is 7:3 and their sum is 50, then find the two numbers.

Solution: Let the first number is $7x$ and the second number is $3x$, then

$$7x + 3x = 50$$

$$\frac{10x}{10} = \frac{50}{10}$$

$$x = 5$$

Therefore, the first number is $7x = 7 \times 5 = 35$ and the second number is

$$3x = 3 \times 5 = 15.$$

3. Two numbers have ratio 13:10. If their difference is 84, then find the larger number.

Solution: let the larger number is $13x$ and the smaller number is $10x$, then

$$13x - 10x = 84$$

$$\frac{3x}{3} = \frac{84}{3}$$

$$x = 28$$

Therefore, the larger number is $13x = 13 \times 28 = 364$.

4. A bag contains red and white balls, the ratio of red to white balls is 3:4. If the bag contains 36 white balls, then find the number of red balls.

Solution: Let the number of red balls is $3x$ and the number of white balls is $4x$

Number of White balls $= 4x = 36$

$$\frac{4x}{4} = \frac{36}{4}$$

$$x = 9$$

Therefore, the number of red balls is $3x = 3 \times 9 = 27$

5. The ratio of the measures of the angles of a triangle is 1:2:3. then find the measure of each angle.

Solution: let the angles of the triangle are $1x, 2x \; and \; 3x$

$1x + 2x + 3x = 180^0 \ldots$ sum of interior angle of a triangle is 180^0

$$\frac{6x}{6} = \frac{180^0}{6}$$

$$x = 30^0$$

Therefore: The first angle is $1x = 1 \times 30^0 = 30^0$

Second angle is $2 \times 30^0 = 60^0$

Third angle is $3x = 3 \times 30^0 = 90^0$

6. If $a, b, and \; c$ are numbers such that $a: b: c = 3: 4: 5$ and $b = 20$,then find $a + b + c$.

Solution: Let $a = 3x, b = 4x \; and \; c = 5x$

$$b = 4x = 20$$

$$\frac{4x}{4} = \frac{20}{4}, x = 5$$

Therefore, $a = 3x = 3 \times 5 = 15, c = 5x = 5 \times 5 = 25$

And $a + b + c = 15 + 20 + 25 = 60.$

Exercise 4.1.2

1. Find two numbers whose ratio is 3 to 5 and whose sum is 192.

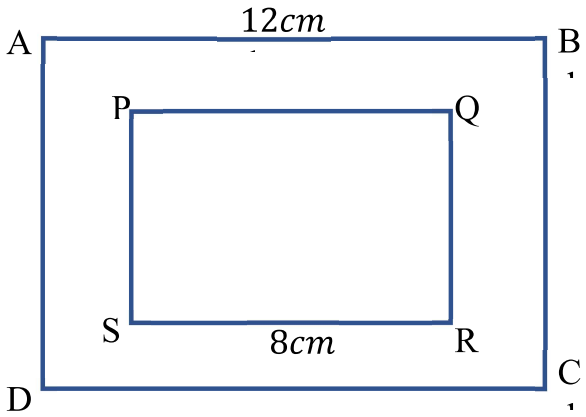
2. A wire of length 240 cm is cut into 3 pieces, in the ratio 1:2:5. Find the length of each piece.

3. Aster, Fatuma, Mohammed and Yared contribute the sum of money to renaissance dam in the ratio 1:3:5:7. If the largest amount contributed is Birr 1050. Calculate the amount contributed by each person.

4. Two numbers have ratio 12:5. Their difference is 98. Find the larger number.

5. If $a:b = 3:2$, find the value of $\frac{a+b}{a-b}$.

6.Find the ratio of the areas of the squares ABCD to PQRS, where AB = 12cm and PQ = 8cm.



7. x, y *and* z are numbers such that $x:y:z = 1:2:4$ and $y = 48$. Find $x + y + z$.

4.1.2 Proportion

Activity 4.1.2

Discuss with your friends

1. Compare the following ratios

   a. 4: 6 and 6: 9 b. $\frac{3}{6}$ and $\frac{40}{80}$

2. What result did you get from above activities?

3. What do you call equality of two ratios?

Definition 4.2: proportion is equality of two ratios.

Example 1: Compare the ratio $6:10$ and $48:80$, do they form proportion?

Solution: The simplest form of $6:10$ is $\frac{6}{10}\frac{\div 2}{\div 2} = \frac{3}{5} = 3:5$

The simplest form of $48:80$ is $\frac{48}{80}\frac{\div 16}{\div 16} = 3:5$

Therefore, $6:10 = 48:80$ is proportion

Example 2: Compare the ratio $1:3$ and $5:10$, do they form proportion?

Solution: $1:3$ is in simplest form

The simplest form of $5:10$ is $\frac{5}{10}\frac{\div 5}{\div 5} = \frac{1}{2} = 1:2$

The two ratios are not equal $(i.e.\ 1:3 \neq 5:10)$. So, they do not form proportion.

Note:

- If the four quantities $a, b, c, and\ d$ are in proportion, then $a:b = c:d$.
- The proportion $a:b = c:d$ can be written as $\frac{a}{b} = \frac{c}{d}$, here $a\ and\ d$ are called extremes (end terms) and $b\ and\ c$ are said to be means (middle terms)
- In proportion the product of means is equal to the product of extremes.

$$(i.e.\ if\ \frac{a}{b} = \frac{c}{d}\ , then\ a \times d = b \times c)$$

You can easily remember this by applying, cross multiplication as follows

$$\text{If } \frac{a}{b} = \frac{c}{d}, \text{ then } a \times d = b \times c \text{ or}$$

$$\text{If } \underbrace{a:b = c:d}_{a \times d}, \text{ then } \underbrace{a \times d}_{b \times c} = b \times c$$

Example 3: Show that $20, 45, 8\ and\ 18$ are in order of proportion.

Solution: To show they are in order of proportion $(i.e.\ 20:45 = 8:18)$

Use cross multiplication (product of means = product of extremes)

Product of means $= 45 \times 8 = 360$

Product of extremes $= 20 \times 18 = 360$

Since, product of means is equal to product of extremes. The numbers $20, 45, 8\ and\ 18$ are in order of proportion.

Example 4: Are the numbers $3, 5, 6\ and\ 12$ in order of proportion?

Solution: check$3:5 = 6:12$?

Product of means $= 5 \times 6 = 30$

Product of extremes $= 3 \times 12 = 36$

Product of means $\neq$ product of extremes, hence the numbers $3, 5, 6\ and\ 12$ are not in order of proportion.

Example 5: If the numbers $3, 5, x\ and\ 20$ are in order of proportion, then find the value of x. Solution: $3:5 = x:20$

Product of means = product extremes

$$\frac{5x}{5} = \frac{3\times20}{5},\ \underline{\underline{x = 12}}$$

Example 6: in proportion $\frac{3}{5} = \frac{9}{y}$, find the value of y

Solution: use cross multiplication

$$\frac{3}{5} = \frac{9}{y}$$

$$\frac{3y}{3} = \frac{5\times9}{3}$$

$$y = 15$$

Example 7: The distance from Addis Ababa to Adama is $75km$. If a map scale is $1:1{,}000{,}000$, then find the distance between the two cities on a map in centimeters.

Solution:

Map scale = Map distance : ground distance

$$1:1{,}000{,}000 = x:7{,}500{,}000\ cm$$

$$\frac{1{,}000{,}000x}{1{,}000{,}000} = \frac{1\times7{,}500{,}000cm}{1{,}000{,}000}$$

$$x = \frac{75\ cm}{10} \qquad x = 7.5cm$$

Therefore, the distance between the two cities on a map is $7.5cm$.

Note:

Map scale=map distance: ground distance

$$1km = 100{,}000cm$$

$$75km = 7{,}500{,}000\ cm$$

Exercise 4.1.3

1. From each pair of ratios below circle the given pair of ratios that are in proportion.
   a. $\frac{2}{3}$ and $\frac{4}{260}$
   b. $\frac{1}{3}$ and $\frac{5}{20}$
   c. $\frac{4}{5}$ and $\frac{12}{20}$
   d. $\frac{8}{4}$ and $\frac{2}{1}$
2. Find the unknown terms in each of the following
   a. $\frac{3}{5} = \frac{x}{20}$
   b. $x: 18 = 2: 9$
   c. $\frac{8}{m} = \frac{3.2}{4}$
   d. $\frac{3}{5} = \frac{18}{n} = \frac{y}{50}$
3. Show that the numbers $14, 21, 2, and\ 3$ are in order of proportion.
4. Given the proportion $10: 18 = 35: 63$, then find
   a. the sum of means.
   b. the product of means.
   c. the sum of extremes.
   d. the product of extremes.
5. A map scale is $1: 1{,}000{,}000$. If the distance between the two cities on a map is $3.5cm$, then find the actual distance between the cities in kilometers.

Direct and inverse proportionality

A. Direct proportionality

Definition 4.3: y is said to be directly proportional to x (written as $y \propto x$), if there is constant k such that $y = kx\ or\ k = \frac{y}{x}$. The number k is called constant of proportionality.

If $y \propto x$, then as x increases y also increases or as x decreases y also decreases. And the ratio $\frac{y}{x}$ is constant.

Example 8: the following table shows the number of kilograms of sugar bought and the price paid in birr.

$Sugar\ in\ kg(x)$	1	2	3	4
$Price\ in\ Birr(y)$	30	60	90	120

a. Is y directly proportional to x, if so what is the constant of proportionality?

b. Write the formula relating y and x.

c. Using the formula calculate the price of 6.5 kg sugar.

d. How much kg sugar will be bought with Birr 315.

Solution:

a. As x increases y also increases and the ratio $\frac{y}{x}$ is constant,

$k = \frac{30}{1} = \frac{60}{2} = \frac{90}{3} = \frac{120}{4} = 30$, hence $y \propto x$

b. $\frac{y}{x} = k$, implies $\frac{y}{x} = 30$ or $y = 30x$

c. y =?, when $x = 6.5kg$

$y = 30x$

$y = 30 \times 6.5$

$y = birr\ 195$

d. x =?, when $y = 315\ Birr$

$y = 30x$

$\frac{315}{30} = \frac{30x}{30}$

$x = 10.5kg$

Example 9: If $y \propto x$, then find the value of m.

x	12	6
y	48	m

Solution: since $y \propto x$, the ratio $\frac{y}{x}$ is constant

$\frac{48}{12} = \frac{m}{6}$

$\frac{12 \times m}{12} = \frac{48 \times 6}{12}$

$\underline{m = 24}$

Example 10: If 5kg sweet potato costs Birr 120, then calculate the cost of 8kg sweet potato.

Solution: As the amount of sweet potato increases, the cost also increases

Hence, the cost of sweet potato is directly proportional to amount of sweet potato.

$Sweet\ potato\ in\ (kg)$	5	8
$Cost\ in\ (Birr)$	120	x

$\frac{120}{5} = \frac{x}{8}$

$\frac{5 \times x}{5} = \frac{120 \times 8}{5}$

$x = 192\ Birr$

Therefore, the cost of 8kg sweet potato is Birr 192

Exercise 4.1.4

1. If y is directly proportional to x; $y = 24$ when $x = 6$,then find
   a. the constant proportionality.
   b. the formula relating $y\ and\ x$
   c. the value y ,when $x = 3$.
   d. the value x ,when $y = 15$.
2. Which of the following table shows direct proportional relationship between x and y?

a.

x	3	7	4
y	6	14	12

b.

x	8	5	6
y	24	15	18

3. What is the value of m from the given table below, if $y \propto x$?

x	14	7
y	21	m

4. y is directly proportional to x, if $x = 20\ when\ y = 160$ then what is the value of $x\ when\ y = 3.2$
5. If a car travels 120 km in 2 hours, then how long will it take for the car to travel 330 km (assuming the car is moving at constant speed)?

B. Inverse proportionality

Definition 4.4: y is said to be inversely proportional to x (written as $y \propto \frac{1}{x}$), if there is constant k such that $y = \frac{k}{x}$ or $k = y \times x$. The number k is called constant of proportionality.

If $y \propto \frac{1}{x}$, then as x increases, y decreases or as x decreases, y increases. And the product $y \times x$ is constant.

Example11: the following table shows the time taken for a car to move a distance of 5000m, at various speed.

$time\ in\ second(x)$	1000	500	250	200
$speed\ in\ m/s(y)$	5	10	20	25

a. Is y inversely proportional to x, if so what is the constant of proportionality?

b. Write the formula relating y and x.

c. Using the formula calculate the time taken by the car to cover the given distance moving at a speed of 8 meter/second.

Solution:

a. As x increases, y decreases and the product $y \times x$ is constant,
$k = 5 \times 1000 = 10 \times 500 = 20 \times 250 = 25 \times 200 = 5000$, hence $y \propto \frac{1}{x}$

b. $y \times x = k$, implies $y \times x = 5000$

c. x =?, when $y = 8\ m/s$

$y \times x = 5000$

$\frac{8 \times x}{8} = \frac{5000}{8}$

$\underline{x = 625\ seconds}$

Example 12: If $y \propto \frac{1}{x}$, then find the value of m.

x	12	6
y	15	n

Solution: Since $y \propto \frac{1}{x}$, the product $y \times x$ is constant

$$\frac{6 \times n}{6} = \frac{12 \times 15}{6}$$

$$\underline{n = 30}$$

Example 13: If 8 persons do a certain job in 9 hours, then how long will it take for 12 persons to do the same job. (assume all person do the job at the same rate).

Solution: as the number of persons increases, the time taken to finish the job decreases.

Hence, the number of persons is inversely proportional to time taken to finish the job

Number of persons	8	12
Time taken to do the job(hr.)	9	x

$$12 \times x = 8 \times 9$$

$$\frac{12 \times x}{12} = \frac{8 \times 9}{12}$$

$$\underline{x = 6\ hr}$$

Therefore, it takes 6 hr for 12 persons to do the job.

Exercise 4.1.5

1. If $y \propto \frac{1}{x}$ and $y = 6$ when $x = 4$,then find the constant of proportionality.
2. y is inversely proportional to x. If $x = 25$, then $y = 8$. What is the value of y when $x = 10$?
3. Which of the following table shows inverse proportional relationship between x and y?

a.

x	2	4	8
y	36	18	9

b.

x	5	10	15
y	20	10	5

4. It takes 8 days for 35 laborers to harvest coffee on a plantation. How long will 20 laborers take to harvest coffee on the same plantation.
5. x is inversely proportional to y.If $x = 25$ then $y = 15$. Find y when $x = 30$
6. 100 men working in a factory produce 6000 articles in 10 days. How long it takes (assume all workers do the job at the same rate)

a. 50 men to produce 6000 articles b. 150 men to produce 6000 articles.

7. A contractor appoints 36 workers to build a wall. They could finish the task in 12 days. How many days will 16 workers take to finish the same task?

4.2 Revision on Percentages

By the end of these section you should be able to:

- *visualizes percent and its relationship to fractions, ratios, and decimal numbers using bar models.*
- *identifies the base, percentage, and rate in a problem.*

Activity 4.2.1

Discuss with your friends

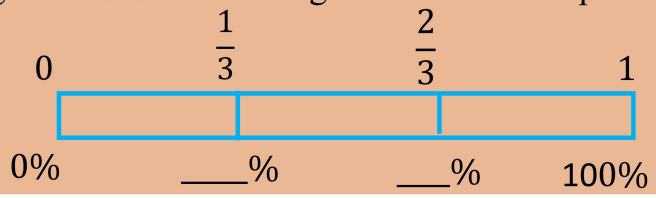
1. Consider the figure below and answer the given questions.

a. Count the green part and the total part.
b. What is the ratio of green part to total part?
c. Express the above ratio in percent form.
d. Define percent in your own words.

2. write the following numbers in percent form

a. $\frac{81}{100}$ b. $\frac{2}{100}$ c. 0.75 d. $\frac{1}{5}$

3. Use the following bar model relate the given fraction and percent.



Definition 4.5: The word percent means "for every hundred" or "per 100". We use the symbol % to denote percent.

For example, 5 percent symbolically written as 5% and 5% means 5 per 100 or $\frac{5}{100}$.

The concept of percent is widely used in a variety of mathematical applications. It helps to perform different valuations listed below.

- Interest
- Sales tax
- Change of Profit and loss
- Inflation

Converting a given percent in fractional form

Example 1: Change 30% in to fraction.

Solution: 30% means 30 per 100

$$\text{So, } 30\% = \frac{30}{100} = \frac{3}{10}$$

Note: To change a given percent to fraction:

- drop the percent sign from the number.
- put 100 as the denominator of the number and write the fraction in the lowest term.

Example 2: Convert each of the following percent form in to fraction**:**

a. 1% b. 45% c. 90% d. 4.8%

Solution:

a. $1\% = \frac{1}{100}$ c. $90\% = \frac{90}{100} = \frac{9}{10}$

b. $45\% = \frac{45\div5}{100\div5} = \frac{9}{20}$ d. $4.8\% = \frac{4.8}{100} = \frac{4.8\times10}{100\times10} = \frac{48\div8}{1000\div8} = \frac{6}{125}$

Converting a given percent in decimal form

To convert a given percent in to decimal form:

First write the percent in fraction form, then convert it in to decimal by division.

Example 3: Express each of the following percent form into decimal form.

a. 30% b. 7.38% c. 114% d. 11%

Solution:

a. $30\% = \frac{30}{100} = 0.3$ c. $114\% = \frac{114}{100} = 1.14$

b. $7.38\% = \frac{7.38}{100} = 0.0738$ d. $11\% = \frac{11}{100} = 0.11$

Exercise 4.2.1

1. Convert each of the following percent forms to fractions?
   a. 60% c. 125% e. $5\frac{3}{4}\%$
   b. 2.6% d. $\frac{3}{4}\%$ f. 0.045%
2. Convert each of the following percent forms in to decimals?
   a. 80% c. 12% e. $5\frac{1}{4}\%$
   b. 26% d. $\frac{2}{5}\%$ f. 0.45%

Converting a given number to percent

Example 4: convert $\frac{2}{5}$ to percent form

Solution: To convert it to percent, try to write equivalent form of the number, so that its denominator is 100.

$\frac{2}{5} = \frac{2\times100}{5\times100} = \frac{40}{100} = 40\%$

Example 5: Convert each of the following fractions as percent form:

a. $\frac{1}{2}$ b. $\frac{4}{5}$ c. $\frac{3}{4}$ d. $\frac{3}{8}$

Solution:

a. $\frac{1}{2} = \frac{1}{2} \times 100\% = \frac{100}{2}\% = 50\%,$

b. $\frac{4}{5} = \frac{4}{5} \times 100\% = \frac{400}{5}\% = 80\%$

c. $\frac{3}{4} = \frac{3}{4} \times 100\% = \frac{300}{4}\% = 75\%$

d. $\frac{3}{8} = \frac{3}{8} \times 100\% = \frac{300}{8}\% = 37.5\%$

Note: To convert a given number to percent form, multiply the given number by $\frac{100}{100}$ *or by* 100%.

Example 6: Express each of the following decimal into percent form

a. 0.5 b. 0.8 c. 0.24 d. 2.09

Solution:

a. $0.5 = 0.5 \times 100\% = 50\%$
b. $0.24 = 0.24 \times 100\% = 24\%$
c. $1.23 = 1.23 \times 100\% = 123\%$
d. $2.09 = 2.09 \times 100\% = 209\%$

Example 7: Express each of the following ratio into percent form

a. 75 seconds to 5minutes
b. 24 hours to 8days
c. 4 km to 500 meters
d. 5litres to 4000mililiters

Solution:

a. 75 seconds to 5minutes$= 75:300 = \frac{75}{300} = \frac{1}{4}$

$(since\ 5\ minutes = 300seconds)$

$= \frac{1}{4} \times 100\% = \frac{100}{4} = 25\%$

b. 24 hours to 8 days $= 24:192 = \frac{24}{192} = \frac{1}{8}$ $(since\ 8\ days\ =\ 192hours)$

$$= \frac{1}{8} \times 100\% = \frac{100}{8} = 12.5\%$$

c. 4 km to 500metres $= 4000:500 = \frac{4000}{500} = 8$ $(since\ 4km = 4000\ meters)$

$$= 8 \times 100\% = 800\%$$

d. 5 liters to 4000 milliliters $= 5000:4000 = \frac{5000}{4000} = \frac{5}{4}$

$(since\ 5L =\ 5000\ ml\ )$

$$= \frac{5}{4} \times 100\% = \frac{500}{4} = 125\%$$

Exercise 4.2.2

1. Convert each of the following fractions in percent form.
   a. $\frac{3}{5}$ c. $2\frac{3}{4}$
   b. $\frac{1}{6}$ d. $\frac{13}{6}$
2. Express each of the following decimals in percent form.

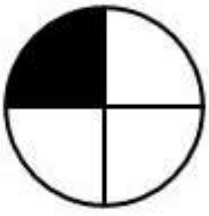
   a. 0.12 b. 7.5 c.3.65 d.0.0012

3. Convert each of the following ratios in percent form.

   a. $2m\ to\ 5m$ c. $50mililiters\ to\ 3liters$

   b. $2hr\ to\ 25minutes$ d. $5kg\ to\ 5000g$

4. Express the shaded part in percent form

a.  b. c.

Calculating base, rate and percentage

Activity 3.2.2

Discuss with your friends

1. Calculate 10% of 200
2. 5% of 60 is calculated as follows

$$5\%\ of\ 60 = \frac{5}{100} \times 60 = \frac{300}{100} = 3$$

In this calculation

a. 5% is called _____ b. 60 is called _____. c. 3 is called _____

Definition 4.6:

Base(B) is the number that represents 100% or total value of something.

Percentage(p) is the number or the amount that represents a part of the whole.

Rate(R) is the ratio of percentage to base, written as percent.

Note:

Base is usually preceded by "of" in the given statement.

Rate is expressed in percent.

Example 8: 20% of 150 is 30. In this statement determine base, rate and percentage.

Solution: The number expressed in percent is rate, hence 20% is rate

The whole part or the number preceded by of is base, hence 150 is base.

The part of whole is percentage, hence 30 is percentage.

Formula relating base, rate and percentage

Example 9: calculate 5% of 120

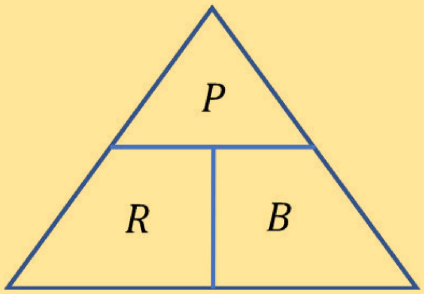
Solution: $5\%\ \text{of}\ 120 = \frac{5}{100} \times 120 = 6$

$$R \times B = P$$

From above example you can see that base, rate and percentage are related by the formula

$$P = R \times B$$

You can also use the following triangle to easily remember the relation between P, R and B



$R = \frac{P}{B} \times 100\%$
$P = R \times B$
$B = \frac{P}{R}$

Note: R is expressed in percent, so $\frac{P}{B}$ is multiplied by 100%

Example 10: calculate the following

a. 80% of $Birr$ 260
b. 9.6% of 7.2 $liters$
c. 90% of 264 cm
d. 34% of 7000kg

Solution:

a. $P = R \times B$
$= 80\% \times 260\ Birr$
$= \frac{80}{100} \times 260\ Birr$
$= 8 \times 26\ Birr$
$= Birr\ 208$

b. $P = R \times B$
$= 9.6\% \times 7.2\ liters$
$= \frac{9.6}{100} \times 7.2\ liters$
$= 0.096 \times 7.2\ liters$
$= 0.6912\ liters$

c. $P = R \times B$
$= 90\% \times 264cm$
$= \frac{90}{100} \times 264\ cm$
$= 0.9 \times 264\ cm$
$= 237.6\ cm$

d. $P = R \times B$
$= 34\% \times 7000kg$
$= \frac{34}{100} \times 7000kg$
$= 34 \times 70kg$
$= 2380kg$

Example 11: Eleni answered 80% of questions correctly in mathematics test. If the test has 25 questions, then how many questions she answered correctly?

Solution: $R = 80\%, B = 25, P = ?$

$$P = R \times B$$
$$= 80\% \times 25$$
$$= \frac{80}{100} \times 25$$
$$= 20$$

Therefore, she answered 20 questions correctly

Exercise 4.2.3

1. Calculate each of the following.
   a. What is 10% of 160?
   b. What is 60% of $Birr$ 300?
   c. Find 4.5% of $9km$.
   d. Find $\frac{3}{4}$% of $Birr$ 5000.
2. In a class there are 60 students. If 20% of the class are girls, then how many girls and boys are there?
3. 25% of people in Addis Ababa city watched the football final game on TV. How many people watched the football final game if the population of the city is 5000 000.

Example 12: calculate the unknown variables in each of the following questions

a. $Birr\ 36\ is\ 9\%\ of\ x$
b. $16\ minutes\ is\ 13\frac{1}{3}\%\ of\ T$

Solution:

a. $R = 9\%, B = x\ and\ P = 36$

$$B = \frac{P}{R}$$
$$= \frac{36}{9\%}$$
$$= \frac{36\ birr}{9/100} = 36\ birr \times \frac{100}{9} = \frac{3600\ birr}{9} = Birr\ 400$$

b. $R = 13\frac{1}{3}\% = \frac{40}{3}\% = \frac{40}{300}$, $B = T\ and\ P = 16\ minutes$

$$B = \frac{P}{R}$$

$$= \frac{16\ minutes}{40/300} = 16\ minutes \times \frac{300}{40} = 120\ minutes$$

Example 13: A women saves 20% of what she earns. If she saves Birr 300 a month, how much does she earn a month?

Solution: $R = 20\%$

A part of money she earns is saved, $P = Birr\ 300$.

The whole amount she earns a month? $i.e.\ B = ?$

$$B = \frac{P}{R}$$

$$B = \frac{300\ birr}{20\%} = \frac{300\ birr}{20/100} = 300\ birr \times \frac{100}{20} = Birr\ 1500$$

Therefore, the woman earns $\underline{Birr\ 1500}$ a month.

Exercise 4.2.4

1. Calculate the following
   a. 6 is 24% of a number, what is the number?
   b. If Birr 700 is 35% of Birr x, then what is the value of x?
   c. If 15% of a number is18, then what is the number?
   d. 40% of a number is 110, what is number?
2. If 30% of a man's salary is Birr 6300, what is the amount of his full salary?
3. In a class where the number of boys is 36 % of the total number of students. If there are 18 boys, then how many students are there in the class?

Example 14: Calculate the following

a. What percent of 400 is 50?
b. What percent of $2kg$ is $400gm$?
c. What percent of $1hr$ is 30 $minutes$?
d. What percent of $60m$ is $72cm$?

Note:

$1kg = 1000gm$

$1hr = 60\ minutes$

$1m = 100cm$

Solution:

a. $B = 400, P = 50, R = ?$

$$R = \frac{P}{B} \times 100\%$$

$$R = \frac{50}{400} \times 100\%$$

$$= \frac{50}{4}\% = 12.5\%$$

b. $B = 2kg = 2000gm, P = 400gm, R = ?$

$$R = \frac{P}{B} \times 100\%$$

$$R = \frac{400gm}{2000gm} \times 100\%$$

$$= \frac{400}{20}\% = 20\%$$

c. $B = 1hr = 60\ miutes, P = 30\ minutes, R = ?$

$$R = \frac{P}{B} \times 100\%$$

$$R = \frac{30\ minutes}{60\ miutes} \times 100\% = \frac{300}{6}\% = 50\%$$

d. $B = 60m = 6000cm, P = 72cm, R = ?$

$$R = \frac{P}{B} \times 100\%$$

$$R = \frac{72cm}{6000cm} \times 100\% = \frac{72}{60}\%$$

$$= 1.2\%$$

Example 15: In a certain village where farmers use two different types of fertilizers, fertilizer type A and fertilizer type B. 10000 farmers were asked which of the fertilizers they use. It was found that 2500 farmers use type A, 4550 farmers use type B and the remaining farmers use both types A and B. Find the percent of the farmers that

a. Use fertilizer type A.
b. Use fertilizer type B.
c. use both types of fertilizers.

Solution:

a. The total 10000 farmers is base, $B = 10000$

2500 is part of total farmers that use type A, hence it is percentage, $P = 2500$

$R = ?$

$$R = \frac{P}{B} \times 100\%$$

$$= \frac{2500}{10000} \times 100\% = 25\%$$

Therefore, 25% of farmers use type A fertilizer

b. Total 10000 farmers is base, $B = 10000$

4550 is part of total farmers that use type B, hence it is percentage, $P = 4550$

$R =?$

$$R = \frac{P}{B} \times 100\%$$

$$= \frac{4550}{10000} \times 100\% = \frac{455}{10}\% = 45.5\%$$

Therefore, 45.5% of farmers use type B fertilizer

c. Total 10000 farmers is base, $B = 10000$

the remaining farmers that use both type of fertilizers is

$10000 - (2500 + 4550) = 2950$, $P = 2950$

$R =?$

$$R = \frac{P}{B} \times 100\%$$

$$= \frac{2950}{10000} \times 100\% = \frac{295}{10}\% = 29.5\%$$

Therefore, 29.5% of farmers use both type of fertilizer

Exercise 4.2.5

1. Evaluate the following
   a. What percent of 60 is 30?
   b. What percent of 24 is 6?
   c. What percent of 800 is 500?
2. A woman saves Br.300 from her monthly salary. If her monthly salary is Br.7500, then find her saving in percent?
3. The following table gives the number of students in a given class according to their age and sex.

Sex	Age		
	13	14	total
Male	15	6	21
Female	18	9	27

   a. What percent of the class are females?

b. What percent of the class are males?
c. What percent of the class are 13 years old?
d. What percent of the class are 14 years old female students?

4. In a class of 48 students 6 of them were absent on Monday. What percent of the class was absent and what percent of class was attended on that day?

Mixed problems containing base, rate and percentage

1. A student scored 16 out of 25 in mathematics mid exam. What is the student's score in percent?
2. There are 50 students in a class. If 8% are absent on a particular day, find the number of students present in the class.
3. In a basket of oranges, 20% of them are defective and 76 are in good condition. Find the total number of oranges in the basket.
4. Hawa was able to sell 220 kilograms of her vegetables before noon. If Hawa had 400 kg of vegetables in the morning, what percent of Kilograms of vegetables was she sold in the morning?
5. A cow gives 24 liters of milk each day. If the milkman sells 75% of the milk in one day, how many liters of milk is left with him?
6. Tolosa sold 540 eggs. If these are 36% of total eggs, then How many eggs are not sold?
7. A factory has 2400 workers, 900 are males and the rest are females. What percent of the workers are females?

4.3 Application of Ratio, Proportion and Percentage

By the end of these section you should be able to:

- *Apply the concept of percentage to solve real life problems*

In our day to day life, we come across a number of situations wherein we use the concept of percent. In the following section, we discuss the application of percentage in different fields like problems in percent change, profit and loss, discount, simple interest, compound interest, Value added tax (VAT), turn of tax (TOT), and income tax.

Percent increase and decrease

Activity 4.3.1

Discuss with your friends

1. A quantity increases from 400 to 600
   a. Find the increased amount (actual change).
   b. Find the ratio of the increased amount to original quantity.
   c. Express the answer in b in the form of percent (find the percent increase)
   d. How do you calculate percent increase?
2. A quantity decreases from 500 to 300
   a. Find the decreased amount (actual change).
   b. Find the ratio of decreased amount to original quantity.
   c. Express the answer in b in the form of percent (find the percent decrease).
   d. How do you calculate percent decrease?

Note:

1. Percent increase is calculated as:

$$percent\ increase = \frac{increased\ amount}{original\ quantity} \times 100\%$$

2. Percent decrease is calculated as:

$$percent\ decrease = \frac{decreased\ amount}{original\ quantity} \times 100\%$$

Example 1: Find the percent increase from 250 to 400.

Solution: Increased amount is $400 - 250 = 150$

Original quantity is 250

$$percent\ increase = \frac{increased\ amount}{original\ quantity} \times 100\%$$

$$percent\ increase = \frac{150}{250} \times 100\% = 60\%$$

Therefore, the percent increase from 250 to 400 is 60%.

Example 2: Find the percent decrease from 400 to 250.

Solution: decreased amount is $400 - 250 = 150$

Original quantity is 400

$$percent\ decrease = \frac{decreased\ amount}{original\ quantity} \times 100\%$$

$$percent\ decrease = \frac{150}{400} \times 100\% = 37.5\%$$

Therefore, the percent decrease from 250 to 400 is 37.5%

Example 3: The students' population of a school is 4000 in 2013 E.C. If the students' population increased to 4620 in 2014 E.C. Then, find the percent increase between the two years.

Solution: Increased amount is $4620 - 4000 = 620$

Increased amount is 620

$$percent\ increase = \frac{increased\ amount}{original\ quantity} \times 100\%$$

$$percent\ increase = \frac{620}{4000} \times 100\% = 15.5\%$$

Example 4: 20% of an article is damaged and thrown away and only 20kg is left. Find its original weight.

Solution: let the original weight is x. The weight of the article decreased from $x\ to\ 20kg$. Percent decrease is 20%.

$$percent\ decrease = \frac{decreased\ amount}{original\ quantity} \times 100\%$$

$20\% = \frac{x-20}{x}$ ….. no need of multiplying by 100%, percent decrease is given.

$$\frac{20}{100} = \frac{x-20}{x},\ 20x = 100(x - 20)$$

$$20x = 100x - 2000$$

$$20x - 100x = -2000$$

$$\frac{-80x}{-80} = \frac{-2000}{-80}$$

$$x = 25kg$$

Therefore, the original weight is $25kg$

Exercise 4.3.1

1. Find the percent change
   a. From 80 to 100
   b. From 800 to 500
   c. From 500 to 300.
   d. From 6000 to 9000
2. While measuring a line segment of length 5cm, it was measured 5.2cm by mistake. Find the percent error in measuring this line segment
3. The number of student fail in mathematics test decreased from 20 to 12. What is the percent decrease?

4. Last year Samuel's salary was Birr 8000. If he gets a 10% increment this year, what is his current salary?
5. A T – shirt, discounted by 30% for a clearance sale. If the discounted price is Birr 399, then calculate
   a. The original price of T – shirt
   b. The amount of discount

4.3.1. Calculating profit and loss percentage

Activity 4.3.2

Discuss with your friends

1. A shop keeper buys a pair of shoes for Birr 500 and sells it for Birr 600. Find
   a. the cost price.
   b. the selling price.
   c. profit
   d. profit percent.
2. The price at which an article is purchased is called _____.
3. The price at which an article is sold is called _____.
4. Profit is made, when selling price is _____than cost price.
5. Loss is made, when selling price is _____than cost price.

Definition 4.7:

Cost Price (C.P): is the Price at which an article is purchased.

Selling Price (S.P): is the Price at which an article is sold.

Profit (Gain): When $S.P > C.P$, then there is profit, and $Profit = S.P - C.P$

Loss: When $S.P < C.P$, then there is loss, and $loss = C.P - S.P$

The formula to calculate profit and loss percent

1. Profit percent:

$$Profit\ \% = \frac{Profit}{cost\ price} \times 100\% \quad or$$

$$Profit\ \% = \frac{sellig\ price - cost\ price}{cost\ price} \times 100\%$$

$$Profit\ \% = \frac{S.P - C.P}{C.P} \times 100\%$$

2. loss percent:

$$Loss\ \% = \frac{Loss}{cost\ price} \times 100\% \quad or$$

$$Loss\ \% = \frac{cost\ price - sellig\ price}{cost\ price} \times 100\%$$

$$Loss\ \% = \frac{C.P - S.P}{C.P} \times 100\%$$

Example 5: A shop keeper buys a jacket for Birr 500, and gives it clean, then sells it for Birr 640. Calculate his profit percent.

Solution: $C.P = 500, S.P = 640, Profit\ \% = ?$

$$Profit\ \% = \frac{S.P - C.P}{C.P} \times 100\%$$
$$Profit\ \% = \frac{640 - 500}{500} \times 100\%$$
$$= \frac{140}{500} \times 100\%$$
$$= 28\%$$

Therefore, his profit percent is 28%.

Example 6: Girma bought 200 eggs for Birr 8 each. 20 eggs are damaged and sold the remaining eggs at Birr 8.5 each, calculate his profit/loss percent.

Solution: The cost of 1 egg is Birr 8, so the cost of 200 eggs is $200 \times 8 = Birr1600 = C.P$

20 eggs are damaged, so $200 - 20 = 180$ eggs are sold at Birr 8.5 each

$S.P = 180 \times 8.5 = Birr\ 1530$

Since, $.P < C.P$, there is loss and his loss percent is:

$$Loss\ \% = \frac{C.P - S.P}{C.P} \times 100\%$$

$$Loss\ \% = \frac{1600 - 1530}{1600} \times 100\%$$

$$= \frac{70}{16} \times 1\%$$

$$\underline{= 4.375\%}$$

Therefore, his loss percent is 4.375%.

Example 7: A dealer gained 10% profit by selling an article for birr 330. What was the original price of the article?

Solution: $profit\ \% = 10\%, S.P = 330, C.P = ?$

$$profit\ \% = \frac{S.P - C.P}{C.P} \times 100\%$$

$10\% = \frac{330 - x}{x}$ ….. no need of multiplying by 100%, profit % is given and $C.P = x$

$$\frac{10}{100} = \frac{330 - x}{x},\ 10x = 100(330 - x)$$

$$10x = 33000 - 100x$$

$$10x + 100x = 33000$$

$$\frac{110x}{110} = \frac{33000}{110}$$

$$\underline{x = 300}$$

Therefore, the original price of the article is $Birr\ 300$.

Exercise 4.3.2

1. Find profit or Loss percent when:
   a. Cost Price = Birr 2500 and Selling Price = Birr 2000
   b. Cost Price = Birr 3500 and Selling Price = Birr 4500
2. Find cost price when:
   a. Selling Price = Birr 7950 and Gain = 6%
   b. Selling Price = Birr 880 and Loss = 12%

3. Find selling price when
   a. Cost Price = Birr 870 and Gain = 10%
   b. Cost Price = Birr 750 and Loss = 15%
4. An article was bought for Birr 2000 and sold by Birr 2200. Find the profit or loss percent.
5. A shop keeper bought a jacket for Birr 1500 and gives it clean, then sold it for Birr 1800.What is his profit percent?
6. Estifanos bought goods by Birr 5200 and sold it at a profit of 12%. Find the selling price.
7. By selling a camera for Birr 24000, a man loss 4%. Find the cost price?
8. A trader bought a TV set for Birr 2500 and sold it at a loss of 8%. what was the selling price?
9. A man bought 100 eggs for Birr 800 and sells them for Birr 10 each, then find his profit percent.

4.3.2 Simple interest

Activity 4.3.3

Discuss with your friends

1. If you deposit Birr 1000 in a bank, then
   a. After a year Birr 1000 increased by a certain amount, why?
   b. By what amount Birr 1000 increases within 1 year, if the bank pays you 10% per year for using your money?
   c. By what amount Birr 1000 increases within 2 years, if a bank pays you 10% per year on money you deposited initially?
2. Define the following by your own words
   a. Interest
   b. Simple interest

When a person has to borrow some money as a loan from his friends, relatives, credit and saving association, bank etc. He promises to return it after a specified time period along with some extra money for using the money of lender.

- The amount of money borrowed or invested is called principal(p).
- The extra money paid for the use of money is called interest(I).
- The time length for which money is borrowed or lent is called Time(T).
- The total amount of money including interest at the end of time period is called Amount(A).
- The percentage charged on the total amount you borrow or save is called interest rate(R).

Example 8: If Birr 2000 is deposited in a bank at a rate of 7% per year, for 1 year then,

a. What is the principal?
b. What is the interest rate?
c. What is the time period?
d. Calculate interest paid in one year

Solution:

a. Money deposited is Birr 2000, so, principal(P) is Birr 2000
b. The interest rate(R) is 7% per year
c. The time period(T) is 1 year
d. The interest(I) in one year is 7% of Birr 2000, so $I = 7\% \times 2000$

$$I = \frac{7}{100} \times 2000 = Birr140$$

The interest paid on original principal only during the whole interest period is called simple interest.

Simple interest is calculated by the formula:

$I = PRT$

Where, $I = simple\ interest$

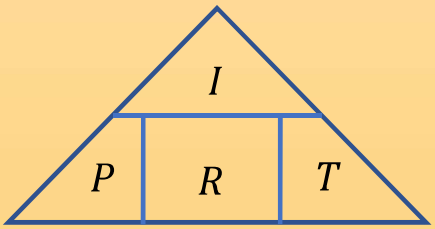
$P = principal$

$T = time$

And amount (A) is given by:

$A = P + I$

You can use the following triangle to easily reme]mber the relation between $I, P, R, and\ T$



$I = PRT$

$P = \frac{I}{RT}$

$T = \frac{I}{PR}$

and $R = \frac{I}{PT} \times 100\%$

Example 9: If Birr 4000 is deposited in a bank at a rate of 7% per year, for 4 years then, calculate

a. Simple interest at the end of 4 years
b. Amount at the end of 4 years

Solution: $P = Birr\ 4000, R = 7\%\ per\ year, T = 4\ years$

a. $I = PRT$

$I = 4000Birr \times \frac{7}{100} \times 4$

$\underline{I = Birr\ 1{,}120}$

b. $A = P + I$

$A = 4000 + 1120$

$\underline{A = Birr\ 5120}$

Example 10: Find the simple interest on Birr 700 at 12 % per year for 3 months.

Solution: $P = Birr700\ , R = 12\%\ per\ year, T = 3\ months$, since the interest rate is per year, the time must be converted to years.

$T = 3\ months = \frac{3}{12}\ year = \frac{1}{4} year$

$I = PRT$

$I = Birr\ 700 \times \frac{12}{100} \times \frac{1}{4}$

$\underline{I = Birr\ 21}$

Example 11: If Birr 1000 grows to 1200 in 4 years at simple interest rate, then calculate the simple interest rate.

Solution: $P = Birr1000, I = 1200 - 1000 = Birr\ 200, T = 4years, R = ?$

$$I = PRT, so\ R = \frac{I}{PT} \times 100\%$$

$$R = \frac{200}{1000 \times 4} \times 100\%$$

$$\underline{R = 5\%\ per\ year}$$

Exercise 4.3.3

1. Michael invests Birr 2000 in the bank that pays simple interest rate of 5% per year for 6 years. Then how much interest will he get in 6 years?
2. Birr 24000 invested at 11% simple interest per annum, then what is the amount after 8 years?
3. How long will it take for Birr 15000 to double itself, if it is invested at simple interest rate of 10% per year.
4. If Birr 20,000 grows to Birr 28000 after 20 years. Then what is the simple interest rate?
5. In how many years will Birr12,000 yield an interest of Birr 1200 at 5% simple interest?
6. An investment earned Birr 2100 interest after 6 years. If the simple interest rate is 7% per year. What was the principal?

4.3.3 Compound interest

Activity 4.3.4

Discuss with your friends

1. If Birr 1000 is invested at a rate of 10% per year, then calculate
   a. The interest at the end of first year
   b. The interest on the second year. (take second year principal to be initial principal plus first year's interest)
   c. The interest on third year. (take third year principal to be initial principal plus first year interest plus second year interest)
   d. The interest in 3 years
2. What is compound interest?

Compound interest is the interest on a loan calculated based on the initial principal plus the accumulated interest from previous periods

Example 12: If Birr 2000 is invested at a rate of 10 % compounded annually for 3 years, then calculate

a. compound interest in 3 years. b. amount at the end of third year.

Solution:

a. Interest for first year I_1

$P_1 = 2000, R = 10\%, T = 1\ year$ ..... for year one only

$I_1 = P_1RT$, so $I_1 = 2000 \times 10\% \times 1$

$I_1 = 2000 \times \frac{10}{100} \times 1$

$I_1 = Birr\ 200$

- Interest for year $2(I_2)$

Principal for second year is initial principal plus interests on previous years.

$P_2 = P_1 + I_1, P_2 = 2000 + 200 = 2200, T = 1$ ..... for year two only

$I_2 = P_2RT, I_2 = 2200 \times 10\% \times 1$

$I_2 = 2200 \times \frac{10}{100} \times 1$

$I_2 = Birr\ 220$

- Interest for year $3(I_3)$

Principal for third year is initial principal plus interests on previous years.

$P_3 = P_1 + I_1 + I_2, P_3 = 2000 + 200 + 220 = 2420,$

$T = 1$ ..... for year three only

$I_3 = P_3RT, I_3 = 2420 \times 10\% \times 1$

$I_3 = 2420 \times \frac{10}{100} \times 1$

$I_3 = Birr\ 242$

Therefore, the interest for 3 years is

$I = I_1 + I_2 + I_3$

$I = 200 + 220 + 242 = Birr\ 662$

b. Amount =?

$A = P + I$

$A = 2000 + 662 = Birr\ 2662$

Note: For compound interest amount can be calculated by the formula

$$A = P(1 + R)^T$$

Where, $A = amount, R = interest\ rate, P = principal\ and$ $T = time$ and

$I = A - P$

Using the formula, the above example is done as follows

$$P = 2000, R = 10\%, T = 3\ years, A = ?, I = ?$$

$$A = P(1 + R)^T$$

$$A = 2000(1 + 10\%)^3$$

$$A = 2000(1 + 0.1)^3 = 2000 \times 1.1^3$$

$$A = 2000 \times 1.331$$

$$A = Birr\ 2662$$

And interest, $I = A - P = 2662 - 2000 = Birr\ 662$

Example 13: what sum of money amounts to Birr 24,200 in 2 years at 10% per annum, compounded annually?

Solution: $P = ?, A = 24{,}200, R = 10\% = 0.1, T = 2$

$$A = P(1 + R)^T$$

$$24{,}200 = P(1 + 0.1)^2$$

$$24{,}200 = P \times 1.21, P = \frac{24200}{1.21} = Birr\ 20{,}000$$

Exercise 4.3.4

1. Find the compound interest on Birr 8000 for 2 years at 5% per annum, compounded annually.
2. Compare the simple interest and compound interest on Birr 8,000 at 10 % per annum for three years if the interest is compounded annually.
3. Find the difference between the simple and the compound interest on Birr 5000 for 2 years at 6% per annum.

4. Robel obtained a loan of Birr 250,000 from the commercial bank of Ethiopia. If the rate of interest is 8% per annum compounded annually and repaid Birr 50, 000 at the end of first year. what amount will he has to pay to the bank after 2 years to discharge his debt?

5. What sum of money will amount to Birr 21296 in 3 years at 10% per annum, compounded annually?

4.3.4 Ethiopian Income Tax, Turn Over Tax, VAT

Activity 4.3.5

Discuss with your friends

1. Why do governments collect tax?
2. List out the different types of tax.

Taxes are imposed by governments on their citizens to generate income for undertaking projects to boost the economy of the country.

Tax raise the revenue of the government to cover cost of

- Administration
- Defense
- Education
- Housing
- Health etc.

We should pay tax!

To boost our country's economy

Taxes help to raise the standard of living in a country.

In this section we will try to see three different types of tax, VAT, turn over tax and income tax.

Value added tax (VAT)

Value added tax (VAT) is a tax imposed by government on sales of some goods and services.

Note:

1. In Ethiopia VAT rate is 15%
2. $amount\ of\ VAT = 15\%\ of\ original\ cost$
3. Cost including VAT= $original\ cost + amount\ of\ VAT$

Example 14: The price of a machine is Birr 3000 before VAT.

a. Calculate the amount of VAT.

b. Calculate the total cost of machine including VAT.

Solution:

a. $VAT = 15\%\ of\ original\ cost$

$= \frac{15}{100} \times 3000$

$= Birr\ 450$

b. $Total\ cost\ of\ machine = 3000 + 450$

$= Birr\ 3450$

Example 15: The cost of TV stand including VAT is Birr 2300.

a. Calculate the cost of TV stand before VAT.

b. Calculate the amount of VAT.

Solution: let the cost before VAT is x

a. cost including VAT= $cost\ before\ VAT + VAT\ amount$

$2300 = x + 15\%\ of\ x$

$2300 = x + 0.15x$

$2300 = 1.15x$

$x = \frac{2300}{1.15} = Birr\ 2000$

Therefore, the cost before VAT is Birr 2000

Note:

$cost\ before\ VAT = \frac{Cost\ \ including\ VAT}{1.15}$

b. VAT amount $= 15\%\ of\ original\ cost$

$= \frac{15}{100} \times 2000$

$= 300\ Birr$

Turn over Tax (TOT)

Turn over tax is imposed on merchants, who are not required to register for VAT, but supply goods and services in the country.

In Ethiopia turn over tax rate is 2% on goods sold and services rendered locally.

Note:

1. A merchant his annual income below Birr 500,000 will be registered to collect turn over tax.
2. A merchant whose annual income above Birr 500,000 will be registered to collect turn over tax.

Example 16: Calculate turn over tax on sales of Birr 10,000

Solution: Turn over tax $= 2\%\ of\ 10{,}000$

$$= \frac{2}{100} \times 10{,}000$$

$$= Birr\ 200$$

Example 17: Elsa stationery has daily sales of Birr 205. The stationery is open 250 days per year. How much is the turn over tax payable by Elsa?

Solution: sales per year $= 205 \times 250 = Birr\ 51{,}250$

Turn over tax $= 2\%\ of\ sales$

$$= \frac{2}{100} \times 51{,}250$$

$$= Birr\ 1{,}025$$

Employment income tax

Employer deduct income tax from the employee before paying monthly salary based on the following tax rate (according to Ethiopian income tax rate)

Employment income (per month) in Birr	Employment income Tax rate
$\boldsymbol{above\ 0\ up\ to\ 600}$	0%
$\boldsymbol{above\ 600\ up\ to\ 1650}$	10%
$\boldsymbol{above\ 1650\ up\ to\ 3200}$	15%
$\boldsymbol{above\ 3200\ up\ to\ 5250}$	20%
$\boldsymbol{above\ 5250\ up\ to\ 7800}$	25%
$\boldsymbol{above\ 7800\ up\ to\ 10900}$	30%
$\boldsymbol{over\ 10,900}$	35%

Example18: The income tax for Birr 850 is calculated as follows

Solution: Birr 850 falls on 10%

- For interval $0\ to\ 600$, the tax rate is 0%, hence $Tax_1 = 0$
- For interval $600\ to\ 850$, the tax rate is 10%, hence tax on this interval is

$$Tax_2 = 10\%\ of\ (850 - 600)$$

$$Tax_2 = \frac{10}{100} \times 250$$

$$Tax_2 = 25$$

Income tax $= Tax_1 + Tax_2 = 0 + 25 = Birr\ 25$

Therefore, the income tax on Birr 850 is $Birr\ 25$.

Note: The above table shows the tax rate for each interval of total income.

Example 19: Alemu earns Birr 4850 per month, calculate

a. the income tax he has to pay.

b. the net income after deducting income tax.

Solution:

a. Birr 4850 falls on 20%

- For interval $0\ to\ 600$, the tax rate is 0%, hence $Tax_1 = 0$
- For interval $600\ to\ 1650$, the tax rate is 10%, hence tax on this interval is

$$Tax_2 = 10\%\ of\ (1650 - 600)$$

$$Tax_2 = \frac{10}{100} \times 1050 = Birr\ 105$$

- For interval $1650\ to\ 3200$, the tax rate is 15%, hence tax on this interval is

$$Tax_3 = 15\%\ of\ (3200 - 1650)$$

$$Tax_3 = \frac{15}{100} \times 1550 = Birr\ 232.5$$

- For interval $3200\ to\ 4850$, the tax rate is 20%, hence tax on this interval is

$$Tax_4 = 20\%\ of\ (4850 - 3200)$$

$$Tax_4 = \frac{20}{100} \times 1650 = Birr\ 330$$

Income tax $= Tax_1 + Tax_2 + Tax_3 + Tax_4$

$$= 0 + 105 + 232.5 + 330$$

$$= Birr\ 667.5$$

Therefore, the income tax on Birr 4850 is $Birr\ 667.5$.

b. net income is $4850 - 667.5 = Birr\ 4182.5$

Example 20: Derive a formula to calculate income tax on Birr x, in terms of x.

Where x falls on the interval $1650\ to\ 3200$

Solution: Birr x falls on 15%

- For interval $0\ to\ 600$, the tax rate is 0%, hence $Tax_1 = 0$
- For interval $601\ to\ 1650$, the tax rate is 10%, hence tax on this interval is

$$Tax_2 = 10\%\ of\ (1650 - 600)$$

$$Tax_2 = \frac{10}{100} \times 1050 = Birr\ 105$$

- For interval $1651\ to\ x$, the tax rate is 15%, hence tax on this interval is

$$Tax_3 = 15\%\ of\ (x - 1650)$$

$$Tax_3 = 0.15x - 247.5$$

Therefore, the income tax

$$Tax = Tax_1 + Tax_2 + Tax_3 = 0 + 105 + 0.15x - 247.5$$

$$Tax = 0.15x - 142.5$$

Exercise 4.3.5

1. Find the income tax and net income of the following employees of commercial Bank of Ethiopia
   a. Ato Ahmed with monthly salary of Birr 7500
   b. w/ro Mekdes with monthly salary Birr 11600
2. A shoe dealer purchased shoe from a shoe company that worth Birr 8000. Find the amount he should pay to the company including VAT

Summary for unit 4

- The method of comparing two or more quantities of the same kind and in the same units by division is known as a ratio.
- proportion is equality of two ratios.
- In proportion the product of means is equal to the product of extremes.

  $(i.e. if\ \frac{a}{b} = \frac{c}{d}\ , then\ a \times d = b \times c)$.
- y is said to be directly proportional to x (written as $y \propto x$), if there is constant k such that $y = kx\ or\ k = \frac{y}{x}$.
- y is said to be inversely proportional to x (written as $y \propto \frac{1}{x}$), if there is constant k such that $y = \frac{k}{x}\ or\ k = y \times x$.
- The word percent means "for every hundred" or "per 100". We use the symbol % to denote percent.
- Base(B) is the number that represents 100% or total value of something.
- Percentage(p) is the number or the amount that represents a part of the whole.
- Rate(R) is the ratio of percentage to base, written as percent.
- Percentage, base and rate are related by the formula

  $P = R \times B, \quad B = \frac{P}{R} \quad \text{and} \quad R = \frac{P}{B}$
- Percent increase is calculated as:

  $$percent\ increase = \frac{increased\ amount}{original\ quantity} \times 100\%$$
- Percent decrease is calculated as:

  $$percent\ decrease = \frac{decreased\ amount}{original\ quantity} \times 100\%$$
- The Price at which an article is purchased, is called its cost price.
- The Price at which an article is sold, is called its selling price.
- When $S.P > C.P$, then there is profit, and $Profit = S.P - C.P$
- When $S.P < C.P$, then there is loss, and $loss = C.P - S.P$

- Profit or loss percent is given by the formula

$$Profit\ \% = \frac{Profit}{cost\ price} \times 100\% \quad \text{and} \quad Loss\ \% = \frac{Loss}{cost\ price} \times 100\%$$

- The interest paid on original principal only during the whole interest period is called simple interest.

$$Interest = principal \times Rate \times Time \text{ Or } I = PRT$$

- Compound interest is the interest on a loan calculated based on the initial principal plus the accumulated interest from previous periods.

$$A = P(1 + R)^T \quad and\ I = A - P$$

- **Value added tax (VAT)** is a tax imposed by government on sales of some goods and services.
- In Ethiopia VAT rate is 15%.
- Turn over tax is imposed on merchant, who are not required to register for VAT, but supply goods and services in the country.
- In Ethiopia turn over tax rate is 2% on goods sold and services rendered locally.

Review exercise for unit 4

I. Write True for the correct statements and False for the incorrect statements.

1. Ratio can be defined as "for every hundred".
2. $4:3x = 32:72\ if\ x = 3$.
3. The ratio of 15 meters to 2000 centimeters is 3:4
4. $Loss\ \% = \frac{loss}{cost\ price} \times 100\%$
5. If 250 divided in the ratio $3:7$, the smaller part is 25.
6. Proportion is the equality of two ratios.
7. If y is inversely proportional to x, then as x increases y also increases.
8. Profit is made when the selling price is less than the cost price.

II. Choose the correct answer from the given alternatives.

9. In a school, 58% of the total number of students is boys. If the number of girls is 840, how many students are there in the class?

   a. 2000 b. 1800 c. 1500 d. 2100

10. $Birr$ 300 is invested at 6% simple interest per annum. How long will it take for the interest to be $Birr$ 180?

    a. 12 years b. 10 years c. 8 years d. 24 years

11. The decimal form of $\frac{3}{4}\%$ is

    a. 0.75 b. 0.075 c. 0.0075 d. 0.057

12. The simple interest on Birr 400 invested for 4 months was Birr 12. What is the annual (yearly) rate of interest?

    a. 36% b. 4.8% c. 7% d. 9%

13. If $a, b\ and\ c$ are numbers such that $a:b:c = 1:3:4$ and $c = 28$ then, the sum of $a + b + c$ is:

    a. 56 b. 60 c. 72 d. 84

14. If $A:B = 7:8$ an $B:C = 12:7$, $A:C$ in its simplest form is:

a. $2:3$ b. $3:2$ c. $19:6$ d. $4:5$

15. Three numbers m, n, p are in the ratio $3:6:4$. Which of the following is the value of $\frac{4m-n}{n+2p}$.

a. $\frac{7}{3}$ b. $\frac{3}{7}$ c. $\frac{2}{3}$ d. $\frac{4}{3}$

16. The ratio of the number of pigs to the number of horses in a farm is 2 to 3. If there are 24 horses, what is the number of pigs?

a. 72 b. 48 c. 36 d. 16

17. If $24, x, 18$ *and* $(x-1)$ are in proportion. Find the value of x.

a. 6 b. 4 c. 12 d. 8

18. If 85% of 'a' is 'b', then which of the following is true?

a. $a = b$ b. $a < b$ c. $20a = 17b$ d. $a > b$

19. which one of the following is not equal to the rest?

a. 2% of 150 b. $\frac{3}{2}$% of 400 c. 5% of 60 d. 6% of 50

20. Which one of the following greater?

a. 20% of 45 b. 25% of 60 c. 2% of 800 d. 74% of 20

III. Work out problems

21. Find $\frac{7}{12}$ of 8400?

22. There are 40 questions on a final exam and Yohannes answered 75% of them correctly. How many questions correctly answered?

23. If 'b' is directly proportional to 'a'and b=30 when a=20, then

a. Find the equation relating a and b

b. Find a when b=42

24. In a given class room, the number of girls is 10 greater than the number of boys. If the ratio of the number of girls to the number of boys is 7:5 then find
    a. The number of girls
    b. The number of boys
    c. The total number of students
25. A company has 420 workers of which 189 are male and the rest are female. What percent of the workers are female?
26. A trader bought a TV set for Birr 22000 and sold it at loss of 5%. what was the selling price?
27. Mussa received a 6% pay raise. He now earns Birr 8268, what was his salary prior to the salary increase?
28. Birr 6000 is invested at a rate of 5% compound interest compounded annually, Find
    a. the amount at the end of 2 years.
    b. the interest at the end of 2 years.
29. A person wants to buy a car from TOYOTA Company. If the price car including VAT is Birr 5,750,000 then,
    a. what is the price of car before VAT?
    b. what is the value of VAT?
30. Calculate the income tax on salary of Birr 5200.
31. Derive a formula to calculate income tax on Birr x, in terms of x. Where x falls on the interval $5250\ to\ 7800$.

Unit 5
Perimeter and Area of Plane figures

Learning Outcomes:

At the end of this unit, students will able to:

- *classifies the different kinds of triangles.*
- *constructs and describe properties of four- sided figures*
- *find the perimeter of triangle, parallelograms, trapezium, rhombus and composite shapes*
- *derive formula for area of triangle, parallelograms, trapezium and rhombus.*
- *calculate areas of triangle, parallelograms, trapezium, rhombus and composite shapes*
- *calculate circumference and area of circle.*
- *apply the concept of area and perimeter of Plane figures in solving real life problems*

Introduction

In this unit you will learn the different kinds of geometric figures and their properties such as triangle, rectangle, square, rhombus, and trapezium. In addition to their properties you will learn, how to calculate the area and perimeter of these geometric figures.

The unit has five sections. The first section deals with revision of triangles, second section deals with four - sided figures, the third section deals with Perimeter and Area of four-sided figures.

The fourth section deals with Perimeters and Areas of triangles. Finally, the last section deals about application of the concept of area and perimeter of Plane figures in solving real life problems.

5.1. Revision of triangles

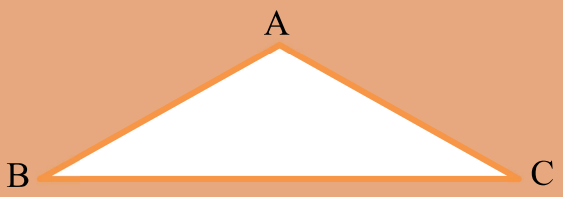
At the end of this section you should be able to:

- *classify the different kinds of triangles.*

Activity 5.1.1

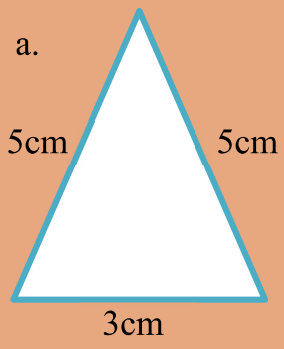
Discuss with your friends/partners

1. Define triangle
2. Name the given triangle.

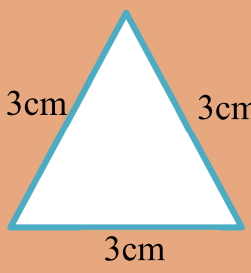


3. Classify the following triangles based on their sides.

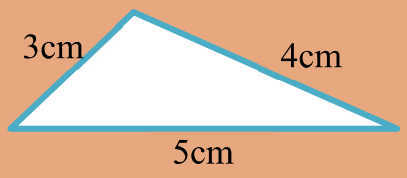
a.



b.

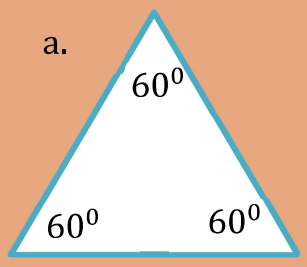


c.

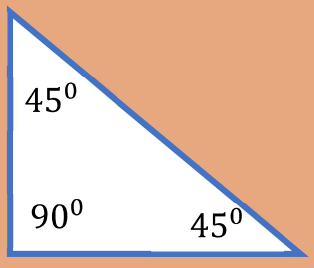


4. Classify the following triangle based on their angles.

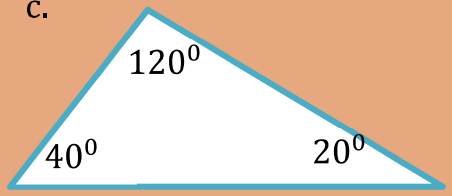
a.



b.

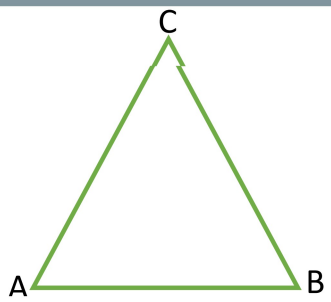


c.



5. Draw a triangle and measure the interior angle of this triangle by using protractor. Find the sum of all interior angles?

Definition 5.1: A triangle is a simple closed plane figure made of three-line segments.



- The line segments forming the triangle are sides of the triangle.

  $\overline{AC}, \overline{AB}$ *and* $\overline{BC}$ are the sides of the above triangle.

- The point where the sides of the triangle meet is called vertex (plural form vertices).

  A, B *and* C are the vertices of the triangle.

- Angles of a triangle are formed by two sides of the triangle meeting at a point (vertex).
- The interior angles of the above triangle are:
  i. The angle at vertex A, $\angle BAC$ *or* $\angle A$
  ii. The angle at vertex B, $\angle ABC$ *or* $\angle B$
  iii. The angle at vertex C, $\angle BCA$ *or* $\angle C$
- The sum of interior angles of a triangle is 180^0.

  $\angle A + \angle B + \angle C = 180^0$
- A triangle is named by using its vertices.

The above triangle is named as triangle ABC, symbolically, ΔABC. It can also be named using any other sequence of vertices like triangle BCA, symbolically, ΔBCA.

Note:

1. A triangle has three sides, three angles, and three vertices.
2. The sum of all interior angles of a triangle is always equal to 180°.

Types of triangles

Triangles can be classified in 2 different ways:

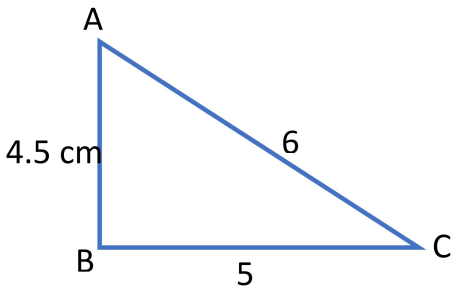
- Classification of triangle according to interior angles.
- Classification of triangle based on sides length.

Classification of triangle based on sides length

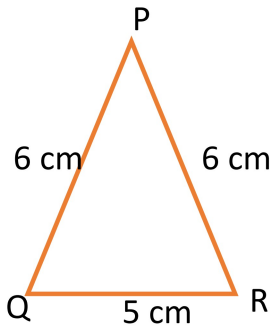
Based on side length, triangles are classified in to three:

1. Scalene Triangle
2. Isosceles Triangle
3. Equilateral Triangle

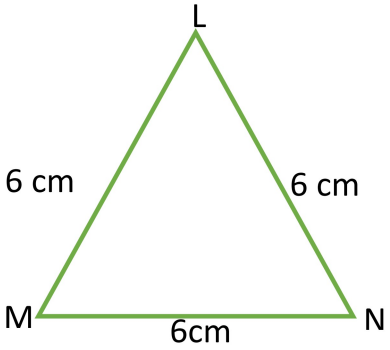
- A triangle in which all the three sides are unequal in length is called a *scalene triangle*.
- A triangle in which two of its sides are equal is called *isosceles triangle*.
- A triangle in which all its three sides are equal in length is called an *equilateral triangle*.



Scalene Triangle



Isosceles Triangle



Equilateral Triangle

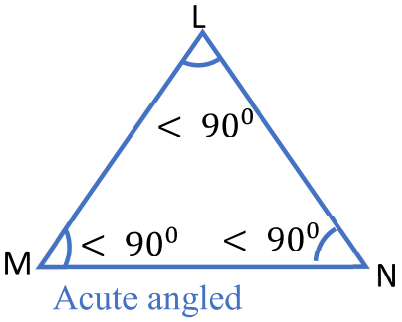
Classification of a triangle according to interior angles

Based on the measure of interior angles, triangles are classified in to three:

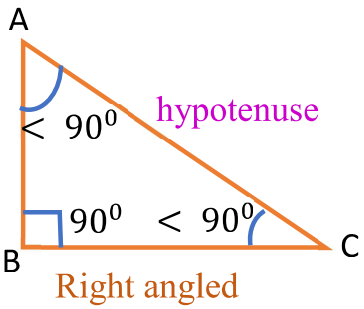
1. Acute Angled Triangle
2. Right-Angled Triangle
3. Obtuse Angled Triangle

- A triangle with all three angles less than 90° is an acute angle triangle.
- A triangle with one angle measuring exactly 90° is a right- angle triangle.
- The other two angles of a right-angle triangle are acute angles.
- The side opposite to the right angle is the largest side of the triangle and is called the hypotenuse.

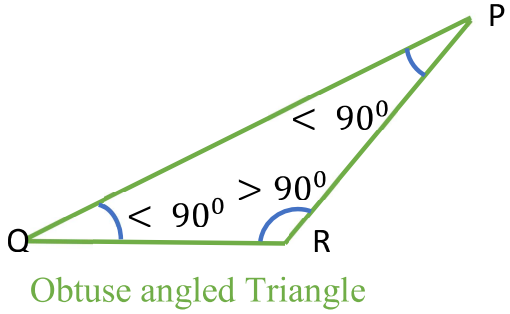
- A triangle with one obtuse angle (greater than 90°) and two acute angles is an obtuse angle triangle.



Acute angled



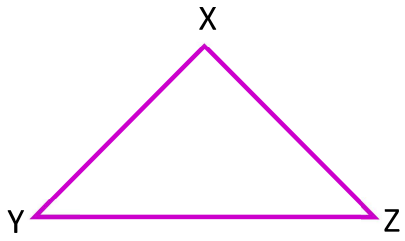
Right angled



Obtuse angled Triangle

Exercise 5.1.1

1. Fill in the blank space with the correct answer.
   a. The triangle in which all its sides are equal is called_____.
   b. _____is a triangle in which all its sides are different length.
   c. Each angle of equilateral triangle is _____.
   d. _____ is a triangle with two equal sides.
   e. In obtuse angled triangle one angle is _____than 90°.
   f. Each angle of an acute triangle is _____than 90°.
   g. The sum of interior angles of a triangle is equal to_____.
2. consider the following triangle.



   i. List the sides, angles and vertices of the triangle
   ii. Name the triangle

3. Classify the following triangle:
   a. Sides of the triangle are 4 cm, 4 cm and 7 cm
   b. Angles of the triangle are 90°, 60° and 30°
   c. Angles of the triangle is 110°, 40° and 30°
   d. Sides of the triangle are 5 cm, 13 cm and 12 cm
   e. Angles of the triangle are 60°, 60° and 60°